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(54) **SOFTWARE DEFINED NETWORK (SDN)-
BASED DOMAIN NAME SYSTEM (DNS)
RESOLUTION FOR A NON-SDN SERVICE**

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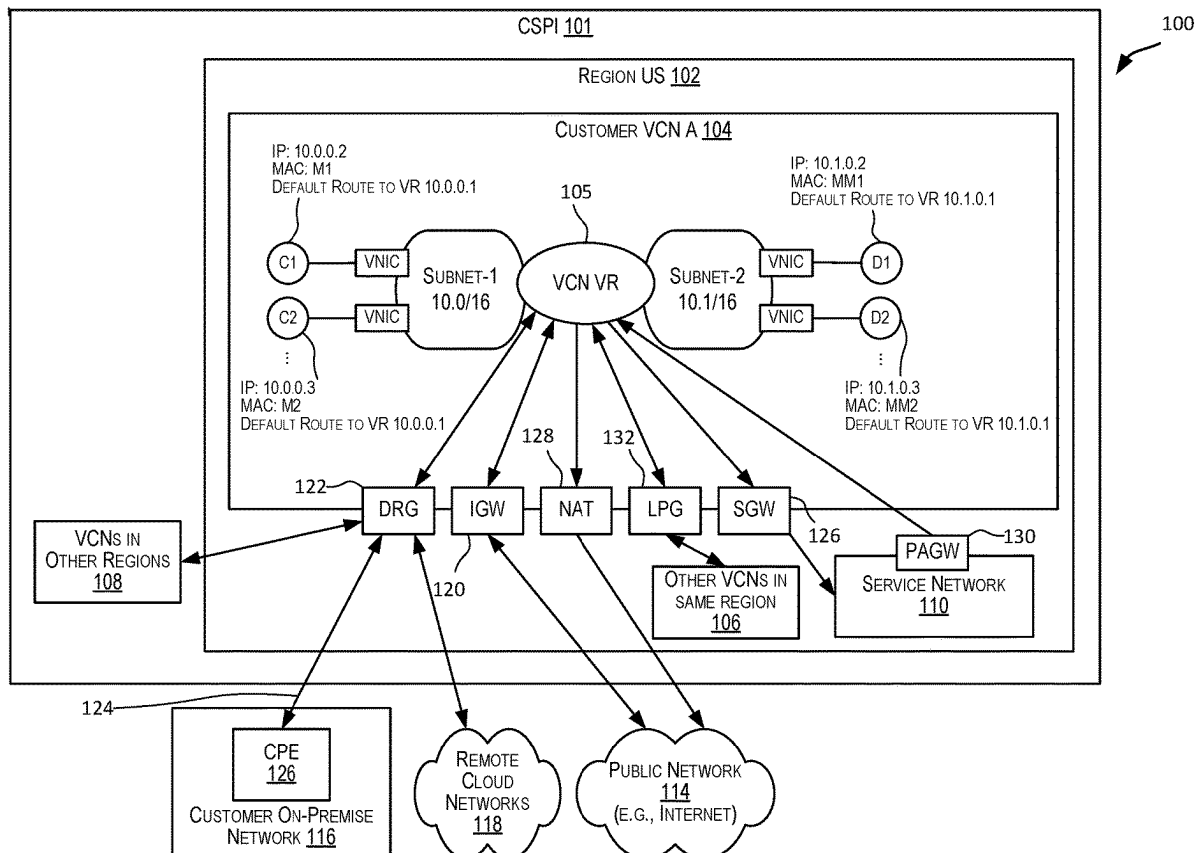
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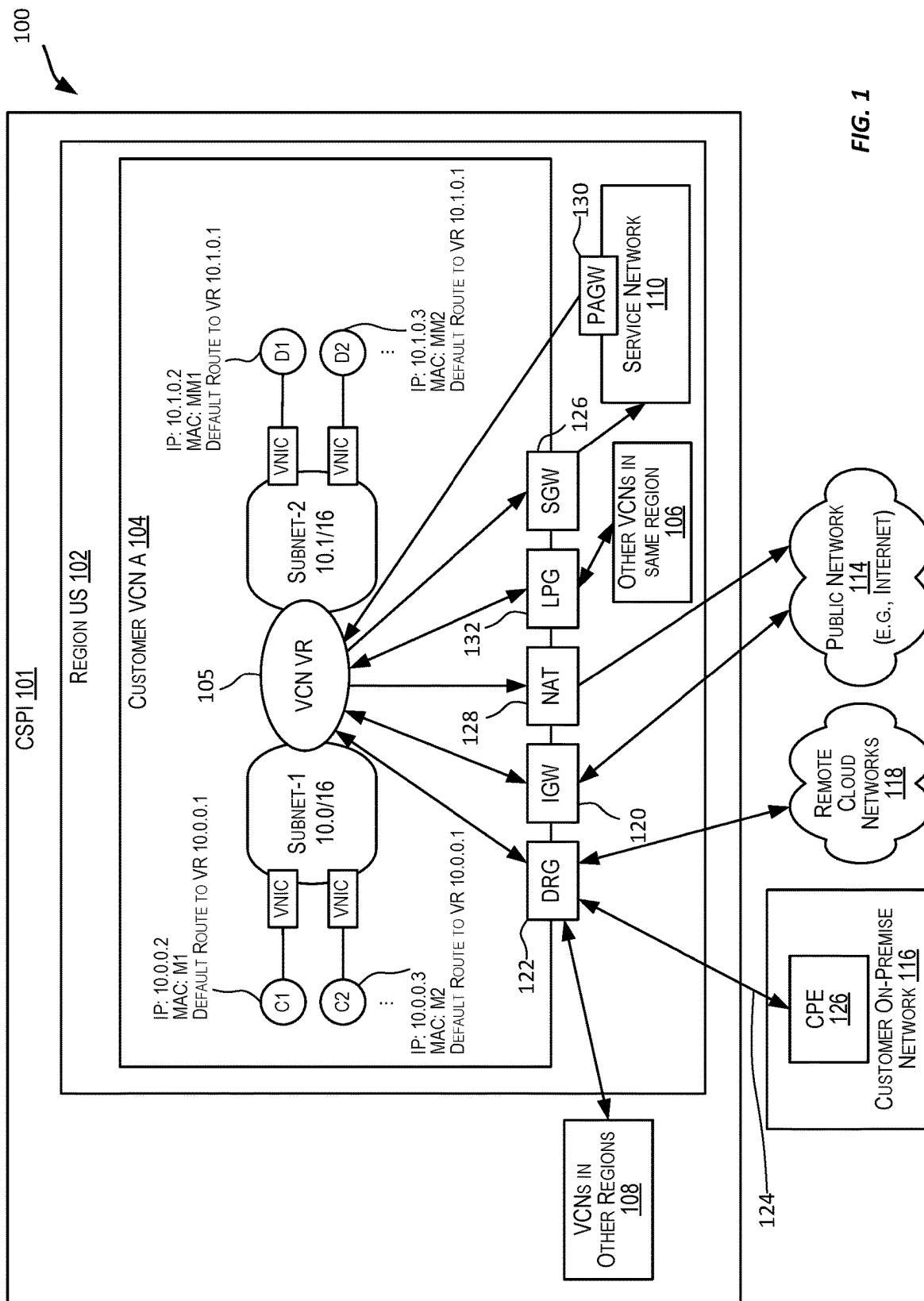
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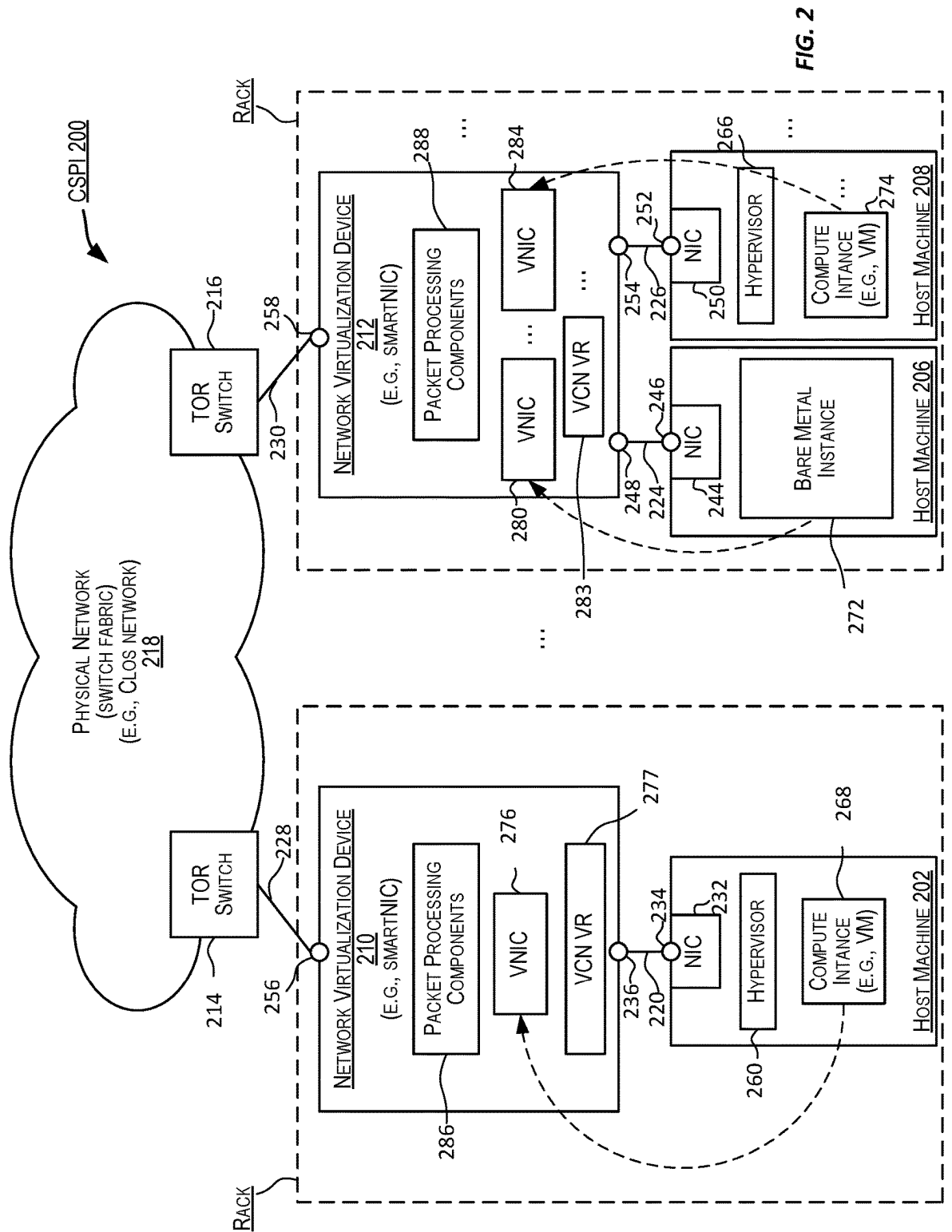
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(57) **ABSTRACT**

The present disclosure relates generally to an SDN providing DNS resolution for a non-SDN service. In particular, the SDN hosts a first endpoint communicatively coupled with the non-SDN service. The SDN also hosts a DNS proxy in support of a second private network of a customer. A second endpoint of the customer can be accessible via the second private network. The non-SDN service can send a DNS resolution request (that includes, for example, an FQDN of the second endpoint) to the first endpoint. The first endpoint forwards this request to the DNS proxy. In turn, the DNS proxy determines the IP address of the second endpoint and returns a reserved IP address corresponding to the DNS proxy and mapped to the second endpoint's IP address. The first endpoint sends this DNS resolution to the non-SDN service that then learns that the FQDN is resolved to the reserved IP address.







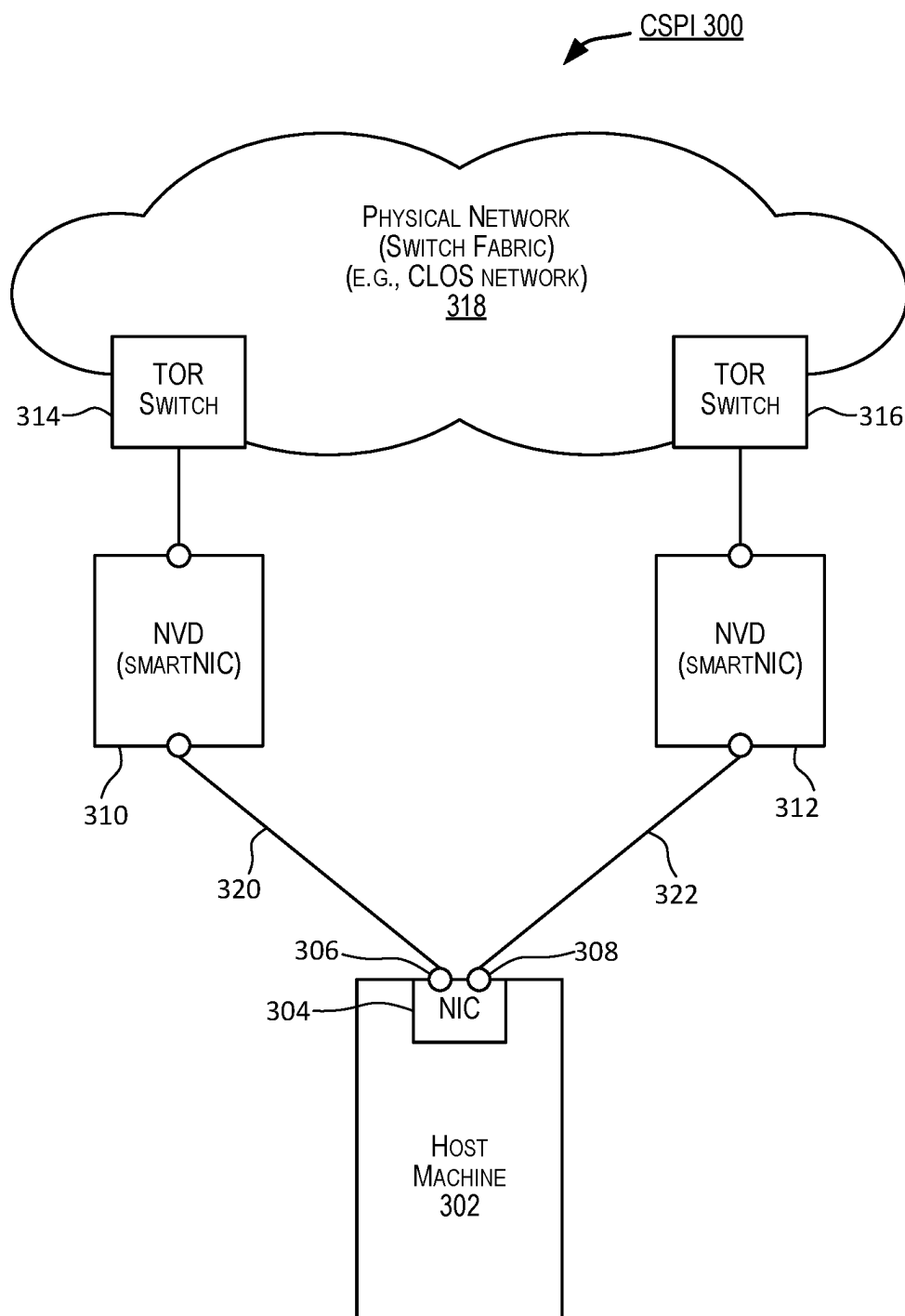


FIG. 3

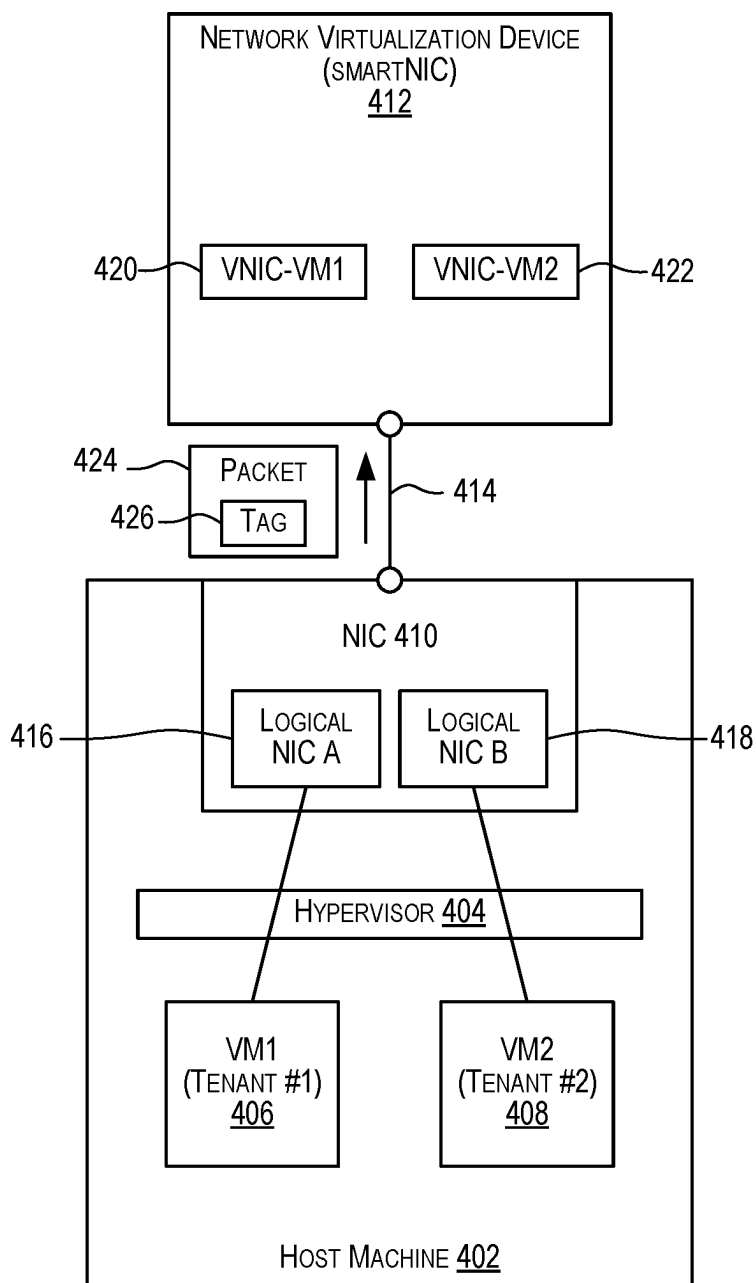


FIG. 4

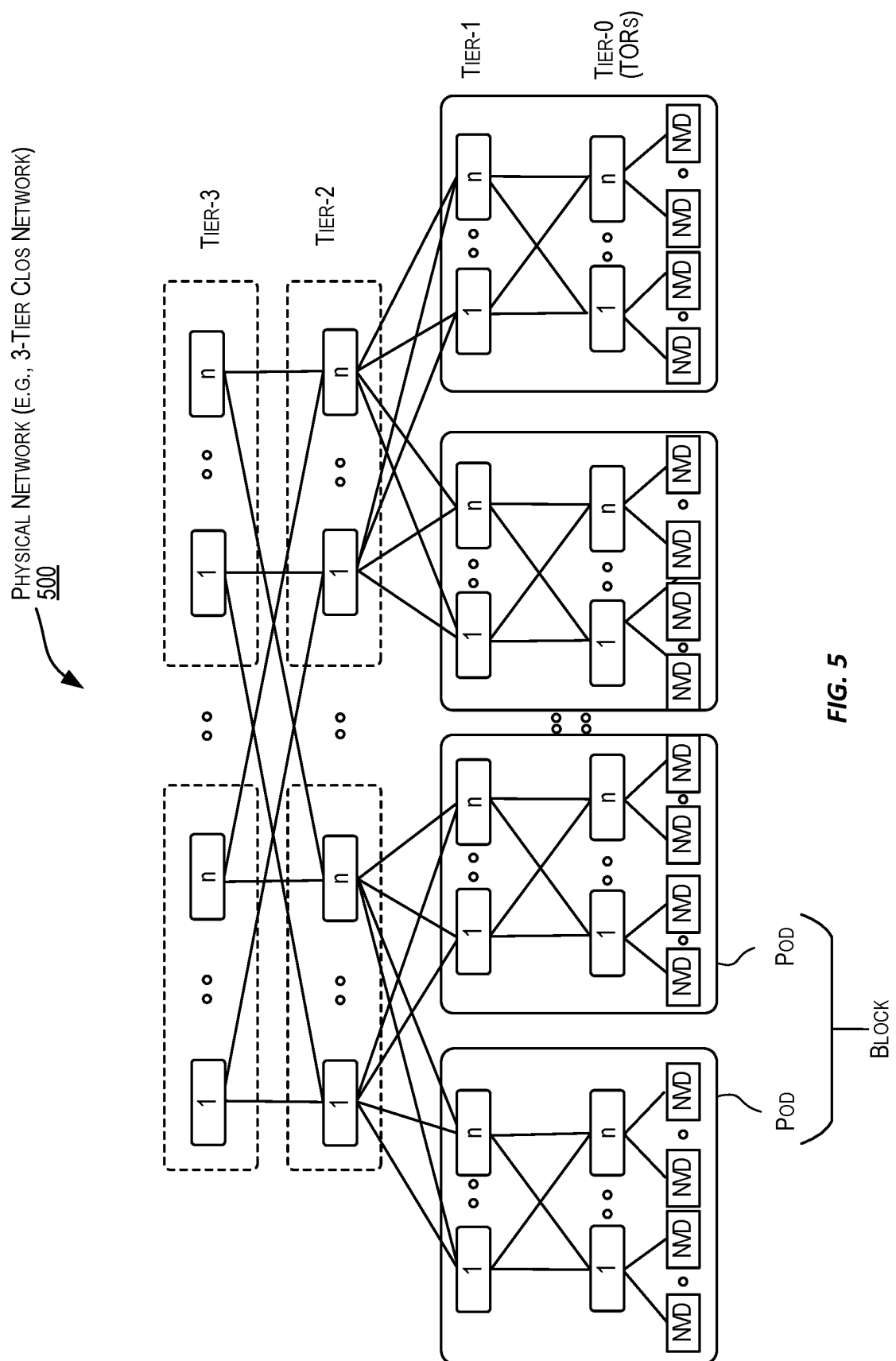


FIG. 5

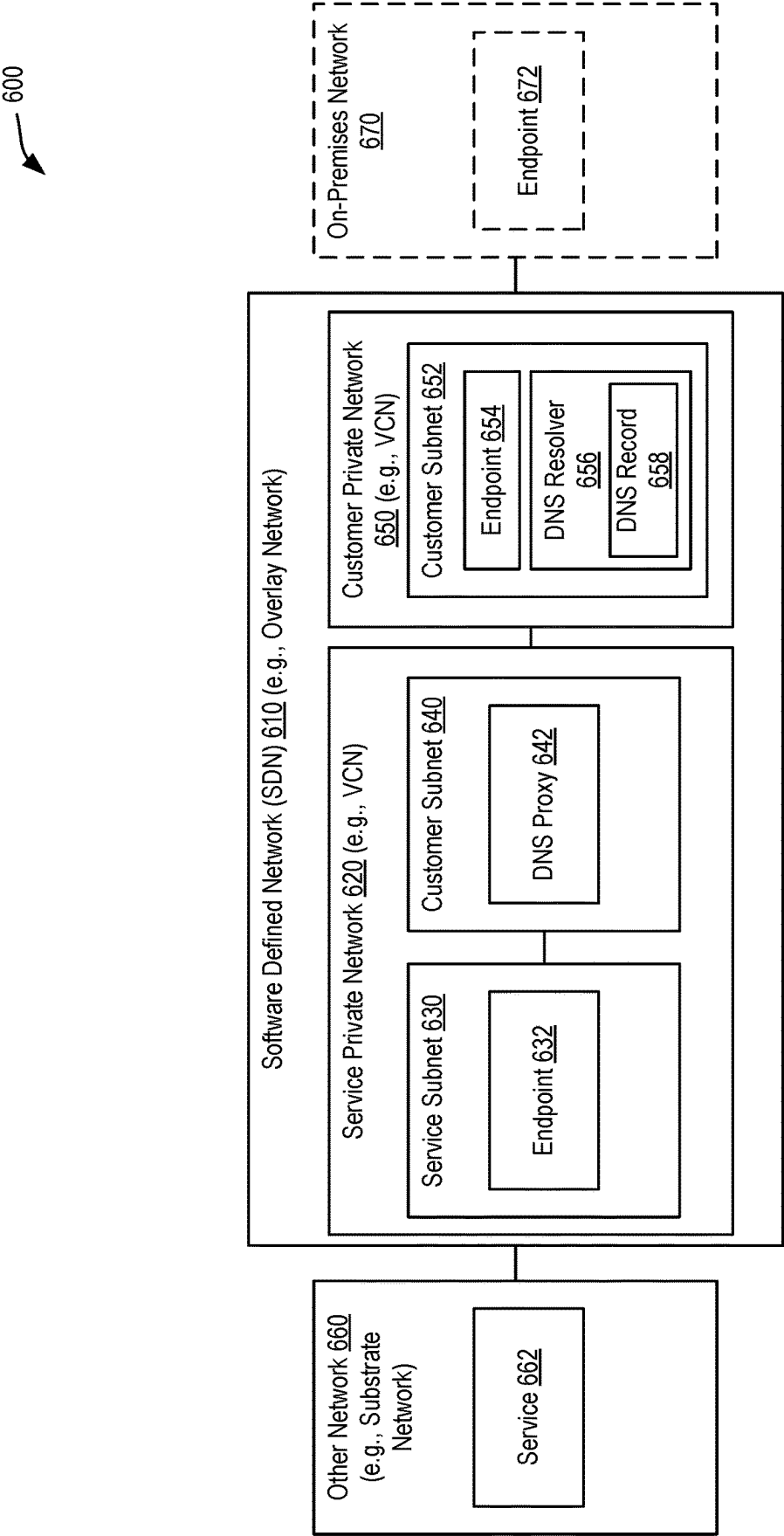


FIG. 6

700

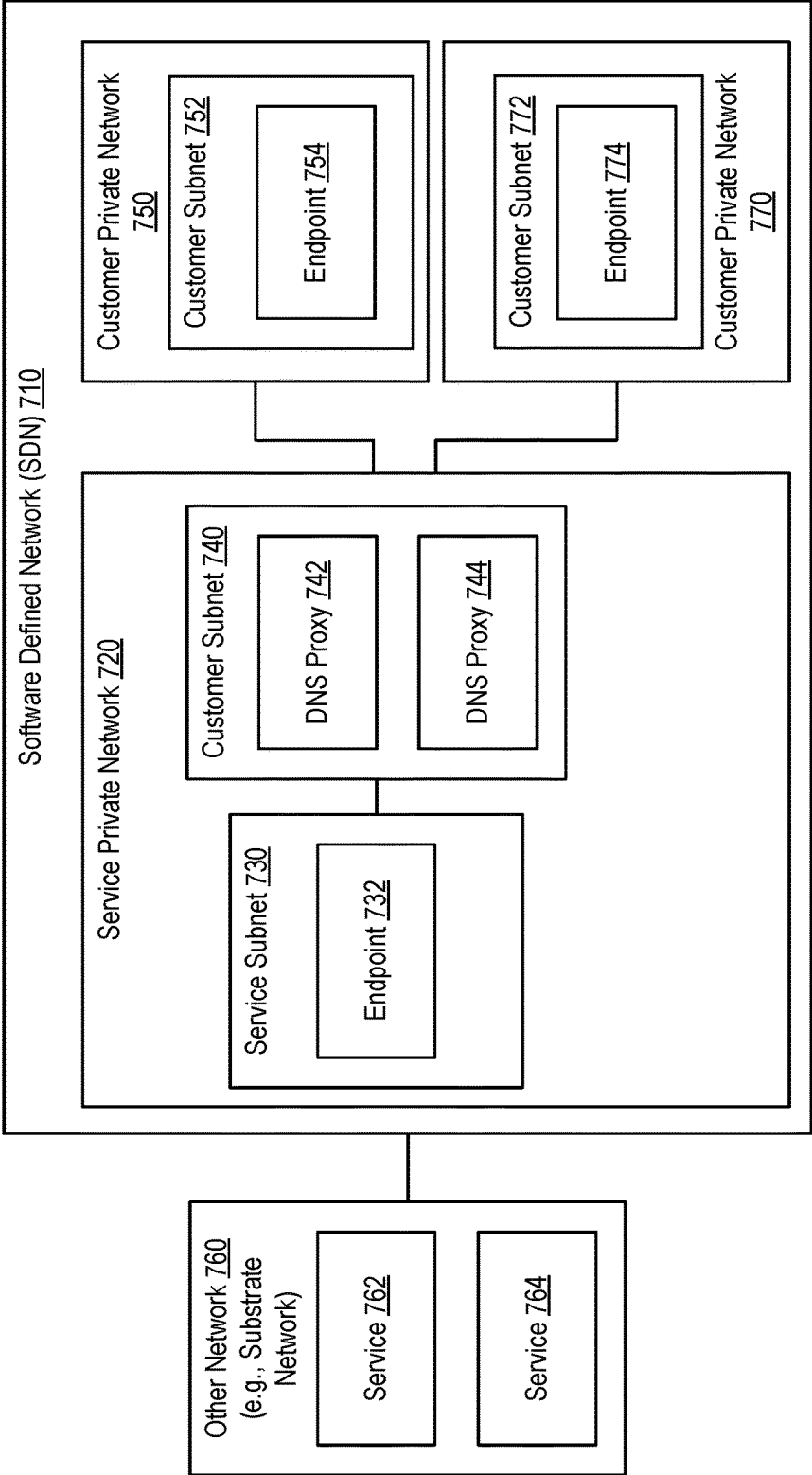


FIG. 7

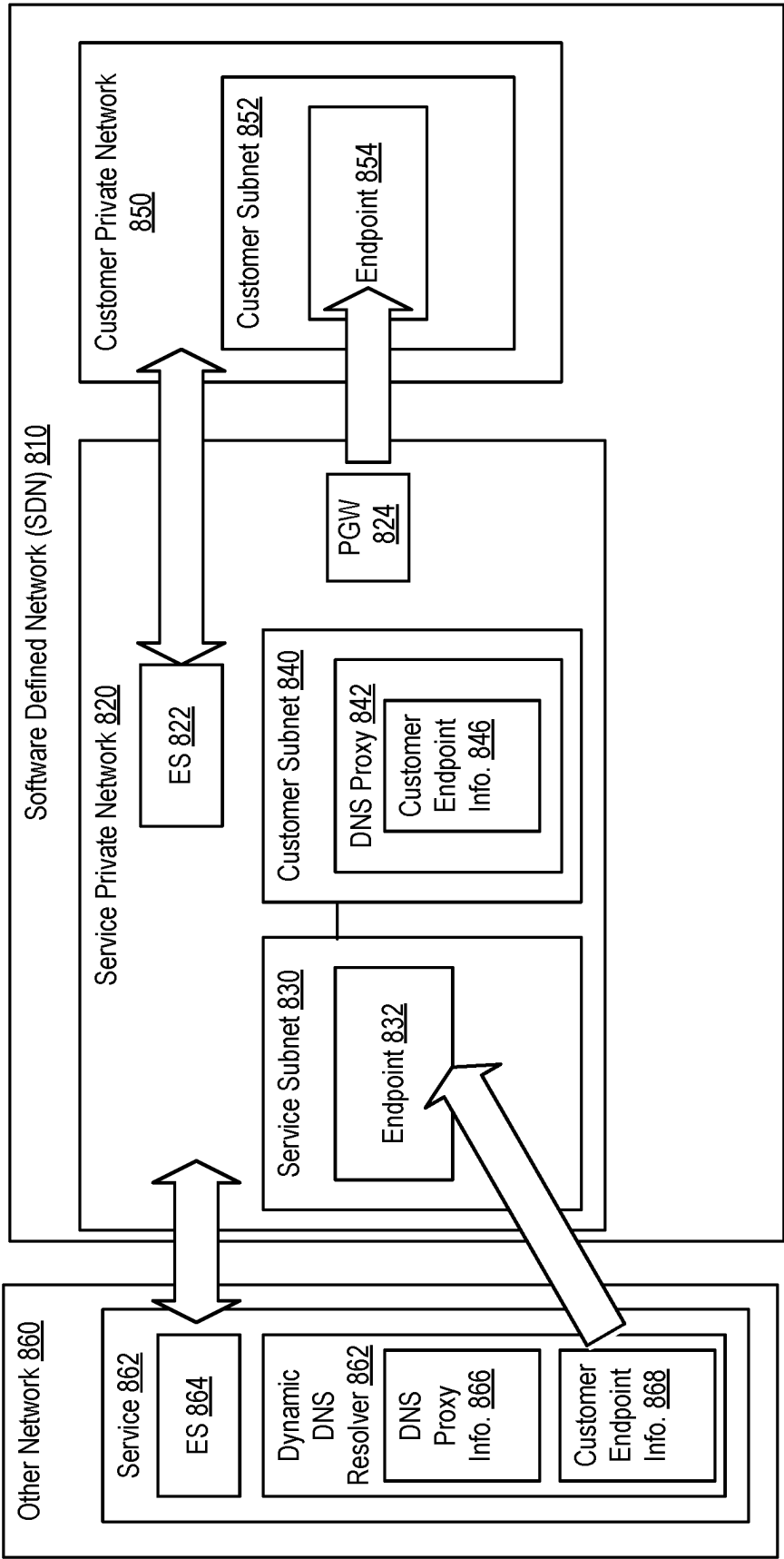


FIG. 8

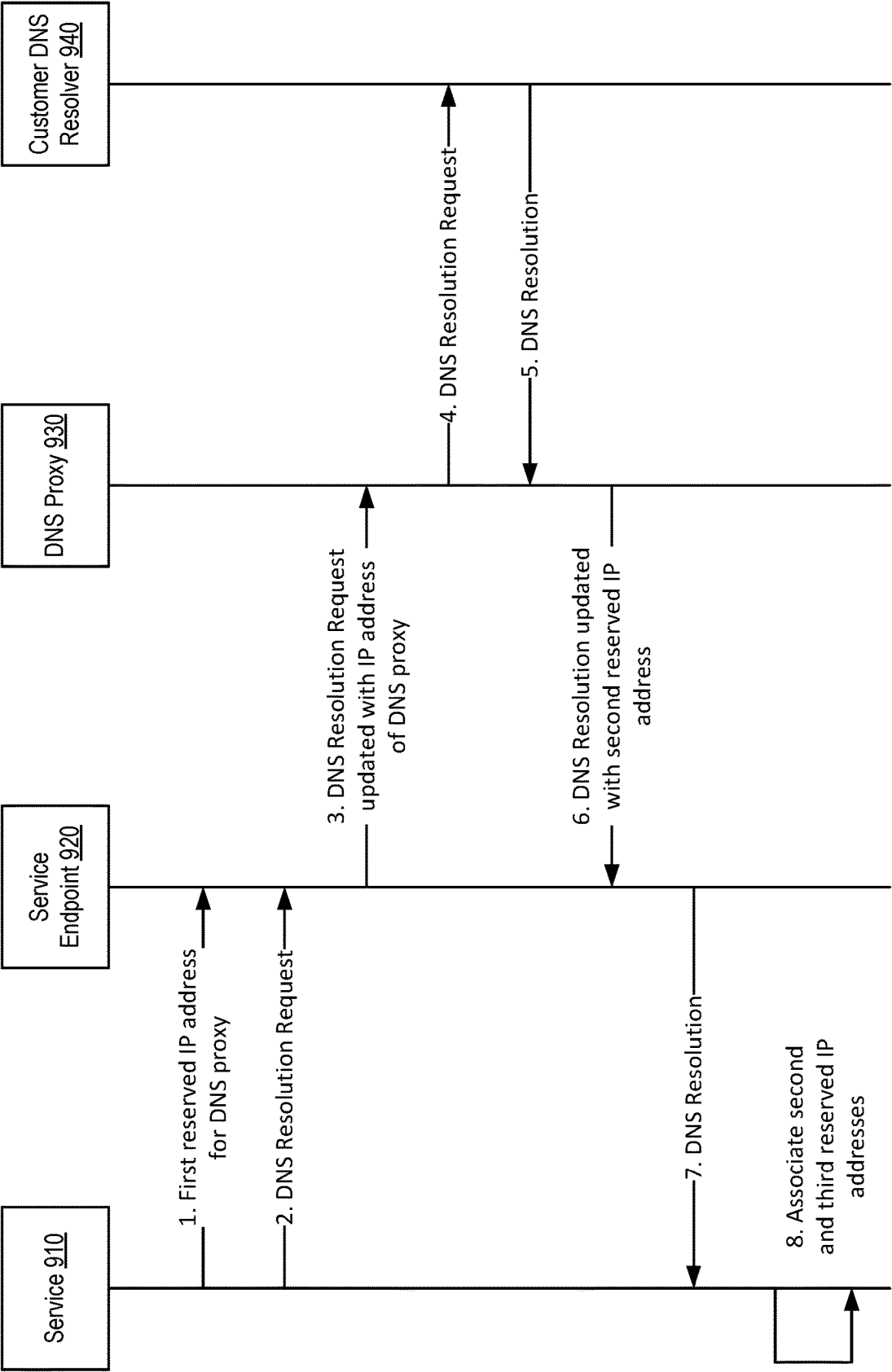


FIG. 9

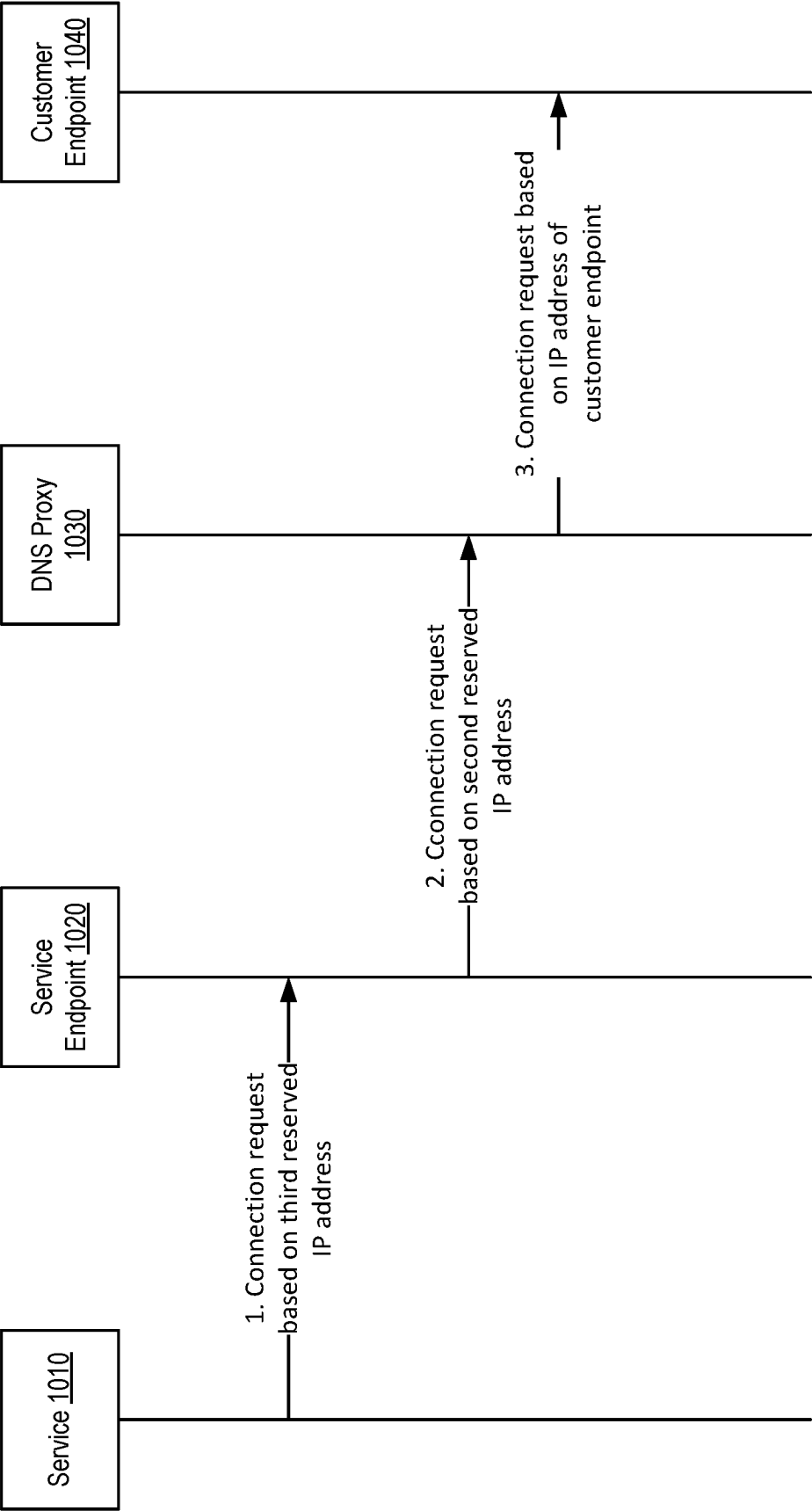


FIG. 10

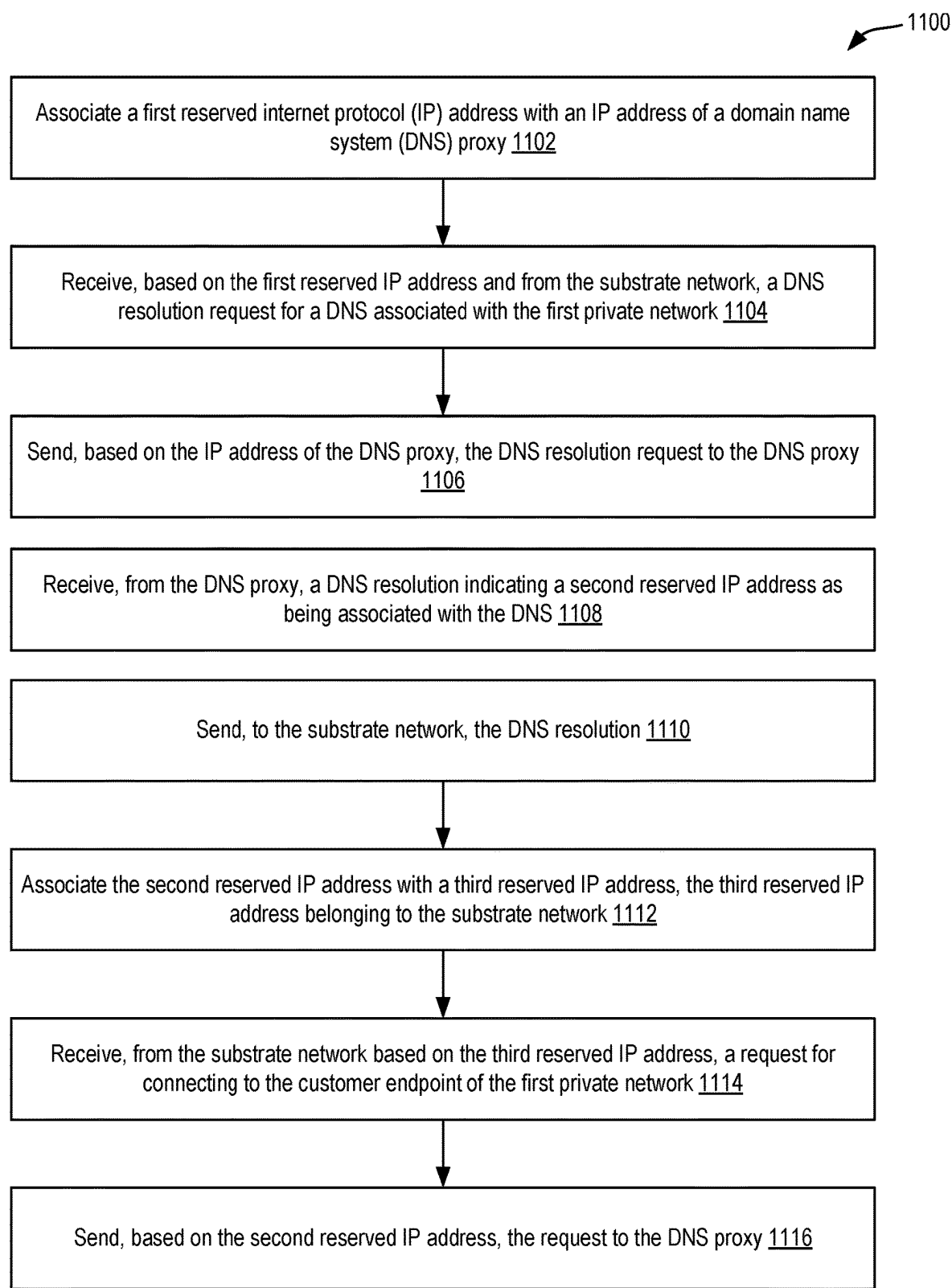


FIG. 11

1200

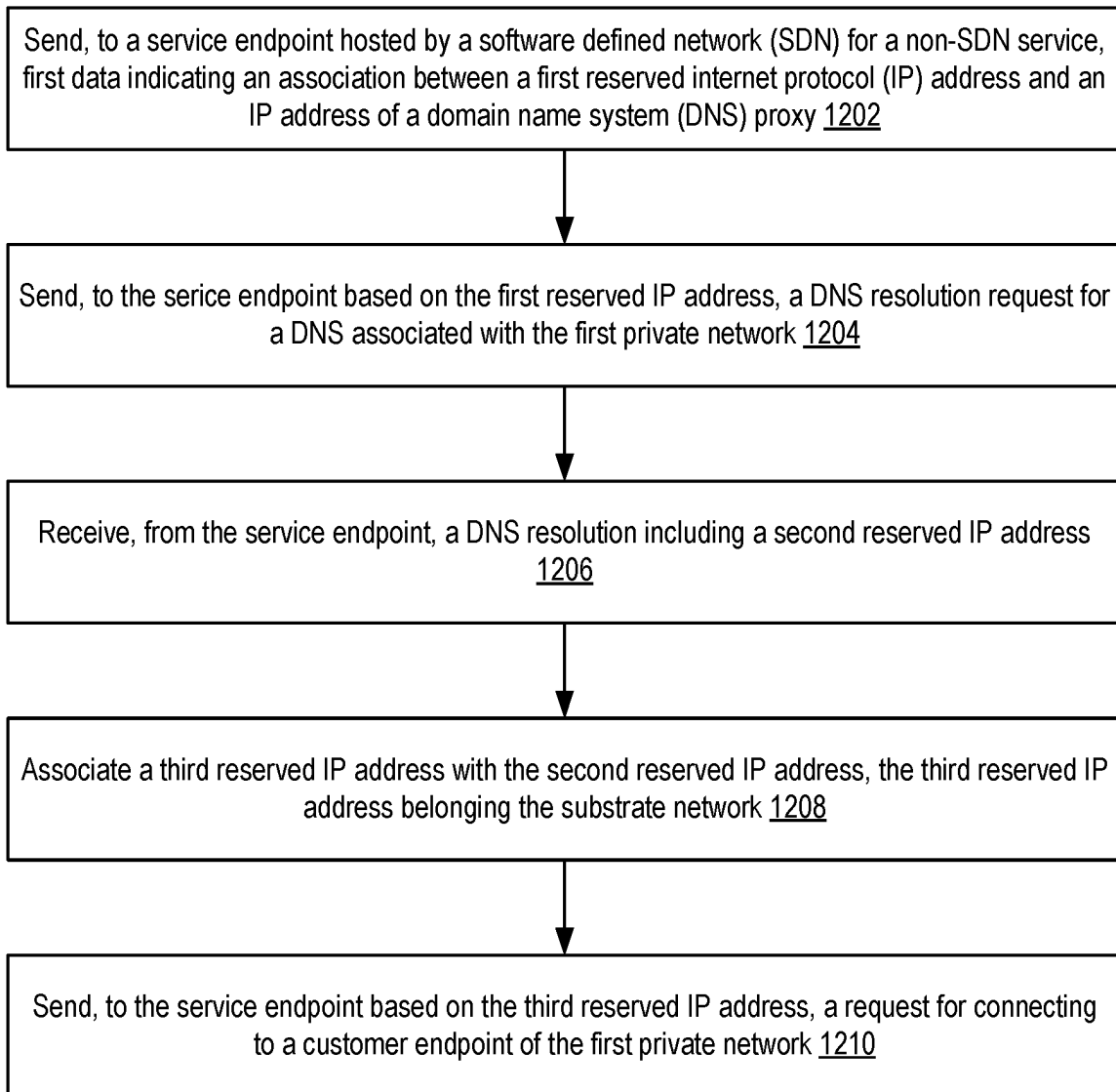


FIG. 12

FIG. 13

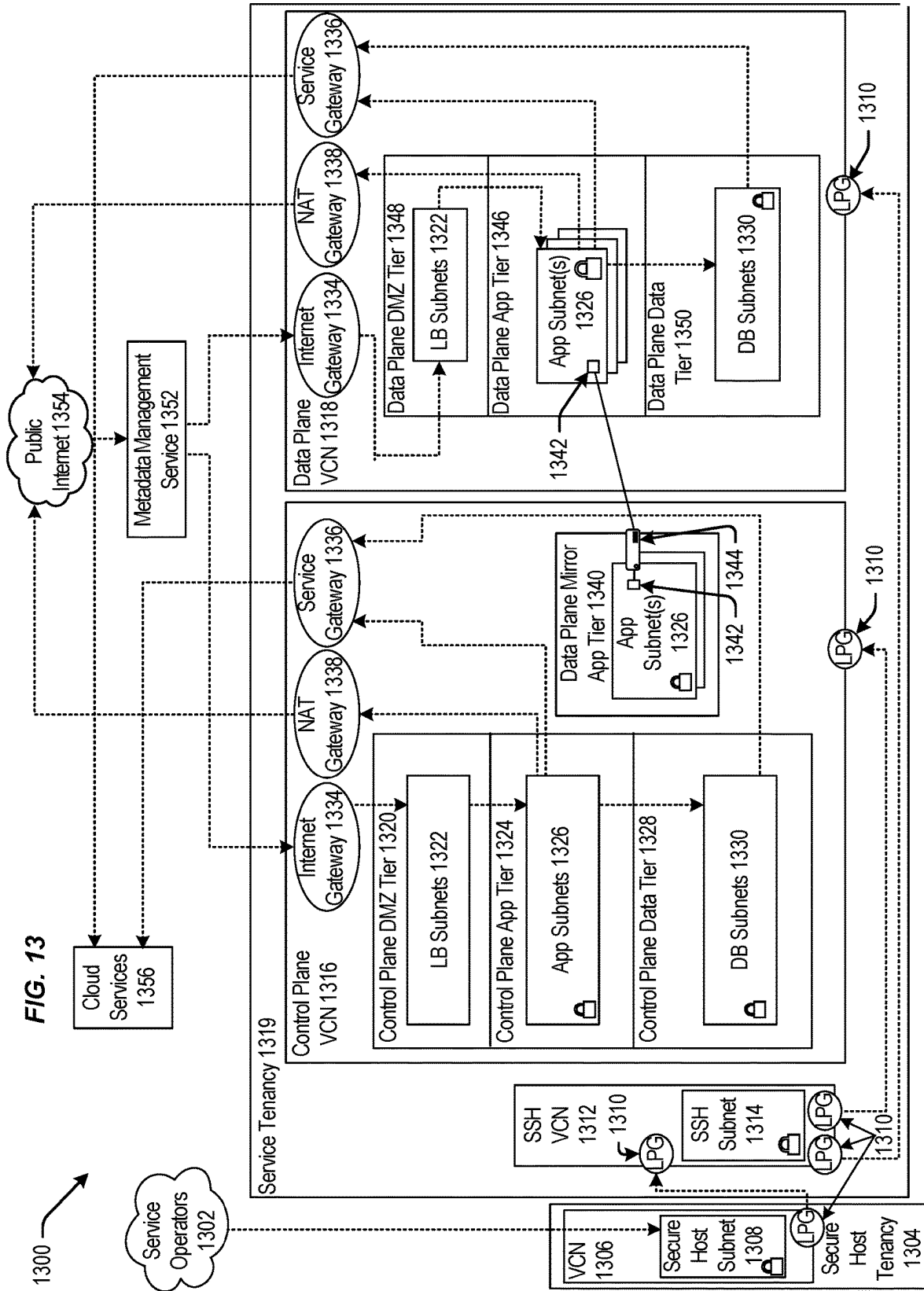
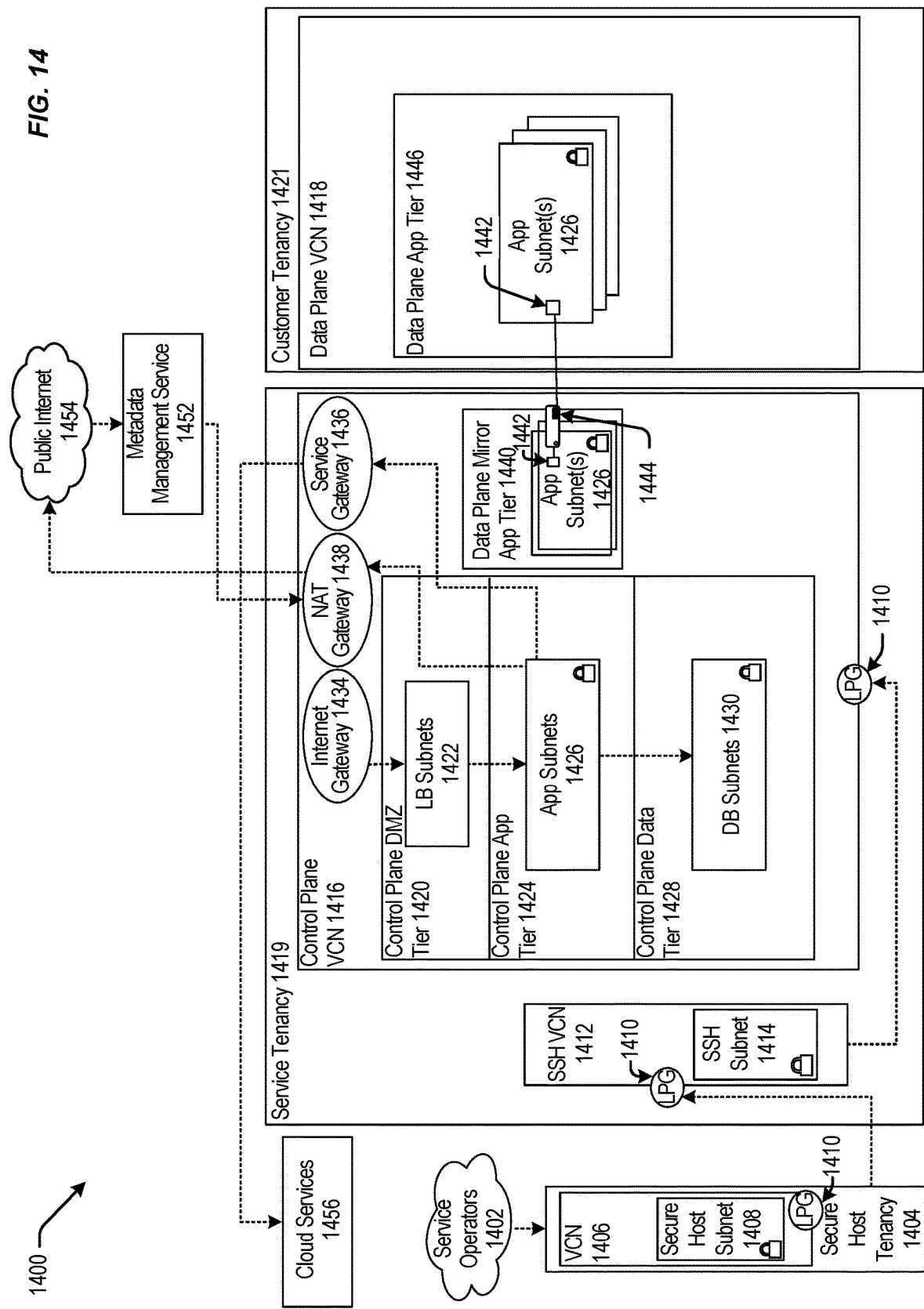


FIG. 14



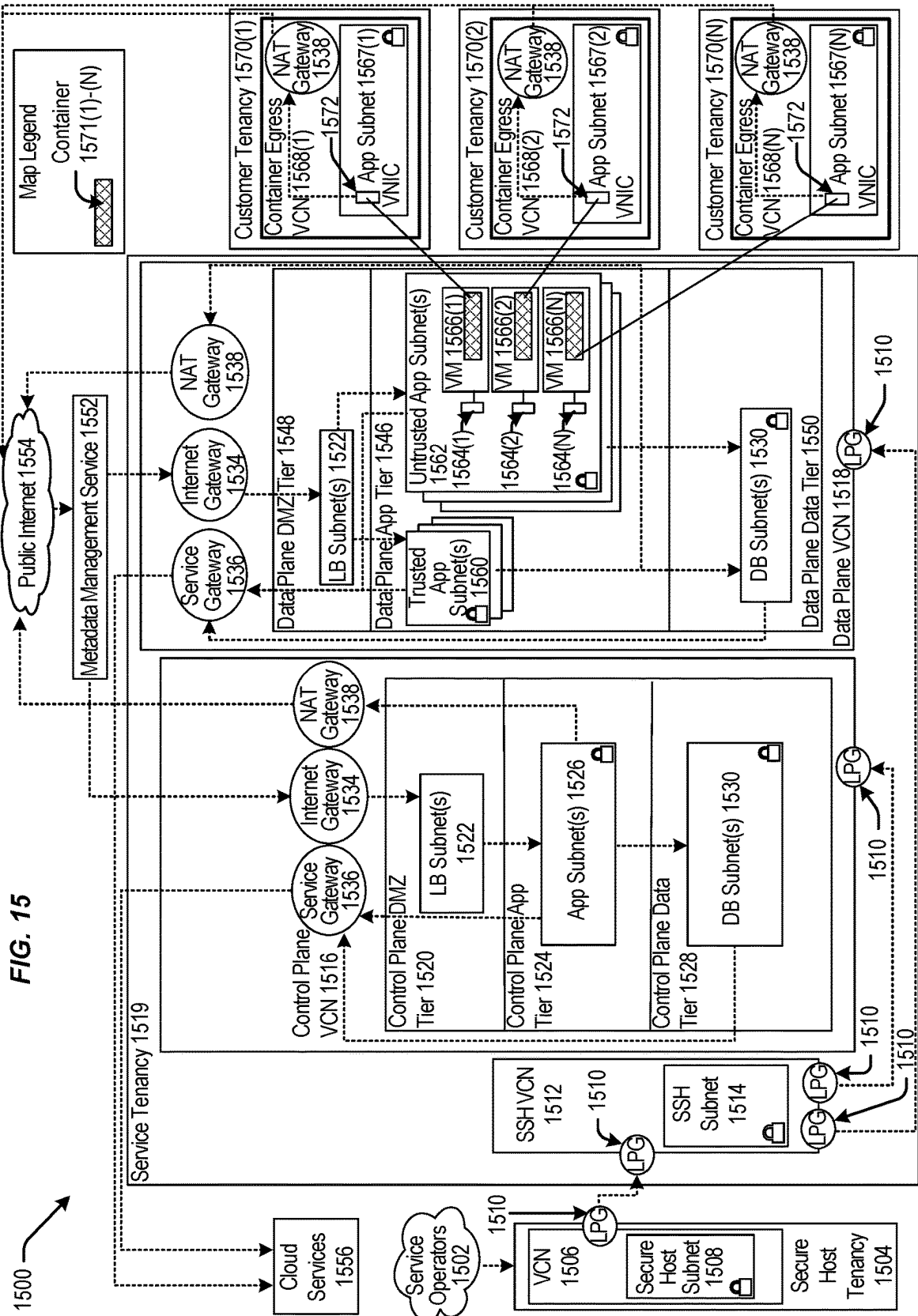
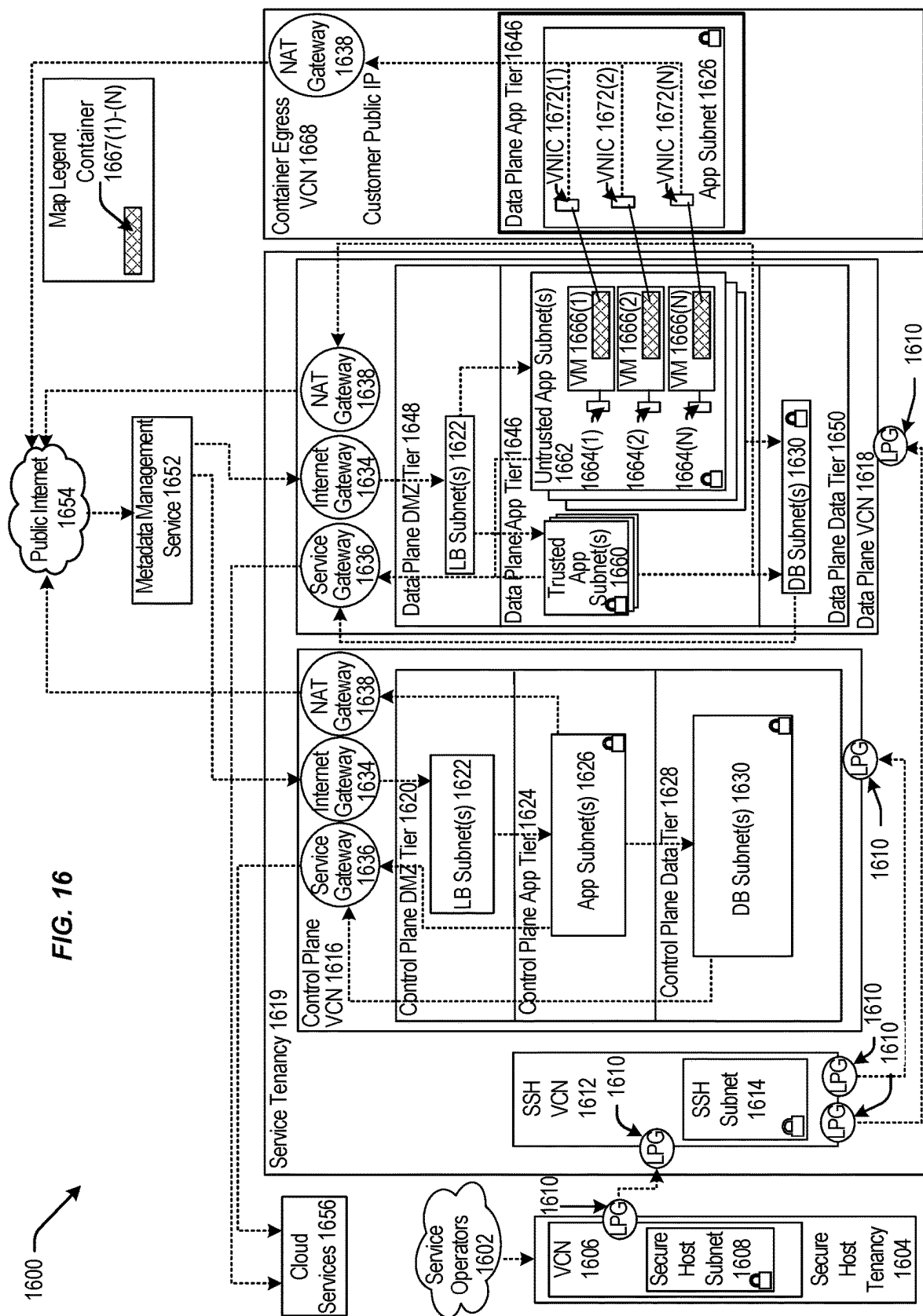


FIG. 16



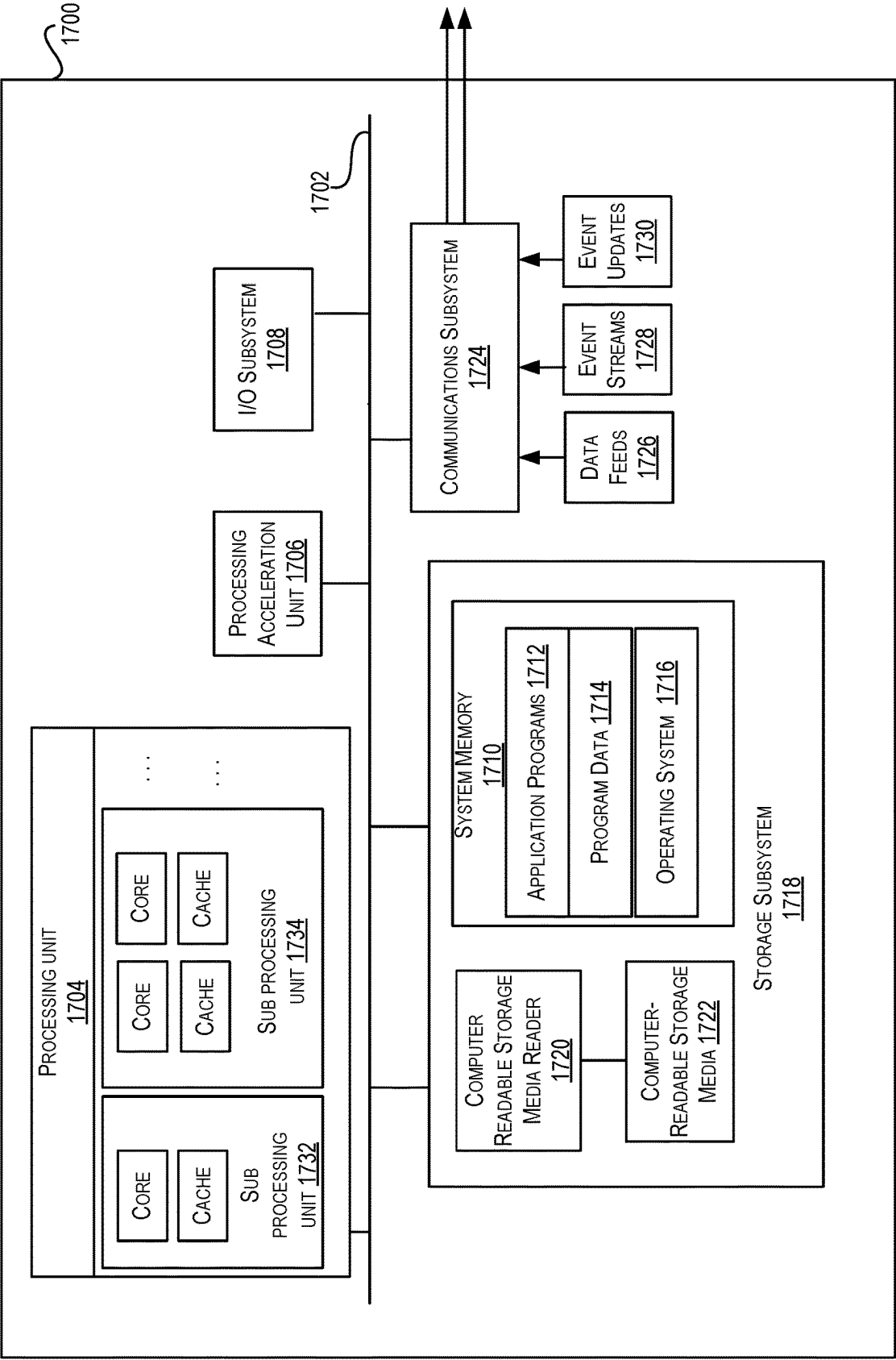


FIG. 17

SOFTWARE DEFINED NETWORK (SDN)- BASED DOMAIN NAME SYSTEM (DNS) RESOLUTION FOR A NON-SDN SERVICE

BACKGROUND

[0001] The last few years have seen a dramatic increase in the adoption of cloud services and this trend is only going to increase. Various different cloud environments are being provided by different cloud service providers (CSPs), each cloud environment providing a set of one or more cloud services. The set of cloud services offered by a cloud environment may include one or more different types of services including but not restricted to Software-as-a-Service (SaaS) services, Infrastructure-as-a-Service (IaaS) services, Platform-as-a-Service (PaaS) services, and others.

[0002] While various different cloud environments are currently available, each cloud environment provides a closed ecosystem for its subscribing customers. As a result, a customer of a cloud environment is restricted to using the services offered by that cloud environment. There is no easy way for a customer subscribing to a cloud environment provided by a CSP to, via that cloud environment, use a service offered provided by a different CSP. Embodiments discussed herein address these and other issues.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Features, embodiments, and advantages of the present disclosure are better understood when the following Detailed Description is read with reference to the accompanying drawings.

[0004] FIG. 1 is a high-level diagram of a distributed environment showing a virtual or overlay cloud network hosted by a cloud service provider infrastructure, according to certain embodiments.

[0005] FIG. 2 depicts a simplified architectural diagram of the physical components in the physical network within a cloud service provider infrastructure (CSPI), according to certain embodiments.

[0006] FIG. 3 shows an example arrangement within CSPI where a host machine is connected to multiple network virtualization devices (NVDs), according to certain embodiments.

[0007] FIG. 4 depicts connectivity between a host machine and an NVD for providing I/O virtualization for supporting multitenancy, according to certain embodiments.

[0008] FIG. 5 depicts a simplified block diagram of a physical network provided by a CSPI, according to certain embodiments.

[0009] FIG. 6 depicts a simplified diagram of a software defined network (SDN) providing domain name system (DNS) resolutions to a non-SDN service, according to certain embodiments.

[0010] FIG. 7 depicts a simplified diagram of an SDN providing DNS resolutions to multiple non-SDN services, according to some embodiments.

[0011] FIG. 8 depicts information used for DNS resolution, according to some embodiments.

[0012] FIG. 9 depicts a sequence diagram for a DNS resolution, according to some embodiments.

[0013] FIG. 10 depicts a sequence diagram for connecting to an endpoint based on a DNS resolution, according to some embodiments.

[0014] FIG. 11 depicts a flow diagram implemented on an SDN for DNS resolution, according to some embodiments.

[0015] FIG. 12 depicts a flow diagram implemented by a non-SDN service for DNS resolution, according to some embodiments.

[0016] FIG. 13 is a block diagram illustrating one pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0017] FIG. 14 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0018] FIG. 15 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0019] FIG. 16 is a block diagram illustrating another pattern for implementing a cloud infrastructure as a service system, according to certain embodiments.

[0020] FIG. 17 is a block diagram illustrating an example computer system, according to certain embodiments.

DETAILED DESCRIPTION

[0021] In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of certain embodiments. However, it will be apparent that various embodiments may be practiced without these specific details. The figures and description are not intended to be restrictive. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs.

[0022] The present disclosure relates generally to a software defined network (SDN) providing domain name system (DNS) resolutions for a non-SDN service. The non-SDN service can be a service implemented on a different network, such as on a substrate network or another non-SDN network. Generally, an SDN refers to an overlay network, such as a cloud network provided by a cloud service provider. DNS resolution is used to support network access to an endpoint, whereby DNS information (e.g., a fully qualified domain name (FQDN)) is resolved to an internet (IP) protocol address.

[0023] In an SDN environment, DNS resolution can be offered as a service of the SDN (e.g., as part of a customer's subnet in the overlay network) to resolve a DNS request for an SDN endpoint or an on-premise endpoint associated with a tenancy at the SDN. However, such a DNS service is available only to SDN services. In other words, when an SDN service sends a DNS resolution request, the DNS service of the SDN can return a DNS resolution. However, a non-SDN service (e.g., a substrate service) may not have access to the DNS service. As such, no DNS resolution can be available to the non-SDN service.

[0024] To extend the DNS resolution to a non-SDN service, embodiments of the present disclosure involve hosting a first endpoint by the SDN, whereby this first endpoint is communicatively coupled with the non-SDN service. The first endpoint can be included in a first private network of the SDN. The first private network can also include a DNS proxy launched in support of a second private network of a customer. A second endpoint of the customer can be accessible via the second private network (e.g., by being included in the second private network or in an on-premises network communicatively coupled with the second private network).

The non-SDN service can send a DNS resolution request (that includes, for example, an FQDN corresponding to the second endpoint) to the first endpoint. The first endpoint forwards this request to the DNS proxy. In turn, the DNS proxy determines the IP address of the second endpoint corresponding to the DNS resolution request (e.g., by querying the second private network). However, this IP address is not sent back to the non-SDN service. Instead, it is masked using NATing techniques. Particularly, the DNS proxy associates the IP address of the second endpoint with a masked IP address (e.g., a reserved IP address corresponding to the DNS proxy) and sends the masked IP address via the first endpoint. The non-SDN service learns that the FQDN is resolved to the masked IP address. The non-SDN service also associates the masked IP address with a non-SDN IP address (e.g., another reserved IP address). From that point on, to connect to the second endpoint, the non-SDN service uses information related to the non-SDN IP address and connects to the second endpoint via the first endpoint and the DNS proxy.

[0025] Herein next, cloud networks are described as SDN examples. Thereafter, SDN-based DNS resolutions are described. Embodiments of the present disclosure are not limited to the cloud environments described herein. Instead, they similarly and equivalently apply to any type of SDN.

Examples of Cloud Networks

[0026] The term cloud service is generally used to refer to a service that is made available by a cloud services provider (CSP) to users or customers on demand (e.g., via a subscription model) using systems and infrastructure (cloud infrastructure) provided by the CSP. Typically, the servers and systems that make up the CSP's infrastructure are separate from the customer's own on-premises servers and systems. Customers can thus avail themselves of cloud services provided by the CSP without having to purchase separate hardware and software resources for the services. Cloud services are designed to provide a subscribing customer easy, scalable access to applications and set of computing resources without the customer having to invest in procuring the infrastructure that is used for providing the services.

[0027] There are several cloud service providers that offer various types of cloud services. There are various different types or models of cloud services including Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), Infrastructure-as-a-Service (IaaS), and others.

[0028] A customer can subscribe to one or more cloud services provided by a CSP. The customer can be any entity such as an individual, an organization, an enterprise, and the like. When a customer subscribes to or registers for a service provided by a CSP, a tenancy or an account is created for that customer. The customer can then, via this account, access the subscribed-to one or more cloud resources associated with the account.

[0029] As noted above, infrastructure as a service (IaaS) is one particular type of cloud computing service. In an IaaS model, the CSP provides infrastructure (referred to as cloud services provider infrastructure or CSPI) that can be used by customers to build their own customizable networks and deploy customer resources. The customer's resources and networks are thus hosted in a distributed environment by infrastructure provided by a CSP. This is different from

traditional computing, where the customer's resources and networks are hosted by infrastructure provided by the customer.

[0030] The CSPI may comprise interconnected high-performance compute resources including various host machines, memory resources, and network resources that form a physical network, which is also referred to as a substrate network or an underlay network. The resources in CSPI may be spread across one or more data centers that may be geographically spread across one or more geographical regions. Virtualization software may be executed by these physical resources to provide a virtualized distributed environment. The virtualization creates an overlay network (also known as a software-based network, a software-defined network, or a virtual network) over the physical network. The CSPI physical network provides the underlying basis for creating one or more overlay or virtual networks on top of the physical network. The physical network (or substrate network or underlay network) comprises physical network devices such as physical switches, routers, computers and host machines, and the like. An overlay network is a logical (or virtual) network that runs on top of a physical substrate network. A given physical network can support one or multiple overlay networks. Overlay networks typically use encapsulation techniques to differentiate between traffic belonging to different overlay networks. A virtual or overlay network is also referred to as a virtual cloud network (VCN). The virtual networks are implemented using software virtualization technologies (e.g., hypervisors, virtualization functions implemented by network virtualization devices (NVDs) (e.g., smartNICs), top-of-rack (TOR) switches, smart TORs that implement one or more functions performed by an NVD, and other mechanisms) to create layers of network abstraction that can be run on top of the physical network. Virtual networks can take on many forms, including peer-to-peer networks, IP networks, and others. Virtual networks are typically either Layer-3 IP networks or Layer-2 VLANs. This method of virtual or overlay networking is often referred to as virtual or overlay Layer-3 networking. Examples of protocols developed for virtual networks include IP-in-IP (or Generic Routing Encapsulation (GRE)) Virtual Extensible LAN (VXLAN-IETF RFC 7348), Virtual Private Networks (VPNs) (e.g., MPLS Layer-3 Virtual Private Networks (RFC 4364)), VMware's NSX, GENEVE (Generic Network Virtualization Encapsulation), and others.

[0031] For IaaS, the infrastructure (CSPI) provided by a CSP can be configured to provide virtualized set of computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing services provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (e.g., billing, monitoring, logging, security, load balancing and clustering, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance. CSPI provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available hosted distributed environment. CSPI offers high-performance compute resources and capabilities and storage

capacity in a flexible virtual network that is securely accessible from various networked locations such as from a customer's on-premises network. When a customer subscribes to or registers for an IaaS service provided by a CSP, the tenancy created for that customer is a secure and isolated partition within the CSPI where the customer can create, organize, and administer their cloud resources.

[0032] Customers can build their own virtual networks using compute, memory, and networking resources provided by CSPI. One or more customer resources or workloads, such as compute instances, can be deployed on these virtual networks. For example, a customer can use resources provided by CSPI to build one or multiple customizable and private virtual network(s) referred to as virtual cloud networks (VCNs). A customer can deploy one or more customer resources, such as compute instances, on a customer VCN. Compute instances can take the form of virtual machines, bare metal instances, and the like. The CSPI thus provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available virtual hosted environment. The customer does not manage or control the underlying physical resources provided by CSPI but has control over operating systems, storage, and deployed applications; and possibly limited control of select networking components (e.g., firewalls).

[0033] The CSP may provide a console that enables customers and network administrators to configure, access, and manage resources deployed in the cloud using CSPI resources. In certain embodiments, the console provides a web-based user interface that can be used to access and manage CSPI. In some implementations, the console is a web-based application provided by the CSP.

[0034] CSPI may support single-tenancy or multi-tenancy architectures. In a single tenancy architecture, a software (e.g., an application, a database) or a hardware component (e.g., a host machine or a server) serves a single customer or tenant. In a multi-tenancy architecture, a software or a hardware component serves multiple customers or tenants. Thus, in a multi-tenancy architecture, CSPI resources are shared between multiple customers or tenants. In a multi-tenancy situation, precautions are taken, and safeguards put in place within CSPI to ensure that each tenant's data is isolated and remains invisible to other tenants.

[0035] In a physical network, a network endpoint ("endpoint") refers to a computing device or system that is connected to a physical network and communicates back and forth with the network to which it is connected. A network endpoint in the physical network may be connected to a Local Area Network (LAN), a Wide Area Network (WAN), or other type of physical network. Examples of traditional endpoints in a physical network include modems, hubs, bridges, switches, routers, and other networking devices, physical computers (or host machines), and the like. Each physical device in the physical network has a fixed network address that can be used to communicate with the device. This fixed network address can be a Layer-2 address (e.g., a MAC address), a fixed Layer-3 address (e.g., an IP address), and the like. In a virtualized environment or in a virtual network, the endpoints can include various virtual endpoints such as virtual machines that are hosted by components of the physical network (e.g., hosted by physical host machines). These endpoints in the virtual network are addressed by overlay addresses such as overlay Layer-2

addresses (e.g., overlay MAC addresses) and overlay Layer-3 addresses (e.g., overlay IP addresses). Network overlays enable flexibility by allowing network managers to move around the overlay addresses associated with network endpoints using software management (e.g., via software implementing a control plane for the virtual network). Accordingly, unlike in a physical network, in a virtual network, an overlay address (e.g., an overlay IP address) can be moved from one endpoint to another using network management software. Since the virtual network is built on top of a physical network, communications between components in the virtual network involves both the virtual network and the underlying physical network. In order to facilitate such communications, the components of CSPI are configured to learn and store mappings that map overlay addresses in the virtual network to actual physical addresses in the substrate network, and vice versa. These mappings are then used to facilitate the communications. Customer traffic is encapsulated to facilitate routing in the virtual network.

[0036] Accordingly, physical addresses (e.g., physical IP addresses) are associated with components in physical networks and overlay addresses (e.g., overlay IP addresses) are associated with entities in virtual or overlay networks. A physical IP address is an IP address associated with a physical device (e.g., a network device) in the substrate or physical network. For example, each NVD has an associated physical IP address. An overlay IP address is an overlay address associated with an entity in an overlay network, such as with a compute instance in a customer's virtual cloud network (VCN). Two different customers or tenants, each with their own private VCNs can potentially use the same overlay IP address in their VCNs without any knowledge of each other. Both the physical IP addresses and overlay IP addresses are types of real IP addresses. These are separate from virtual IP addresses. A virtual IP address is typically a single IP address that represents or maps to multiple real IP addresses. A virtual IP address provides a 1-to-many mapping between the virtual IP address and multiple real IP addresses. For example, a load balancer may use a VIP to map to or represent multiple servers, each server having its own real IP address.

[0037] The cloud infrastructure or CSPI is physically hosted in one or more data centers in one or more regions around the world. The CSPI may include components in the physical or substrate network and virtualized components (e.g., virtual networks, compute instances, virtual machines, etc.) that are in a virtual network built on top of the physical network components. In certain embodiments, the CSPI is organized and hosted in realms, regions, and availability domains. A region is typically a localized geographic area that contains one or more data centers. Regions are generally independent of each other and can be separated by vast distances, for example, across countries or even continents. For example, a first region may be in Australia, another one in Japan, yet another one in India, and the like. CSPI resources are divided among regions such that each region has its own independent subset of CSPI resources. Each region may provide a set of core infrastructure services and resources, such as, compute resources (e.g., bare metal servers, virtual machine, containers and related infrastructure, etc.); storage resources (e.g., block volume storage, file storage, object storage, archive storage); networking resources (e.g., virtual cloud networks (VCNs), load balancing resources, connections to on-premise networks),

database resources; edge networking resources (e.g., DNS); and access management and monitoring resources, and others. Each region generally has multiple paths connecting it to other regions in the realm.

[0038] Generally, an application is deployed in a region (i.e., deployed on infrastructure associated with that region) where it is most heavily used, because using nearby resources is faster than using distant resources. Applications can also be deployed in different regions for various reasons, such as redundancy to mitigate the risk of region-wide events such as large weather systems or earthquakes, to meet varying requirements for legal jurisdictions, tax domains, and other business or social criteria, and the like.

[0039] The data centers within a region can be further organized and subdivided into availability domains (ADs). An availability domain may correspond to one or more data centers located within a region. A region can be composed of one or more availability domains. In such a distributed environment, CSPI resources are either region-specific, such as a virtual cloud network (VCN), or availability domain-specific, such as a compute instance.

[0040] ADs within a region are isolated from each other, fault tolerant, and are configured such that they are very unlikely to fail simultaneously. This is achieved by the ADs not sharing critical infrastructure resources such as networking, physical cables, cable paths, cable entry points, etc., such that a failure at one AD within a region is unlikely to impact the availability of the other ADs within the same region. The ADs within the same region may be connected to each other by a low latency, high bandwidth network, which makes it possible to provide high-availability connectivity to other networks (e.g., the Internet, customers' on-premises networks, etc.) and to build replicated systems in multiple ADs for both high-availability and disaster recovery. Cloud services use multiple ADs to ensure high availability and to protect against resource failure. As the infrastructure provided by the IaaS provider grows, more regions and ADs may be added with additional capacity. Traffic between availability domains is usually encrypted.

[0041] In certain embodiments, regions are grouped into realms. A realm is a logical collection of regions. Realms are isolated from each other and do not share any data. Regions in the same realm may communicate with each other, but regions in different realms cannot. A customer's tenancy or account with the CSP exists in a single realm and can be spread across one or more regions that belong to that realm. Typically, when a customer subscribes to an IaaS service, a tenancy or account is created for that customer in the customer-specified region (referred to as the "home" region) within a realm. A customer can extend the customer's tenancy across one or more other regions within the realm. A customer cannot access regions that are not in the realm where the customer's tenancy exists.

[0042] An IaaS provider can provide multiple realms, each realm catered to a particular set of customers or users. For example, a commercial realm may be provided for commercial customers. As another example, a realm may be provided for a specific country for customers within that country. As yet another example, a government realm may be provided for a government, and the like. For example, the government realm may be catered for a specific government and may have a heightened level of security than a commercial realm. For example, Oracle Cloud Infrastructure (OCI) currently offers a realm for commercial regions and

two realms (e.g., FedRAMP authorized and IL5 authorized) for government cloud regions.

[0043] In certain embodiments, an AD can be subdivided into one or more fault domains. A fault domain is a grouping of infrastructure resources within an AD to provide anti-affinity. Fault domains allow for the distribution of compute instances such that the instances are not on the same physical hardware within a single AD. This is known as anti-affinity. A fault domain refers to a set of hardware components (computers, switches, and more) that share a single point of failure. A compute pool is logically divided up into fault domains. Due to this, a hardware failure or compute hardware maintenance event that affects one fault domain does not affect instances in other fault domains. Depending on the embodiment, the number of fault domains for each AD may vary. For instance, in certain embodiments each AD contains three fault domains. A fault domain acts as a logical data center within an AD.

[0044] When a customer subscribes to an IaaS service, resources from CSPI are provisioned for the customer and associated with the customer's tenancy. The customer can use these provisioned resources to build private networks and deploy resources on these networks. The customer networks that are hosted in the cloud by the CSPI are referred to as virtual cloud networks (VCNs). A customer can set up one or more virtual cloud networks (VCNs) using CSPI resources allocated for the customer. A VCN is a virtual or software defined private network. The customer resources that are deployed in the customer's VCN can include compute instances (e.g., virtual machines, bare-metal instances) and other resources. These compute instances may represent various customer workloads such as applications, load balancers, databases, and the like. A compute instance deployed on a VCN can communicate with publicly accessible endpoints ("public endpoints") over a public network such as the Internet, with other instances in the same VCN or other VCNs (e.g., the customer's other VCNs, or VCNs not belonging to the customer), with the customer's on-premise data centers or networks, and with service endpoints, and other types of endpoints.

[0045] The CSP may provide various services using the CSPI. In some instances, customers of CSPI may themselves act like service providers and provide services using CSPI resources. A service provider may expose a service endpoint, which is characterized by identification information (e.g., an IP Address, a DNS name and port). A customer's resource (e.g., a compute instance) can consume a particular service by accessing a service endpoint exposed by the service for that particular service. These service endpoints are generally endpoints that are publicly accessible by users using public IP addresses associated with the endpoints via a public communication network such as the Internet. Network endpoints that are publicly accessible are also sometimes referred to as public endpoints.

[0046] In certain embodiments, a service provider may expose a service via an endpoint (sometimes referred to as a service endpoint) for the service. Customers of the service can then use this service endpoint to access the service. In certain implementations, a service endpoint provided for a service can be accessed by multiple customers that intend to consume that service. In other implementations, a dedicated service endpoint may be provided for a customer such that only that customer can access the service using that dedicated service endpoint.

[0047] In certain embodiments, when a VCN is created, it is associated with a private overlay Classless Inter-Domain Routing (CIDR) address space, which is a range of private overlay IP addresses that are assigned to the VCN (e.g., 10.0/16). A VCN includes associated subnets, route tables, and gateways. A VCN resides within a single region but can span one or more or all of the region's availability domains. A gateway is a virtual interface that is configured for a VCN and enables communication of traffic to and from the VCN to one or more endpoints outside the VCN. One or more different types of gateways may be configured for a VCN to enable communication to and from different types of endpoints.

[0048] A VCN can be subdivided into one or more subnetworks such as one or more subnets. A subnet is thus a unit of configuration or a subdivision that can be created within a VCN. A VCN can have one or multiple subnets. Each subnet within a VCN is associated with a contiguous range of overlay IP addresses (e.g., 10.0.0.0/24 and 10.0.1.0/24) that do not overlap with other subnets in that VCN, and which represent an address space subset within the address space of the VCN.

[0049] Each compute instance is associated with a virtual network interface card (VNIC), that enables the compute instance to participate in a subnet of a VCN. A VNIC is a logical representation of physical Network Interface Card (NIC). In general, a VNIC is an interface between an entity (e.g., a compute instance, a service) and a virtual network. A VNIC exists in a subnet, has one or more associated IP addresses, and associated security rules or policies. A VNIC is equivalent to a Layer-2 port on a switch. A VNIC is attached to a compute instance and to a subnet within a VCN. A VNIC associated with a compute instance enables the compute instance to be a part of a subnet of a VCN and enables the compute instance to communicate (e.g., send and receive packets) with endpoints that are on the same subnet as the compute instance, with endpoints in different subnets in the VCN, or with endpoints outside the VCN. The VNIC associated with a compute instance thus determines how the compute instance connects with endpoints inside and outside the VCN. A VNIC for a compute instance is created and associated with that compute instance when the compute instance is created and added to a subnet within a VCN. For a subnet comprising a set of compute instances, the subnet contains the VNICs corresponding to the set of compute instances, each VNIC attached to a compute instance within the set of computer instances.

[0050] Each compute instance is assigned a private overlay IP address via the VNIC associated with the compute instance. This private overlay IP address is assigned to the VNIC that is associated with the compute instance when the compute instance is created and used for routing traffic to and from the compute instance. All VNICs in a given subnet use the same route table, security lists, and DHCP options. As described above, each subnet within a VCN is associated with a contiguous range of overlay IP addresses (e.g., 10.0.0.0/24 and 10.0.1.0/24) that do not overlap with other subnets in that VCN, and which represent an address space subset within the address space of the VCN. For a VNIC on a particular subnet of a VCN, the private overlay IP address that is assigned to the VNIC is an address from the contiguous range of overlay IP addresses allocated for the subnet.

[0051] In certain embodiments, a compute instance may optionally be assigned additional overlay IP addresses in addition to the private overlay IP address, such as, for example, one or more public IP addresses if in a public subnet. These multiple addresses are assigned either on the same VNIC or over multiple VNICs that are associated with the compute instance. Each instance however has a primary VNIC that is created during instance launch and is associated with the overlay private IP address assigned to the instance—this primary VNIC cannot be removed. Additional VNICs, referred to as secondary VNICs, can be added to an existing instance in the same availability domain as the primary VNIC. All the VNICs are in the same availability domain as the instance. A secondary VNIC can be in a subnet in the same VCN as the primary VNIC, or in a different subnet that is either in the same VCN or a different one.

[0052] A compute instance may optionally be assigned a public IP address if it is in a public subnet. A subnet can be designated as either a public subnet or a private subnet at the time the subnet is created. A private subnet means that the resources (e.g., compute instances) and associated VNICs in the subnet cannot have public overlay IP addresses. A public subnet means that the resources and associated VNICs in the subnet can have public IP addresses. A customer can designate a subnet to exist either in a single availability domain or across multiple availability domains in a region or realm.

[0053] As described above, a VCN may be subdivided into one or more subnets. In certain embodiments, a Virtual Router (VR) configured for the VCN (referred to as the VCN VR or just VR) enables communications between the subnets of the VCN. For a subnet within a VCN, the VR represents a logical gateway for that subnet that enables the subnet (i.e., the compute instances on that subnet) to communicate with endpoints on other subnets within the VCN, and with other endpoints outside the VCN. The VCN VR is a logical entity that is configured to route traffic between VNICs in the VCN and virtual gateways ("gateways") associated with the VCN. A VCN VR is a Layer-3/IP Layer concept. In one embodiment, there is one VCN VR for a VCN where the VCN VR has potentially an unlimited number of ports addressed by IP addresses, with one port for each subnet of the VCN. In this manner, the VCN VR has a different IP address for each subnet in the VCN that the VCN VR is attached to. The VR is also connected to the various gateways configured for a VCN. In certain embodiments, a particular overlay IP address from the overlay IP address range for a subnet is reserved for a port of the VCN VR for that subnet. For example, consider a VCN having two subnets with associated address ranges 10.0/16 and 10.1/16, respectively. For the first subnet within the VCN with address range 10.0/16, an address from this range is reserved for a port of the VCN VR for that subnet. In some instances, the first IP address from the range may be reserved for the VCN VR. For example, for the subnet with overlay IP address range 10.0/16, IP address 10.0.0.1 may be reserved for a port of the VCN VR for that subnet. For the second subnet within the same VCN with address range 10.1/16, the VCN VR may have a port for that second subnet with IP address 10.1.0.1. The VCN VR has a different IP address for each of the subnets in the VCN.

[0054] In some other embodiments, each subnet within a VCN may have its own associated VR that is addressable by the subnet using a reserved or default IP address associated

with the VR. The reserved or default IP address may, for example, be the first IP address from the range of IP addresses associated with that subnet. The VNICs in the subnet can communicate (e.g., send and receive packets) with the VR associated with the subnet using this default or reserved IP address. In such an embodiment, the VR is the ingress/egress point for that subnet. The VR associated with a subnet within the VCN can communicate with other VRs associated with other subnets within the VCN. The VRs can also communicate with gateways associated with the VCN. The VR function for a subnet is running on or executed by one or more NVDs executing VNICs functionality for VNICs in the subnet.

[0055] Route tables, security rules, and DHCP options may be configured for a VCN. Route tables are virtual route tables for the VCN and include rules to route traffic from subnets within the VCN to destinations outside the VCN by way of gateways or specially configured instances. A VCN's route tables can be customized to control how packets are forwarded/routed to and from the VCN. DHCP options refers to configuration information that is automatically provided to the instances when they boot up.

[0056] Security rules configured for a VCN represent overlay firewall rules for the VCN. The security rules can include ingress and egress rules, and specify the types of traffic (e.g., based upon protocol and port) that is allowed in and out of the instances within the VCN. The customer can choose whether a given rule is stateful or stateless. For instance, the customer can allow incoming SSH traffic from anywhere to a set of instances by setting up a stateful ingress rule with source CIDR 0.0.0.0/0, and destination TCP port 22. Security rules can be implemented using network security groups or security lists. A network security group consists of a set of security rules that apply only to the resources in that group. A security list, on the other hand, includes rules that apply to all the resources in any subnet that uses the security list. A VCN may be provided with a default security list with default security rules. DHCP options configured for a VCN provide configuration information that is automatically provided to the instances in the VCN when the instances boot up.

[0057] In certain embodiments, the configuration information for a VCN is determined and stored by a VCN Control Plane. The configuration information for a VCN may include, for example, information about the address range associated with the VCN, subnets within the VCN and associated information, one or more VRs associated with the VCN, compute instances in the VCN and associated VNICs, NVDs executing the various virtualization network functions (e.g., VNICs, VRs, gateways) associated with the VCN, state information for the VCN, and other VCN-related information. In certain embodiments, a VCN Distribution Service publishes the configuration information stored by the VCN Control Plane, or portions thereof, to the NVDs. The distributed information may be used to update information (e.g., forwarding tables, routing tables, etc.) stored and used by the NVDs to forward packets to and from the compute instances in the VCN.

[0058] In certain embodiments, the creation of VCNs and subnets are handled by a VCN Control Plane (CP), and the launching of compute instances is handled by a Compute Control Plane. The Compute Control Plane is responsible for allocating the physical resources for the compute instance and then calls the VCN Control Plane to create and attach

VNICs to the compute instance. The VCN CP also sends VCN data mappings to the VCN data plane that is configured to perform packet forwarding and routing functions. In certain embodiments, the VCN CP provides a distribution service that is responsible for providing updates to the VCN data plane. Examples of a VCN Control Plane are also depicted in FIGS. 6, 7, 8, and 9 (see references 616, 716, 816, and 916) and described below.

[0059] A customer may create one or more VCNs using resources hosted by CSPI. A compute instance deployed on a customer VCN may communicate with different endpoints. These endpoints can include endpoints that are hosted by CSPI and endpoints outside CSPI.

[0060] Various different architectures for implementing cloud-based service using CSPI are depicted in FIGS. 1-5 and 13-17, and are described below. FIG. 1 is a high-level diagram of a distributed environment 100 showing an overlay or customer VCN hosted by CSPI according to certain embodiments. The distributed environment depicted in FIG. 1 includes multiple components in the overlay network. Distributed environment 100 depicted in FIG. 1 is merely an example and is not intended to unduly limit the scope of claimed embodiments. Many variations, alternatives, and modifications are possible. For example, in some implementations, the distributed environment depicted in FIG. 1 may have more or fewer systems or components than those shown in FIG. 1, may combine two or more systems, or may have a different configuration or arrangement of systems.

[0061] As shown in the example depicted in FIG. 1, distributed environment 100 comprises CSPI 101 that provides services and resources that customers can subscribe to and use to build their virtual cloud networks (VCNs). In certain embodiments, CSPI 101 offers IaaS services to subscribing customers. The data centers within CSPI 101 may be organized into one or more regions. One example region "Region US" 102 is shown in FIG. 1. A customer has configured a customer VCN c/o Oracle International Corporation for region 102. The customer may deploy various compute instances on VCN 104, where the compute instances may include virtual machines or bare metal instances. Examples of instances include applications, database, load balancers, and the like.

[0062] In the embodiment depicted in FIG. 1, customer VCN 104 comprises two subnets, namely, "Subnet-1" and "Subnet-2", each subnet with its own CIDR IP address range. In FIG. 1, the overlay IP address range for Subnet-1 is 10.0/16 and the address range for Subnet-2 is 10.1/16. A VCN Virtual Router 105 represents a logical gateway for the VCN that enables communications between subnets of the VCN 104, and with other endpoints outside the VCN. VCN VR 105 is configured to route traffic between VNICs in VCN 104 and gateways associated with VCN 104. VCN VR 105 provides a port for each subnet of VCN 104. For example, VR 105 may provide a port with IP address 10.0.0.1 for Subnet-1 and a port with IP address 10.1.0.1 for Subnet-2.

[0063] Multiple compute instances may be deployed on each subnet, where the compute instances can be virtual machine instances, and/or bare metal instances. The compute instances in a subnet may be hosted by one or more host machines within CSPI 101. A compute instance participates in a subnet via a VNIC associated with the compute instance. For example, as shown in FIG. 1, a compute instance C1 is part of Subnet-1 via a VNIC associated with the compute instance. Likewise, compute instance C2 is part of Subnet-1

via a VNIC associated with C2. In a similar manner, multiple compute instances, which may be virtual machine instances or bare metal instances, may be part of Subnet-1. Via its associated VNIC, each compute instance is assigned a private overlay IP address and a MAC address. For example, in FIG. 1, compute instance C1 has an overlay IP address of 10.0.0.2 and a MAC address of M1, while compute instance C2 has a private overlay IP address of 10.0.0.3 and a MAC address of M2. Each compute instance in Subnet-1, including compute instances C1 and C2, has a default route to VCN VR 105 using IP address 10.0.0.1, which is the IP address for a port of VCN VR 105 for Subnet-1.

[0064] Subnet-2 can have multiple compute instances deployed on it, including virtual machine instances and/or bare metal instances. For example, as shown in FIG. 1, compute instances D1 and D2 are part of Subnet-2 via VNICs associated with the respective compute instances. In the embodiment depicted in FIG. 1, compute instance D1 has an overlay IP address of 10.1.0.2 and a MAC address of MM1, while compute instance D2 has a private overlay IP address of 10.1.0.3 and a MAC address of MM2. Each compute instance in Subnet-2, including compute instances D1 and D2, has a default route to VCN VR 105 using IP address 10.1.0.1, which is the IP address for a port of VCN VR 105 for Subnet-2.

[0065] VCN A 104 may also include one or more load balancers. For example, a load balancer may be provided for a subnet and may be configured to load balance traffic across multiple compute instances on the subnet. A load balancer may also be provided to load balance traffic across subnets in the VCN.

[0066] A particular compute instance deployed on VCN 104 can communicate with various different endpoints. These endpoints may include endpoints that are hosted by CSPI 200 and endpoints outside CSPI 200. Endpoints that are hosted by CSPI 101 may include: an endpoint on the same subnet as the particular compute instance (e.g., communications between two compute instances in Subnet-1); an endpoint on a different subnet but within the same VCN (e.g., communication between a compute instance in Subnet-1 and a compute instance in Subnet-2); an endpoint in a different VCN in the same region (e.g., communications between a compute instance in Subnet-1 and an endpoint in a VCN in the same region 106 or 110, communications between a compute instance in Subnet-1 and an endpoint in service network 110 in the same region); or an endpoint in a VCN in a different region (e.g., communications between a compute instance in Subnet-1 and an endpoint in a VCN in a different region 108). A compute instance in a subnet hosted by CSPI 101 may also communicate with endpoints that are not hosted by CSPI 101 (i.e., are outside CSPI 101). These outside endpoints include endpoints in the customer's on-premises network 116, endpoints within other remote cloud hosted networks 118, public endpoints 114 accessible via a public network such as the Internet, and other endpoints.

[0067] Communications between compute instances on the same subnet are facilitated using VNICs associated with the source compute instance and the destination compute instance. For example, compute instance C1 in Subnet-1 may want to send packets to compute instance C2 in Subnet-1. For a packet originating at a source compute instance and whose destination is another compute instance in the same subnet, the packet is first processed by the VNIC

associated with the source compute instance. Processing performed by the VNIC associated with the source compute instance can include determining destination information for the packet from the packet headers, identifying any policies (e.g., security lists) configured for the VNIC associated with the source compute instance, determining a next hop for the packet, performing any packet encapsulation/decapsulation functions as needed, and then forwarding/routing the packet to the next hop with the goal of facilitating communication of the packet to its intended destination. When the destination compute instance is in the same subnet as the source compute instance, the VNIC associated with the source compute instance is configured to identify the VNIC associated with the destination compute instance and forward the packet to that VNIC for processing. The VNIC associated with the destination compute instance is then executed and forwards the packet to the destination compute instance.

[0068] For a packet to be communicated from a compute instance in a subnet to an endpoint in a different subnet in the same VCN, the communication is facilitated by the VNICs associated with the source and destination compute instances and the VCN VR. For example, if compute instance C1 in Subnet-1 in FIG. 1 wants to send a packet to compute instance D1 in Subnet-2, the packet is first processed by the VNIC associated with compute instance C1. The VNIC associated with compute instance C1 is configured to route the packet to the VCN VR 105 using default route or port 10.0.0.1 of the VCN VR. VCN VR 105 is configured to route the packet to Subnet-2 using port 10.1.0.1. The packet is then received and processed by the VNIC associated with D1 and the VNIC forwards the packet to compute instance D1.

[0069] For a packet to be communicated from a compute instance in VCN 104 to an endpoint that is outside VCN 104, the communication is facilitated by the VNIC associated with the source compute instance, VCN VR 105, and gateways associated with VCN 104. One or more types of gateways may be associated with VCN 104. A gateway is an interface between a VCN and another endpoint, where another endpoint is outside the VCN. A gateway is a Layer-3/IP layer concept and enables a VCN to communicate with endpoints outside the VCN. A gateway thus facilitates traffic flow between a VCN and other VCNs or networks. Various different types of gateways may be configured for a VCN to facilitate different types of communications with different types of endpoints. Depending upon the gateway, the communications may be over public networks (e.g., the Internet) or over private networks. Various communication protocols may be used for these communications.

[0070] For example, compute instance C1 may want to communicate with an endpoint outside VCN 104. The packet may be first processed by the VNIC associated with source compute instance C1. The VNIC processing determines that the destination for the packet is outside the Subnet-1 of C1. The VNIC associated with C1 may forward the packet to VCN VR 105 for VCN 104. VCN VR 105 then processes the packet and as part of the processing, based upon the destination for the packet, determines a particular gateway associated with VCN 104 as the next hop for the packet. VCN VR 105 may then forward the packet to the particular identified gateway. For example, if the destination is an endpoint within the customer's on-premise network, then the packet may be forwarded by VCN VR 105 to

Dynamic Routing Gateway (DRG) gateway **122** configured for VCN **104**. The packet may then be forwarded from the gateway to a next hop to facilitate communication of the packet to its final intended destination.

[0071] Various different types of gateways may be configured for a VCN. Examples of gateways that may be configured for a VCN are depicted in FIG. 1 and described below. Examples of gateways associated with a VCN are also depicted in FIGS. 18, 19, 20, and 21 (for example, gateways referenced by reference numbers **1834**, **1836**, **1838**, **1934**, **1936**, **1938**, **2034**, **2036**, **2038**, **2134**, **2136**, and **2138**) and described below. As shown in the embodiment depicted in FIG. 1, a Dynamic Routing Gateway (DRG) **122** may be added to or be associated with customer VCN **104** and provides a path for private network traffic communication between customer VCN **104** and another endpoint, where the another endpoint can be the customer's on-premise network **116**, a VCN **108** in a different region of CSPI **101**, or other remote cloud networks **118** not hosted by CSPI **101**. Customer on-premise network **116** may be a customer network or a customer data center built using the customer's resources. Access to customer on-premise network **116** is generally very restricted. For a customer that has both a customer on-premise network **116** and one or more VCNs **104** deployed or hosted in the cloud by CSPI **101**, the customer may want their on-premise network **116** and their cloud based VCN **104** to be able to communicate with each other. This enables a customer to build an extended hybrid environment encompassing the customer's VCN **104** hosted by CSPI **101** and their on-premises network **116**. DRG **122** enables this communication. To enable such communications, a communication channel **124** is set up where one endpoint of the channel is in customer on-premise network **116** and the other endpoint is in CSPI **101** and connected to customer VCN **104**. Communication channel **124** can be over public communication networks such as the Internet or private communication networks. Various different communication protocols may be used such as IPsec VPN technology over a public communication network such as the Internet, Oracle's FastConnect technology that uses a private network instead of a public network, and others. The device or equipment in customer on-premise network **116** that forms one end point for communication channel **124** is referred to as the customer premise equipment (CPE), such as CPE **126** depicted in FIG. 1. On the CSPI **101** side, the endpoint may be a host machine executing DRG **122**.

[0072] In certain embodiments, a Remote Peering Connection (RPC) can be added to a DRG, which allows a customer to peer one VCN with another VCN in a different region. Using such an RPC, customer VCN **104** can use DRG **122** to connect with a VCN **108** in another region. DRG **122** may also be used to communicate with other remote cloud networks **118**, not hosted by CSPI **101** such as a Microsoft AZURE cloud, Amazon AWS cloud, and others.

[0073] As shown in FIG. 1, an Internet Gateway (IGW) **120** may be configured for customer VCN **104** that enables a compute instance on VCN **104** to communicate with public endpoints **114** accessible over a public network such as the Internet. IGW **120** is a gateway that connects a VCN to a public network such as the Internet. IGW **120** enables a public subnet (where the resources in the public subnet have public overlay IP addresses) within a VCN, such as VCN **104**, direct access to public endpoints **112** on a public

network **114** such as the Internet. Using IGW **120**, connections can be initiated from a subnet within VCN **104** or from the Internet.

[0074] A Network Address Translation (NAT) gateway **128** can be configured for customer's VCN **104** and enables cloud resources in the customer's VCN, which do not have dedicated public overlay IP addresses, access to the Internet and it does so without exposing those resources to direct incoming Internet connections (e.g., L4-L7 connections). This enables a private subnet within a VCN, such as private Subnet-1 in VCN **104**, with private access to public endpoints on the Internet. In NAT gateways, connections can be initiated only from the private subnet to the public Internet and not from the Internet to the private subnet.

[0075] In certain embodiments, a Service Gateway (SGW) **126** can be configured for customer VCN **104** and provides a path for private network traffic between VCN **104** and supported services endpoints in a service network **110**. In certain embodiments, service network **110** may be provided by the CSP and may provide various services. An example of such a service network is Oracle's Services Network, which provides various services that can be used by customers. For example, a compute instance (e.g., a database system) in a private subnet of customer VCN **104** can back up data to a service endpoint (e.g., Object Storage) without needing public IP addresses or access to the Internet. In certain embodiments, a VCN can have only one SGW, and connections can only be initiated from a subnet within the VCN and not from service network **110**. If a VCN is peered with another, resources in the other VCN typically cannot access the SGW. Resources in on-premises networks that are connected to a VCN with FastConnect or VPN Connect can also use the service gateway configured for that VCN.

[0076] In certain implementations, SGW **126** uses the concept of a service Classless Inter-Domain Routing (CIDR) label, which is a string that represents all the regional public IP address ranges for the service or group of services of interest. The customer uses the service CIDR label when they configure the SGW and related route rules to control traffic to the service. The customer can optionally utilize it when configuring security rules without needing to adjust them if the service's public IP addresses change in the future.

[0077] A Local Peering Gateway (LPG) **132** is a gateway that can be added to customer VCN **104** and enables VCN **104** to peer with another VCN in the same region. Peering means that the VCNs communicate using private IP addresses, without the traffic traversing a public network such as the Internet or without routing the traffic through the customer's on-premises network **116**. In preferred embodiments, a VCN has a separate LPG for each peering it establishes. Local Peering or VCN Peering is a common practice used to establish network connectivity between different applications or infrastructure management functions.

[0078] Service providers, such as providers of services in service network **110**, may provide access to services using different access models. According to a public access model, services may be exposed as public endpoints that are publicly accessible by compute instance in a customer VCN via a public network such as the Internet and or may be privately accessible via SGW **126**. According to a specific private access model, services are made accessible as private IP endpoints in a private subnet in the customer's VCN. This is

referred to as a Private Endpoint (PE) access and enables a service provider to expose their service as an instance in the customer's private network. A Private Endpoint resource represents a service within the customer's VCN. Each PE manifests as a VNIC (referred to as a PE-VNIC, with one or more private IPs) in a subnet chosen by the customer in the customer's VCN. A PE thus provides a way to present a service within a private customer VCN subnet using a VNIC. Since the endpoint is exposed as a VNIC, all the features associated with a VNIC such as routing rules, security lists, etc., are now available for the PE VNIC.

[0079] A service provider can register their service to enable access through a PE. The provider can associate policies with the service that restricts the service's visibility to the customer tenancies. A provider can register multiple services under a single virtual IP address (VIP), especially for multi-tenant services. There may be multiple such private endpoints (in multiple VCNs) that represent the same service.

[0080] Compute instances in the private subnet can then use the PE VNIC's private IP address or the service DNS name to access the service. Compute instances in the customer VCN can access the service by sending traffic to the private IP address of the PE in the customer VCN. A Private Access Gateway (PAGW) **130** is a gateway resource that can be attached to a service provider VCN (e.g., a VCN in service network **110**) that acts as an ingress/egress point for all traffic from/to customer subnet private endpoints. PAGW **130** enables a provider to scale the number of PE connections without utilizing its internal IP address resources. A provider needs only configure one PAGW for any number of services registered in a single VCN. Providers can represent a service as a private endpoint in multiple VCNs of one or more customers. From the customer's perspective, the PE VNIC, which, instead of being attached to a customer's instance, appears attached to the service with which the customer wishes to interact. The traffic destined to the private endpoint is routed via PAGW **130** to the service. These are referred to as customer-to-service private connections (C2S connections).

[0081] The PE concept can also be used to extend the private access for the service to customer's on-premises networks and data centers, by allowing the traffic to flow through FastConnect/IPsec links and the private endpoint in the customer VCN. Private access for the service can also be extended to the customer's peered VCNs, by allowing the traffic to flow between LPG **132** and the PE in the customer's VCN.

[0082] A customer can control routing in a VCN at the subnet level, so the customer can specify which subnets in the customer's VCN, such as VCN **104**, use each gateway. A VCN's route tables are used to decide if traffic is allowed out of a VCN through a particular gateway. For example, in a particular instance, a route table for a public subnet within customer VCN **104** may send non-local traffic through IGW **120**. The route table for a private subnet within the same customer VCN **104** may send traffic destined for CSP services through SGW **126**. All remaining traffic may be sent via the NAT gateway **128**. Route tables only control traffic going out of a VCN.

[0083] Security lists associated with a VCN are used to control traffic that comes into a VCN via a gateway via inbound connections. All resources in a subnet use the same route table and security lists. Security lists may be used to

control specific types of traffic allowed in and out of instances in a subnet of a VCN. Security list rules may comprise ingress (inbound) and egress (outbound) rules. For example, an ingress rule may specify an allowed source address range, while an egress rule may specify an allowed destination address range. Security rules may specify a particular protocol (e.g., TCP, ICMP), a particular port (e.g., **22** for SSH, **3389** for Windows RDP), etc. In certain implementations, an instance's operating system may enforce its own firewall rules that are aligned with the security list rules. Rules may be stateful (e.g., a connection is tracked, and the response is automatically allowed without an explicit security list rule for the response traffic) or stateless.

[0084] Access from a customer VCN (i.e., by a resource or compute instance deployed on VCN **104**) can be categorized as public access, private access, or dedicated access. Public access refers to an access model where a public IP address or a NAT is used to access a public endpoint. Private access enables customer workloads in VCN **104** with private IP addresses (e.g., resources in a private subnet) to access services without traversing a public network such as the Internet. In certain embodiments, CSPI **101** enables customer VCN workloads with private IP addresses to access the (public service endpoints of) services using a service gateway. A service gateway thus offers a private access model by establishing a virtual link between the customer's VCN and the service's public endpoint residing outside the customer's private network.

[0085] Additionally, CSPI may offer dedicated public access using technologies such as FastConnect public peering where customer on-premises instances can access one or more services in a customer VCN using a FastConnect connection and without traversing a public network such as the Internet. CSPI also may offer dedicated private access using FastConnect private peering where customer on-premises instances with private IP addresses can access the customer's VCN workloads using a FastConnect connection. FastConnect is a network connectivity alternative to using the public Internet to connect a customer's on-premise network to CSPI and its services. FastConnect provides an easy, elastic, and economical way to create a dedicated and private connection with higher bandwidth options and a more reliable and consistent networking experience when compared to Internet-based connections.

[0086] FIG. 1 and the accompanying description above describes various virtualized components in an example virtual network. As described above, the virtual network is built on the underlying physical or substrate network. FIG. 2 depicts a simplified architectural diagram of the physical components in the physical network within CSPI **200** that provide the underlay for the virtual network according to certain embodiments. As shown, CSPI **200** provides a distributed environment comprising components and resources (e.g., compute, memory, and networking resources) provided by a cloud service provider (CSP). These components and resources are used to provide cloud services (e.g., IaaS services) to subscribing customers, i.e., customers that have subscribed to one or more services provided by the CSP. Based upon the services subscribed to by a customer, a subset of resources (e.g., compute, memory, and networking resources) of CSPI **200** are provisioned for the customer. Customers can then build their own cloud-based (i.e., CSPI-hosted) customizable and private virtual networks using

physical compute, memory, and networking resources provided by CSPI 200. As previously indicated, these customer networks are referred to as virtual cloud networks (VCNs). A customer can deploy one or more customer resources, such as compute instances, on these customer VCNs. Compute instances can be in the form of virtual machines, bare metal instances, and the like. CSPI 200 provides infrastructure and a set of complementary cloud services that enable customers to build and run a wide range of applications and services in a highly available hosted environment.

[0087] In the example embodiment depicted in FIG. 2, the physical components of CSPI 200 include one or more physical host machines or physical servers (e.g., 202, 206, 208), network virtualization devices (NVDs) (e.g., 210, 212), top-of-rack (TOR) switches (e.g., 214, 216), and a physical network (e.g., 218), and switches in physical network 218. The physical host machines or servers may host and execute various compute instances that participate in one or more subnets of a VCN. The compute instances may include virtual machine instances, and bare metal instances. For example, the various compute instances depicted in FIG. 1 may be hosted by the physical host machines depicted in FIG. 2. The virtual machine compute instances in a VCN may be executed by one host machine or by multiple different host machines. The physical host machines may also host virtual host machines, container-based hosts or functions, and the like. The VNICs and VCN VR depicted in FIG. 1 may be executed by the NVDs depicted in FIG. 2. The gateways depicted in FIG. 1 may be executed by the host machines and/or by the NVDs depicted in FIG. 2.

[0088] The host machines or servers may execute a hypervisor (also referred to as a virtual machine monitor or VMM) that creates and enables a virtualized environment on the host machines. The virtualization or virtualized environment facilitates cloud-based computing. One or more compute instances may be created, executed, and managed on a host machine by a hypervisor on that host machine. The hypervisor on a host machine enables the physical set of computing resources of the host machine (e.g., compute, memory, and networking resources) to be shared between the various compute instances executed by the host machine.

[0089] For example, as depicted in FIG. 2, host machines 202 and 208 execute hypervisors 260 and 266, respectively. These hypervisors may be implemented using software, firmware, or hardware, or combinations thereof. Typically, a hypervisor is a process or a software layer that sits on top of the host machine's operating system (OS), which in turn executes on the hardware processors of the host machine. The hypervisor provides a virtualized environment by enabling the physical set of computing resources (e.g., processing resources such as processors/cores, memory resources, networking resources) of the host machine to be shared among the various virtual machine compute instances executed by the host machine. For example, in FIG. 2, hypervisor 260 may sit on top of the OS of host machine 202 and enables the set of computing resources (e.g., processing, memory, and networking resources) of host machine 202 to be shared between compute instances (e.g., virtual machines) executed by host machine 202. A virtual machine can have its own operating system (referred to as a guest operating system), which may be the same as or different from the OS of the host machine. The operating system of a virtual machine executed by a host machine may be the same as or different from the operating system of another

virtual machine executed by the same host machine. A hypervisor thus enables multiple operating systems to be executed alongside each other while sharing the same set of computing resources of the host machine. The host machines depicted in FIG. 2 may have the same or different types of hypervisors.

[0090] A compute instance can be a virtual machine instance or a bare metal instance. In FIG. 2, compute instances 268 on host machine 202 and 274 on host machine 208 are examples of virtual machine instances. Host machine 206 is an example of a bare metal instance that is provided to a customer.

[0091] In certain instances, an entire host machine may be provisioned to a single customer, and all of the one or more compute instances (either virtual machines or bare metal instance) hosted by that host machine belong to that same customer. In other instances, a host machine may be shared between multiple customers (i.e., multiple tenants). In such a multi-tenancy scenario, a host machine may host virtual machine compute instances belonging to different customers. These compute instances may be members of different VCNs of different customers. In certain embodiments, a bare metal compute instance is hosted by a bare metal server without a hypervisor. When a bare metal compute instance is provisioned, a single customer or tenant maintains control of the physical CPU, memory, and network interfaces of the host machine hosting the bare metal instance and the host machine is not shared with other customers or tenants.

[0092] As previously described, each compute instance that is part of a VCN is associated with a VNIC that enables the compute instance to become a member of a subnet of the VCN. The VNIC associated with a compute instance facilitates the communication of packets or frames to and from the compute instance. A VNIC is associated with a compute instance when the compute instance is created. In certain embodiments, for a compute instance executed by a host machine, the VNIC associated with that compute instance is executed by an NVD connected to the host machine. For example, in FIG. 2, host machine 202 executes a virtual machine compute instance 268 that is associated with VNIC 276, and VNIC 276 is executed by NVD 210 connected to host machine 202. As another example, bare metal instance 272 hosted by host machine 206 is associated with VNIC 280 that is executed by NVD 212 connected to host machine 206. As yet another example, VNIC 284 is associated with compute instance 274 executed by host machine 208, and VNIC 284 is executed by NVD 212 connected to host machine 208.

[0093] For compute instances hosted by a host machine, an NVD connected to that host machine also executes VCN VRs corresponding to VCNs of which the compute instances are members. For example, in the embodiment depicted in FIG. 2, NVD 210 executes VCN VR 277 corresponding to the VCN of which compute instance 268 is a member. NVD 212 may also execute one or more VCN VRs 283 corresponding to VCNs corresponding to the compute instances hosted by host machines 206 and 208.

[0094] A host machine may include one or more network interface cards (NIC) that enable the host machine to be connected to other devices. A NIC on a host machine may provide one or more ports (or interfaces) that enable the host machine to be communicatively connected to another device. For example, a host machine may be connected to an NVD using one or more ports (or interfaces) provided on the

host machine and on the NVD. A host machine may also be connected to other devices such as another host machine.

[0095] For example, in FIG. 2, host machine 202 is connected to NVD 210 using link 220 that extends between a port 234 provided by a NIC 232 of host machine 202 and between a port 236 of NVD 210. Host machine 206 is connected to NVD 212 using link 224 that extends between a port 246 provided by a NIC 244 of host machine 206 and between a port 248 of NVD 212. Host machine 208 is connected to NVD 212 using link 226 that extends between a port 252 provided by a NIC 250 of host machine 208 and between a port 254 of NVD 212.

[0096] The NVDs are in turn connected via communication links to top-of-the-rack (TOR) switches, which are connected to physical network 218 (also referred to as the switch fabric). In certain embodiments, the links between a host machine and an NVD, and between an NVD and a TOR switch are Ethernet links. For example, in FIG. 2, NVDs 210 and 212 are connected to TOR switches 214 and 216, respectively, using links 228 and 230. In certain embodiments, the links 220, 224, 226, 228, and 230 are Ethernet links. The collection of host machines and NVDs that are connected to a TOR is sometimes referred to as a rack.

[0097] Physical network 218 provides a communication fabric that enables TOR switches to communicate with each other. Physical network 218 can be a multi-tiered network. In certain implementations, physical network 218 is a multi-tiered Clos network of switches, with TOR switches 214 and 216 representing the leaf level nodes of the multi-tiered and multi-node physical switching network 218. Different Clos network configurations are possible including but not limited to a 2-tier network, a 3-tier network, a 4-tier network, a 5-tier network, and in general a “n”-tiered network. An example of a Clos network is depicted in FIG. 5 and described below.

[0098] Various different connection configurations are possible between host machines and NVDs such as one-to-one configuration, many-to-one configuration, one-to-many configuration, and others. In a one-to-one configuration implementation, each host machine is connected to its own separate NVD. For example, in FIG. 2, host machine 202 is connected to NVD 210 via NIC 232 of host machine 202. In a many-to-one configuration, multiple host machines are connected to one NVD. For example, in FIG. 2, host machines 206 and 208 are connected to the same NVD 212 via NICs 244 and 250, respectively.

[0099] In a one-to-many configuration, one host machine is connected to multiple NVDs. FIG. 3 shows an example within CSPI 300 where a host machine is connected to multiple NVDs. As shown in FIG. 3, host machine 302 comprises a network interface card (NIC) 304 that includes multiple ports 306 and 308. Host machine 300 is connected to a first NVD 310 via port 306 and link 320 and connected to a second NVD 312 via port 308 and link 322. Ports 306 and 308 may be Ethernet ports and the links 320 and 322 between host machine 302 and NVDs 310 and 312 may be Ethernet links. NVD 310 is in turn connected to a first TOR switch 314 and NVD 312 is connected to a second TOR switch 316. The links between NVDs 310 and 312, and TOR switches 314 and 316 may be Ethernet links. TOR switches 314 and 316 represent the Tier-0 switching devices in multi-tiered physical network 318.

[0100] The arrangement depicted in FIG. 3 provides two separate physical network paths to and from physical switch

network 318 to host machine 302: a first path traversing TOR switch 314 to NVD 310 to host machine 302, and a second path traversing TOR switch 316 to NVD 312 to host machine 302. The separate paths provide for enhanced availability (referred to as high availability) of host machine 302. If there are problems in one of the paths (e.g., a link in one of the paths goes down) or devices (e.g., a particular NVD is not functioning), then the other path may be used for communications to/from host machine 302.

[0101] In the configuration depicted in FIG. 3, the host machine is connected to two different NVDs using two different ports provided by a NIC of the host machine. In other embodiments, a host machine may include multiple NICs that enable connectivity of the host machine to multiple NVDs.

[0102] Referring back to FIG. 2, an NVD is a physical device or component that performs one or more network and/or storage virtualization functions. An NVD may be any device with one or more processing units (e.g., CPUs, Network Processing Units (NPU), FPGAs, packet processing pipelines, etc.), memory including cache, and ports. The various virtualization functions may be performed by software/firmware executed by the one or more processing units of the NVD.

[0103] An NVD may be implemented in various different forms. For example, in certain embodiments, an NVD is implemented as an interface card referred to as a smartNIC or an intelligent NIC with an embedded processor onboard. A smartNIC is a separate device from the NICs on the host machines. In FIG. 2, the NVDs 210 and 212 may be implemented as smartNICs that are connected to host machines 202, and host machines 206 and 208, respectively.

[0104] A smartNIC is however just one example of an NVD implementation. Various other implementations are possible. For example, in some other implementations, an NVD or one or more functions performed by the NVD may be incorporated into or performed by one or more host machines, one or more TOR switches, and other components of CSPI 200. For example, an NVD may be embodied in a host machine where the functions performed by an NVD are performed by the host machine. As another example, an NVD may be part of a TOR switch, or a TOR switch may be configured to perform functions performed by an NVD that enables the TOR switch to perform various complex packet transformations that are used for a public cloud. A TOR that performs the functions of an NVD is sometimes referred to as a smart TOR. In yet other implementations, where virtual machines (VMs) instances, but not bare metal (BM) instances, are offered to customers, functions performed by an NVD may be implemented inside a hypervisor of the host machine. In some other implementations, some of the functions of the NVD may be offloaded to a centralized service running on a fleet of host machines.

[0105] In certain embodiments, such as when implemented as a smartNIC as shown in FIG. 2, an NVD may comprise multiple physical ports that enable it to be connected to one or more host machines and to one or more TOR switches. A port on an NVD can be classified as a host-facing port (also referred to as a “south port”) or a network-facing or TOR-facing port (also referred to as a “north port”). A host-facing port of an NVD is a port that is used to connect the NVD to a host machine. Examples of host-facing ports in FIG. 2 include port 236 on NVD 210, and ports 248 and 254 on NVD 212. A network-facing port

of an NVD is a port that is used to connect the NVD to a TOR switch. Examples of network-facing ports in FIG. 2 include port **256** on NVD **210**, and port **258** on NVD **212**. As shown in FIG. 2, NVD **210** is connected to TOR switch **214** using link **228** that extends from port **256** of NVD **210** to the TOR switch **214**. Likewise, NVD **212** is connected to TOR switch **216** using link **230** that extends from port **258** of NVD **212** to the TOR switch **216**.

[0106] An NVD receives packets and frames from a host machine (e.g., packets and frames generated by a compute instance hosted by the host machine) via a host-facing port and, after performing the necessary packet processing, may forward the packets and frames to a TOR switch via a network-facing port of the NVD. An NVD may receive packets and frames from a TOR switch via a network-facing port of the NVD and, after performing the necessary packet processing, may forward the packets and frames to a host machine via a host-facing port of the NVD.

[0107] In certain embodiments, there may be multiple ports and associated links between an NVD and a TOR switch. These ports and links may be aggregated to form a link aggregator group of multiple ports or links (referred to as a LAG). Link aggregation allows multiple physical links between two endpoints (e.g., between an NVD and a TOR switch) to be treated as a single logical link. All the physical links in a given LAG may operate in full-duplex mode at the same speed. LAGs help increase the bandwidth and reliability of the connection between two endpoints. If one of the physical links in the LAG goes down, traffic is dynamically and transparently reassigned to one of the other physical links in the LAG. The aggregated physical links deliver higher bandwidth than each individual link. The multiple ports associated with a LAG are treated as a single logical port. Traffic can be load-balanced across the multiple physical links of a LAG. One or more LAGs may be configured between two endpoints. The two endpoints may be between an NVD and a TOR switch, between a host machine and an NVD, and the like.

[0108] An NVD implements or performs network virtualization functions. These functions are performed by software/firmware executed by the NVD. Examples of network virtualization functions include without limitation: packet encapsulation and de-capsulation functions; functions for creating a VCN network; functions for implementing network policies such as VCN security list (firewall) functionality; functions that facilitate the routing and forwarding of packets to and from compute instances in a VCN; and the like. In certain embodiments, upon receiving a packet, an NVD is configured to execute a packet processing pipeline for processing the packet and determining how the packet is to be forwarded or routed. As part of this packet processing pipeline, the NVD may execute one or more virtual functions associated with the overlay network such as executing VNICs associated with compute instances in the VCN, executing a Virtual Router (VR) associated with the VCN, the encapsulation and decapsulation of packets to facilitate forwarding or routing in the virtual network, execution of certain gateways (e.g., the Local Peering Gateway), the implementation of Security Lists, Network Security Groups, network address translation (NAT) functionality (e.g., the translation of Public IP to Private IP on a host by host basis), throttling functions, and other functions.

[0109] In certain embodiments, the packet processing data path in an NVD may comprise multiple packet pipelines,

each composed of a series of packet transformation stages. In certain implementations, upon receiving a packet, the packet is parsed and classified to a single pipeline. The packet is then processed in a linear fashion, one stage after another, until the packet is either dropped or sent out over an interface of the NVD. These stages provide basic functional packet processing building blocks (e.g., validating headers, enforcing throttle, inserting new Layer-2 headers, enforcing L4 firewall, VCN encapsulation/decapsulation, etc.) so that new pipelines can be constructed by composing existing stages, and new functionality can be added by creating new stages and inserting them into existing pipelines.

[0110] An NVD may perform both control plane and data plane functions corresponding to a control plane and a data plane of a VCN. Examples of a VCN Control Plane are also depicted in FIGS. **18**, **19**, **20**, and **21** (see references **1816**, **1916**, **2016**, and **2116**) and described below. Examples of a VCN Data Plane are depicted in FIGS. **18**, **19**, **20**, and **21** (see references **1818**, **1918**, **2018**, and **2118**) and described below. The control plane functions include functions used for configuring a network (e.g., setting up routes and route tables, configuring VNICs, etc.) that controls how data is to be forwarded. In certain embodiments, a VCN Control Plane is provided that computes all the overlay-to-substrate mappings centrally and publishes them to the NVDs and to the virtual network edge devices such as various gateways such as the DRG, the SGW, the IGW, etc. Firewall rules may also be published using the same mechanism. In certain embodiments, an NVD only gets the mappings that are relevant for that NVD. The data plane functions include functions for the actual routing/forwarding of a packet based upon configuration set up using control plane. A VCN data plane is implemented by encapsulating the customer's network packets before they traverse the substrate network. The encapsulation/decapsulation functionality is implemented on the NVDs. In certain embodiments, an NVD is configured to intercept all network packets in and out of host machines and perform network virtualization functions.

[0111] As indicated above, an NVD executes various virtualization functions including VNICs and VCN VRs. An NVD may execute VNICs associated with the compute instances hosted by one or more host machines connected to the VNIC. For example, as depicted in FIG. 2, NVD **210** executes the functionality for VNIC **276** that is associated with compute instance **268** hosted by host machine **202** connected to NVD **210**. As another example, NVD **212** executes VNIC **280** that is associated with bare metal compute instance **272** hosted by host machine **206** and executes VNIC **284** that is associated with compute instance **274** hosted by host machine **208**. A host machine may host compute instances belonging to different VCNs, which belong to different customers, and the NVD connected to the host machine may execute the VNICs (i.e., execute VNICs-relate functionality) corresponding to the compute instances.

[0112] An NVD also executes VCN Virtual Routers corresponding to the VCNs of the compute instances. For example, in the embodiment depicted in FIG. 2, NVD **210** executes VCN VR **277** corresponding to the VCN to which compute instance **268** belongs. NVD **212** executes one or more VCN VRs **283** corresponding to one or more VCNs to which compute instances hosted by host machines **206** and **208** belong. In certain embodiments, the VCN VR corresponding to that VCN is executed by all the NVDs connected to host machines that host at least one compute

instance belonging to that VCN. If a host machine hosts compute instances belonging to different VCNs, an NVD connected to that host machine may execute VCN VRs corresponding to those different VCNs.

[0113] In addition to VNICs and VCN VRs, an NVD may execute various software (e.g., daemons) and include one or more hardware components that facilitate the various network virtualization functions performed by the NVD. For purposes of simplicity, these various components are grouped together as “packet processing components” shown in FIG. 2. For example, NVD 210 comprises packet processing components 286 and NVD 212 comprises packet processing components 288. For example, the packet processing components for an NVD may include a packet processor that is configured to interact with the NVD’s ports and hardware interfaces to monitor all packets received by and communicated using the NVD and store network information. The network information may, for example, include network flow information identifying different network flows handled by the NVD and per flow information (e.g., per flow statistics). In certain embodiments, network flows information may be stored on a per VNIC basis. The packet processor may perform packet-by-packet manipulations as well as implement stateful NAT and L4 firewall (FW). As another example, the packet processing components may include a replication agent that is configured to replicate information stored by the NVD to one or more different replication target stores. As yet another example, the packet processing components may include a logging agent that is configured to perform logging functions for the NVD. The packet processing components may also include software for monitoring the performance and health of the NVD and, also possibly of monitoring the state and health of other components connected to the NVD.

[0114] FIG. 1 shows the components of an example virtual or overlay network including a VCN, subnets within the VCN, compute instances deployed on subnets, VNICs associated with the compute instances, a VR for a VCN, and a set of gateways configured for the VCN. The overlay components depicted in FIG. 1 may be executed or hosted by one or more of the physical components depicted in FIG. 2. For example, the compute instances in a VCN may be executed or hosted by one or more host machines depicted in FIG. 2. For a compute instance hosted by a host machine, the VNIC associated with that compute instance is typically executed by an NVD connected to that host machine (i.e., the VNIC functionality is provided by the NVD connected to that host machine). The VCN VR function for a VCN is executed by all the NVDs that are connected to host machines hosting or executing the compute instances that are part of that VCN. The gateways associated with a VCN may be executed by one or more different types of NVDs. For example, certain gateways may be executed by smartNICs, while others may be executed by one or more host machines or other implementations of NVDs.

[0115] As described above, a compute instance in a customer VCN may communicate with various different endpoints, where the endpoints can be within the same subnet as the source compute instance, in a different subnet but within the same VCN as the source compute instance, or with an endpoint that is outside the VCN of the source compute instance. These communications are facilitated using VNICs associated with the compute instances, the VCN VRs, and the gateways associated with the VCNs.

[0116] For communications between two compute instances on the same subnet in a VCN, the communication is facilitated using VNICs associated with the source and destination compute instances. The source and destination compute instances may be hosted by the same host machine or by different host machines. A packet originating from a source compute instance may be forwarded from a host machine hosting the source compute instance to an NVD connected to that host machine. On the NVD, the packet is processed using a packet processing pipeline, which can include execution of the VNIC associated with the source compute instance. Since the destination endpoint for the packet is within the same subnet, execution of the VNIC associated with the source compute instance results in the packet being forwarded to an NVD executing the VNIC associated with the destination compute instance, which then processes and forwards the packet to the destination compute instance. The VNICs associated with the source and destination compute instances may be executed on the same NVD (e.g., when both the source and destination compute instances are hosted by the same host machine) or on different NVDs (e.g., when the source and destination compute instances are hosted by different host machines connected to different NVDs). The VNICs may use routing/forwarding tables stored by the NVD to determine the next hop for the packet.

[0117] For a packet to be communicated from a compute instance in a subnet to an endpoint in a different subnet in the same VCN, the packet originating from the source compute instance is communicated from the host machine hosting the source compute instance to the NVD connected to that host machine. On the NVD, the packet is processed using a packet processing pipeline, which can include execution of one or more VNICs, and the VR associated with the VCN. For example, as part of the packet processing pipeline, the NVD executes or invokes functionality corresponding to the VNIC (also referred to as executes the VNIC) associated with source compute instance. The functionality performed by the VNIC may include looking at the VLAN tag on the packet. Since the packet’s destination is outside the subnet, the VCN VR functionality is next invoked and executed by the NVD. The VCN VR then routes the packet to the NVD executing the VNIC associated with the destination compute instance. The VNIC associated with the destination compute instance then processes the packet and forwards the packet to the destination compute instance. The VNICs associated with the source and destination compute instances may be executed on the same NVD (e.g., when both the source and destination compute instances are hosted by the same host machine) or on different NVDs (e.g., when the source and destination compute instances are hosted by different host machines connected to different NVDs).

[0118] If the destination for the packet is outside the VCN of the source compute instance, then the packet originating from the source compute instance is communicated from the host machine hosting the source compute instance to the NVD connected to that host machine. The NVD executes the VNIC associated with the source compute instance. Since the destination end point of the packet is outside the VCN, the packet is then processed by the VCN VR for that VCN. The NVD invokes the VCN VR functionality, which may result in the packet being forwarded to an NVD executing the appropriate gateway associated with the VCN. For example, if the destination is an endpoint within the cus-

tomers' on-premise network, then the packet may be forwarded by the VCN VR to the NVD executing the DRG gateway configured for the VCN. The VCN VR may be executed on the same NVD as the NVD executing the VNIC associated with the source compute instance or by a different NVD. The gateway may be executed by an NVD, which may be a smartNIC, a host machine, or other NVD implementation. The packet is then processed by the gateway and forwarded to a next hop that facilitates communication of the packet to its intended destination endpoint. For example, in the embodiment depicted in FIG. 2, a packet originating from compute instance 268 may be communicated from host machine 202 to NVD 210 over link 220 (using NIC 232). On NVD 210, VNIC 276 is invoked since it is the VNIC associated with source compute instance 268. VNIC 276 is configured to examine the encapsulated information in the packet and determine a next hop for forwarding the packet with the goal of facilitating communication of the packet to its intended destination endpoint, and then forward the packet to the determined next hop.

[0119] A compute instance deployed on a VCN can communicate with various different endpoints. These endpoints may include endpoints that are hosted by CSPI 200 and endpoints outside CSPI 200. Endpoints hosted by CSPI 200 may include instances in the same VCN or other VCNs, which may be the customer's VCNs, or VCNs not belonging to the customer. Communications between endpoints hosted by CSPI 200 may be performed over physical network 218. A compute instance may also communicate with endpoints that are not hosted by CSPI 200 or are outside CSPI 200. Examples of these endpoints include endpoints within a customer's on-premise network or data center, or public endpoints accessible over a public network such as the Internet. Communications with endpoints outside CSPI 200 may be performed over public networks (e.g., the Internet) (not shown in FIG. 2) or private networks (not shown in FIG. 2) using various communication protocols.

[0120] The architecture of CSPI 200 depicted in FIG. 2 is merely an example and is not intended to be limiting. Variations, alternatives, and modifications are possible in alternative embodiments. For example, in some implementations, CSPI 200 may have more or fewer systems or components than those shown in FIG. 2, may combine two or more systems, or may have a different configuration or arrangement of systems. The systems, subsystems, and other components depicted in FIG. 2 may be implemented in software (e.g., code, instructions, program) executed by one or more processing units (e.g., processors, cores) of the respective systems, using hardware, or combinations thereof. The software may be stored on a non-transitory storage medium (e.g., on a memory device).

[0121] FIG. 4 depicts connectivity between a host machine and an NVD for providing I/O virtualization for supporting multitendency according to certain embodiments. As depicted in FIG. 4, host machine 402 executes a hypervisor 404 that provides a virtualized environment. Host machine 402 executes two virtual machine instances, VM1 406 belonging to customer/tenant #1 and VM2 408 belonging to customer/tenant #2. Host machine 402 comprises a physical NIC 410 that is connected to an NVD 412 via link 414. Each of the compute instances is attached to a VNIC that is executed by NVD 412. In the embodiment in FIG. 4, VM1 406 is attached to VNIC-VM1 420 and VM2 408 is attached to VNIC-VM2 422.

[0122] As shown in FIG. 4, NIC 410 comprises two logical NICs, logical NIC A 416 and logical NIC B 418. Each virtual machine is attached to and configured to work with its own logical NIC. For example, VM1 406 is attached to logical NIC A 416 and VM2 408 is attached to logical NIC B 418. Even though host machine 402 comprises only one physical NIC 410 that is shared by the multiple tenants, due to the logical NICs, each tenant's virtual machine believes they have their own host machine and NIC.

[0123] In certain embodiments, each logical NIC is assigned its own VLAN ID. Thus, a specific VLAN ID is assigned to logical NIC A 416 for Tenant #1 and a separate VLAN ID is assigned to logical NIC B 418 for Tenant #2. When a packet is communicated from VM1 406, a tag assigned to Tenant #1 is attached to the packet by the hypervisor and the packet is then communicated from host machine 402 to NVD 412 over link 414. In a similar manner, when a packet is communicated from VM2 408, a tag assigned to Tenant #2 is attached to the packet by the hypervisor and the packet is then communicated from host machine 402 to NVD 412 over link 414. Accordingly, a packet 424 communicated from host machine 402 to NVD 412 has an associated tag 426 that identifies a specific tenant and associated VM. On the NVD, for a packet 424 received from host machine 402, the tag 426 associated with the packet is used to determine whether the packet is to be processed by VNIC-VM1 420 or by VNIC-VM2 422. The packet is then processed by the corresponding VNIC. The configuration depicted in FIG. 4 enables each tenant's compute instance to believe that they own their own host machine and NIC. The setup depicted in FIG. 4 provides for I/O virtualization for supporting multi-tenancy.

[0124] FIG. 5 depicts a simplified block diagram of a physical network 500 according to certain embodiments. The embodiment depicted in FIG. 5 is structured as a Clos network. A Clos network is a particular type of network topology designed to provide connection redundancy while maintaining high bisection bandwidth and maximum resource utilization. A Clos network is a type of non-blocking, multistage or multi-tiered switching network, where the number of stages or tiers can be two, three, four, five, etc. The embodiment depicted in FIG. 5 is a 3-tiered network comprising tiers 1, 2, and 3. The TOR switches 504 represent Tier-0 switches in the Clos network. One or more NVDs are connected to the TOR switches. Tier-0 switches are also referred to as edge devices of the physical network. The Tier-0 switches are connected to Tier-1 switches, which are also referred to as leaf switches. In the embodiment depicted in FIG. 5, a set of "n" Tier-0 TOR switches are connected to a set of "n" Tier-1 switches and together form a pod. Each Tier-0 switch in a pod is interconnected to all the Tier-1 switches in the pod, but there is no connectivity of switches between pods. In certain implementations, two pods are referred to as a block. Each block is served by or connected to a set of "n" Tier-2 switches (sometimes referred to as spine switches). There can be several blocks in the physical network topology. The Tier-2 switches are in turn connected to "n" Tier-3 switches (sometimes referred to as super-spine switches). Communication of packets over physical network 500 is typically performed using one or more Layer-3 communication protocols. Typically, all the layers of the physical network, except for the TORs layer are n-ways redundant thus allowing for high availability. Policies may be specified for pods and blocks to control the

visibility of switches to each other in the physical network so as to enable scaling of the physical network.

[0125] A feature of a Clos network is that the maximum hop count to reach from one Tier-0 switch to another Tier-0 switch (or from an NVD connected to a Tier-0 switch to another NVD connected to a Tier-0 switch) is fixed. For example, in a 3-Tiered Clos network at most seven hops are needed for a packet to reach from one NVD to another NVD, where the source and target NVDs are connected to the leaf tier of the Clos network. Likewise, in a 4-tiered Clos network, at most nine hops are needed for a packet to reach from one NVD to another NVD, where the source and target NVDs are connected to the leaf tier of the Clos network. Thus, a Clos network architecture maintains consistent latency throughout the network, which is important for communication within and between data centers. A Clos topology scales horizontally and is cost effective. The bandwidth/throughput capacity of the network can be easily increased by adding more switches at the various tiers (e.g., more leaf and spine switches) and by increasing the number of links between the switches at adjacent tiers.

[0126] In certain embodiments, each resource within CSPI is assigned a unique identifier called a Cloud Identifier (CID). This identifier is included as part of the resource's information and can be used to manage the resource, for example, via a Console or through APIs. An example syntax for a CID is:

[0127] `ocid1.<RESOURCE TYPE>.<REALM>.[REGION][.FUTURE USE].<UNIQUE ID>`

[0128] where,

[0129] `ocid1`: The literal string indicating the version of the CID;

[0130] `resource type`: The type of resource (for example, instance, volume, VCN, subnet, user, group, and so on);

[0131] `realm`: The realm the resource is in. Example values are "c1" for the commercial realm, "c2" for the Government Cloud realm, or "c3" for the Federal Government Cloud realm, etc. Each realm may have its own domain name;

[0132] `region`: The region the resource is in. If the region is not applicable to the resource, this part might be blank;

[0133] `future use`: Reserved for future use.

[0134] `unique ID`: The unique portion of the ID. The format may vary depending on the type of resource or service.

SDN-Based DNS Resolution

[0135] As briefly described herein above, an SDN-based resolution involves a DNS resolution at an SDN, where the DNS resolution is provided to a non-SDN service. In the next figures, examples of such an SDN-based resolution are described. In these examples, the non-SDN service is illustrated to belong to a substrate network. However, the embodiments are not limited as such. Instead, the embodiments similarly and equivalently apply to any non-SDN service. For example, whereas the DNS resolution is performed at a first SDN, the non-SDN service can be a service of a second SDN.

[0136] FIG. 6 depicts a simplified diagram 600 of an SDN 610 providing DNS resolutions to a non-SDN service, according to certain embodiments. In an example, the SDN 610 can be an overlay network, such as a cloud network

provided by a CSPI. The non-SDN service is illustrated as a service 662 of another network 660. This other network can be, for example, a substrate network. The SDN 610 is configured to return DNS resolutions to the service 662, where these DNS resolutions relate to resources of a customer private network 650 of a customer and/or resource of an on-premises network 670 of the customer. The on-premises network 670 can be communicatively coupled with the customer private network 650 (e.g., via a FastConnect peering). The on-premises network 670 is illustrated with a dashed rectangle to indicate that the use of such a network is optional and can depend on the customer's configuration.

[0137] In an example, the SDN 610 includes a service private network 620, in addition to the customer private network 650. The service private network 620 can be implemented for the service 662 of the other network 660. In comparison, the customer private network 650 can be implemented for the customer. Each of the two private networks 620 and 650 can be a VCN. The two private networks 620 and 650 are communicatively coupled (e.g., at the virtual level and the underlying substrate level using virtualization technology).

[0138] The service private network 620 can include two subnets: a service subnet 630 and a customer subnet 640. The service subnet 630 is configured for the service 662 and includes an endpoint 632. The endpoint 632 can be implemented as a private endpoint in the service private network 620 for the service 662. In comparison, the customer subnet 640 is configured for the customer and, as such, is associated with the customer private network 650. The customer subnet 640 includes a DNS proxy 642. The DNS proxy 642 provides DNS resolutions for IP address that existing with the customer private network 650 and/or on premises network 670. The endpoint 632 is configured to bridge the service 662 with the DNS proxy 642 such that the DNS resolutions can be provided from the DNS proxy 642 for IP addresses of the customer private network 650 and/or on premises network 670 to the service 662 on the other network 660.

[0139] As further illustrated in FIG. 6, the customer network 650 includes a customer subnet 652. An endpoint 654 of the customer private network 650 can have an IP address that belongs to the customer subnet 652. In particular, the customer can define (e.g., based on input via a control plane) the IP address from the customer subnet 652 to use for the endpoint 654. The endpoint 654 can also be a private endpoint of the customer private network 652. The customer can also define (e.g., based on input via the control plane) and FQDN to use for accessing the endpoint 654. The FQDN and the IP address can be associated in a DNS record 658 of a DNS resolver 656 of the customer private network 650. In addition to or alternative to customer input defining the IP address and/or the FQDN, the control plane can define the IP address and/or FQDN during the configuration of the customer private network 650 and/or the launch of the endpoint 654.

[0140] Similarly, the on-premises network 670 can include an endpoint 672. This endpoint 672 can also be private endpoint and can have an IP address and an FQDN. A DNS record can associate the IP address and the FQDN. Such a record can be maintained by a DNS resolver (not shown in FIG. 6) of the on-premises network 670 and/or the DNS resolver 656 of the customer private network 650.

[0141] In an example, the service 662 can be implemented in hardware on the substrate network. For instance, the service 662 can represent a key management system that manages cryptographic keys (e.g., rotation, expiration, etc.) for the endpoint 654, the endpoint 672, and/or other endpoints of the customer accessible via the customer private network 650. Other examples of the service 662 are possible. For instance, the service 662 can be an object storage service (e.g., hosted on the substrate network or another SDN) that can store objects for the endpoint 654, the endpoint 672, and/or other endpoints of the customer accessible via the customer private network 650.

[0142] Furthermore, the service 662 can be a single tenant service. Particularly, the customer private network 650 is implemented based on a tenancy of the customer at the SDN 610. Other customers can be associated with other tenancies at the SDN 610. For each of such customers, one or more customer private networks can be deployed in association with their tenancy. In this illustration, the service 662 is deployed in support of only the customer's tenancy. Another service would be deployed in support of a tenancy of another customer. Alternatively, the service 662 can be a multi-tenant service. Particularly, the service 662 is deployed in support of multiple tenancies, each corresponding to a different customer.

[0143] Generally, the service 662 sends a DNS resolution request to the endpoint 632. This request can include, for example, the FQDN of the endpoint 654. As further described herein above, this request can be sent using a first reserved IP address (e.g., a first class E IP address) belonging to the other network 660 (e.g., reserved for a component of the other network 660 such as for the service 662). Given an association between the first reserved IP address and an actual IP address (e.g., an overlay IP address) of the DNS proxy 642, the endpoint 632 sends the DNS resolution request to the DNS proxy 642. This association can be stored by the endpoint 632.

[0144] Next, the DNS proxy 642 interacts with (e.g., queries or performs a DNS look-up using the FQDN from the DNS resolution request) the DNS resolver 656 to obtain the DNS record 658 (or information therefrom) and, accordingly, return a DNS resolution to the endpoint 632. As further described herein below, the DNS resolution can mask the actual IP address of the endpoint 654. Instead, a masked IP address (e.g., a second reserved IP address, such as a second class E IP address, reserved for the DNS proxy 642) is returned. The DNS proxy 642 can store an association between the masked IP address and the actual IP address.

[0145] In turn, the endpoint 632 sends the DNS resolution to service 662, where the DNS resolution indicates the masked IP address. As further described herein below, the service 662 can associate the masked IP address with yet another IP address (e.g., a third reserved IP address, such as a third class E IP address, reserved for a component of the other network 660, such as the service 662). The association between the two IP addresses and the FQDN can be stored as part of a DNS record at the service 662 for the endpoint 654. Thereafter, to connect to the endpoint 654, the service uses the association (or DNS record) to send a connection request (e.g., by using the third reserved IP address) to the endpoint 632. The endpoint 632 then determines the second reserved IP address of the DNS proxy 642 and sends the connection request thereto. The DNS proxy then determines the actual IP address of the endpoint 654 and send the

connection request thereto. A connection can then be established between the service 662 and the endpoint 654. The endpoint 632 and the DNS proxy 642 may be network hops that support this connection.

[0146] A similar approach can be used for DNS resolution and subsequent connection to the endpoint 672 of the on-premises network 670. As described herein above, the DNS resolver 656 of the customer private network 650 and/or another DNS resolver of the on-premises network 670 maintains a DNS record of the endpoint 672. Depending on the implementation, the DNS proxy 642 can query the DNS resolver 656 or the DNS resolver of the on-premises network 670.

[0147] FIG. 7 depicts a simplified diagram 700 of an SDN 710 providing DNS resolutions to multiple non-SDN services, according to some embodiments. Some components of the SDN 710 are similar to those of SDN 610 of FIG. 6. In the interest of brevity, the description of similar components is not repeated in FIG. 7. Instead, the description provided in connection with FIG. 6 applies to the similar components of FIG. 7. Furthermore, FIG. 7 does not depict an on-premises network. Nonetheless, one or more on-premises networks can be deployed, each corresponding to a customer having a customer private network 750 in the SDN 710. Similar to FIG. 6, here also, a non-SDN service can obtain a DSN resolution for an endpoint in an on-premises network.

[0148] As illustrated, the SDN 710 includes a service private network 720. The service private network 720 is similar to the service private network 620 of FIG. 6. In particular, the service private network 720 includes a service subnet 730 (similar to the first service subnet 630 of FIG. 6) and a customer subnet 740 (similar to the customer subnet 640 of FIG. 6). In turn, the service subnet 730 includes an endpoint 732, similar to the endpoint 632 of FIG. 6. The customer subnet 740 is similar to the DNS proxy 742. Here though, the customer subnet includes a plurality DNS proxies, such as a first DNS proxy 742 and a second DNS proxy 744 (although more than two DNS proxies are possible).

[0149] The SDN 710 also includes a plurality of customer private networks, such as a first customer private network 750 and a second private customer network 770 (although more than two DNS customer private networks are possible). In turn, each one of the private customer private networks is similar to the customer private network 650 of FIG. 6, is associated with a customer tenancy, and includes a customer subnet (simile to the customer subnet 652 of FIG. 6) and an endpoint 754 (similar to the endpoint 654 of FIG. 6), among other components (e.g., a DNS resolver storing DNS records, and possibly multiple customer subnets). Each one of the DNS proxies corresponds to one of the customer private networks or a customer subnet within a customer private network. In particular, a DNS resolver within a customer private network (e.g., in a customer subnet thereof) is proxied by a DNS proxy within the service private network 720 (e.g., in the customer subnet 740 thereof). In the illustration of FIG. 7, the DNS proxy 742 proxies DNS resolution for the endpoint 754 in the customer private network 750. In comparison, the DNS proxy 744 proxies DNS resolution for the endpoint 774 in the customer private network 770.

[0150] Another network 760, which is similar to the other network 660 of FIG. 7, interfaces with the SDN 710 and includes a plurality of services, such as a first service 762

and a second service **764** (although more than two services are possible). Each one of the services **762** and **764** can be referred to as a non-SDN service **710** (as a way to indicate that this service is not hosted by the SDN **710**) and can be similar to the service **662** of FIG. 6.

[0151] In an example, each one of the services **762** and **764** is a single tenant service. In particular, the service **762** is associated with the customer tenancy corresponding to the customer private network **750**. Said differently, the service **762** can have access to the endpoint **754** (and, possibly, other resources of the customer private network **750**) but not the endpoint **774** or other resources of the customer private network **770**. The access can be based on a DNS resolution provided by the DNS proxy **742** (given its association with the customer private network **750**) and bridged by the endpoint **732** of the service subnet **730**. In comparison, the service **764** is associated with the customer tenancy corresponding to the customer private network **770**. Accordingly, the service **764** can have access to the endpoint **774** (and, possibly, other resources of the customer private network **770**) but not the endpoint **754** or other resources of the customer private network **750**. The access can be based on a DNS resolution provided by the DNS proxy **744** (given its association with the customer private network **770**) and bridged by the endpoint **732** of the service subnet **730**. The same endpoint **732** is deployed for both services **762** and **764** as a bridge for the DNS resolution, where the DNS resolution for each one of the services **762** and **764** differs. The difference is, among other things, in the DNS proxying (e.g., using DNS proxy **742** for the service **762** and the DNS proxy **744** for the service **764**) and the particular endpoint/resource for which the DNS resolution can be returned (e.g., endpoint **754** for the service **762** but not the service **764**, and endpoint **774** for the service **764** but not the service **762**).

[0152] As described in FIG. 6, each one of the DNS proxies **742** and **744** stores a DNS resolution of the relevant FQDN and IP address and masks this IP address by using a reserved IP address associated with the DNS proxy. Similarly, each one of the services **762** and **764** stores a DNS resolution of the relevant FQDN and the reserved IP address of the associated DNS proxy and associates this reserved IP address with another reserved IP address. The endpoint **732** stores associations between the reserved IP addresses to facilitate the bridging.

[0153] For example, the service **762** sends a DNS resolution request to the endpoint **732**. This request can include, for example, the FQDN of the endpoint **754**. As further described herein above, this request can be sent using a first reserved IP address (e.g., a first class E IP address) belonging to the other network **760**. Given an association between the first reserved IP address and a first actual IP address (e.g., an overlay IP address) of the DNS proxy **742**, the endpoint **732** sends the DNS resolution request to the DNS proxy **742**. Next, the DNS proxy **742** interacts with the DNS resolver of the customer private network **750** to obtain the DNS record and, accordingly, return a DNS resolution to the endpoint **732**. As further described herein below, the DNS proxy **742** can mask the actual IP address of the endpoint **754**. In particular, a second reserved IP address, such as a second class E IP address, reserved for the DNS proxy **742** (instead of the actual IP address) is returned in the DNS resolution to the endpoint **732**. In turn, the endpoint **732** sends the DNS resolution to the service **762**, where the DNS resolution indicates the second reserved IP address. As further

described herein below, the service **762** can associate the second reserved IP address with yet another IP address (e.g., a third reserved IP address, such as a third class E IP address). The association between the two IP addresses and the FQDN can be stored as part of a DNS record at the service **762** for the endpoint **754**.

[0154] In comparison, the service **764** sends a DNS resolution request to also the endpoint **732**. This request can include, for example, the FQDN of the endpoint **774**. As further described herein above, this request can be sent using a fourth reserved IP address (e.g., a fourth class E IP address) belonging to the other network **760**. Given an association between the fourth reserved IP address and a second actual IP address (e.g., an overlay IP address) of the DNS proxy **744**, the endpoint **732** sends the DNS resolution request to the DNS proxy **744**. Next, the DNS proxy **744** interacts with the DNS resolver of the customer private network **770** to obtain the DNS record and, accordingly, return a DNS resolution to the endpoint **732**. As further described herein below, the DNS proxy **744** can mask the actual IP address of the endpoint **774**. In particular, a fifth reserved IP address, such as a fifth class E IP address, reserved for the DNS proxy **744** (instead of the actual IP address) is returned in the DNS resolution to the endpoint **732**. In turn, the endpoint **732** sends the DNS resolution to the service **764**, where the DNS resolution indicates the fifth reserved IP address. As further described herein below, the service **764** can associate the fifth reserved IP address with yet another IP address (e.g., a sixth reserved IP address, such as a sixth class E IP address). The association between the two IP addresses and the FQDN can be stored as part of a DNS record at the service **764** for the endpoint **774**.

[0155] FIG. 8 depicts information used for DNS resolution, according to some embodiments. In the illustration a FIG. 8, an SDN **810** and another network **860** communicatively coupled with the SDN **810** are shown. The SDN **810** is similar to any one of the SDNs **610** or **710** of FIG. 6 or FIG. 6, whereas the other network **860** is similar to any of the other network **660** or **760** of FIG. 6 or 7. In the interest of brevity, the description of similar components is not repeated in FIG. 7.

[0156] Instead, the description provided in connection with FIGS. 6-7 applies to the similar components of FIG. 8. In FIG. 8, the configuration of an endpoint **832** within a service subnet **830** of a service private network **820** that belongs to the SDN **810** is described. The use of the endpoint **832** for DNS resolution and subsequent traffic is also described.

[0157] In an example, the other network **860** (e.g., a substrate network) includes the service **862** (e.g., similar to the service **662** of FIG. 6 or the service **762** of FIG. 7). In turn, the service **862** includes multiple components that enable access to an endpoint **854** within a customer subnet **852** of a customer private network **850** that corresponds to a tenancy of a customer and that is implemented in the SDN **810**. In the illustration of FIG. 8, the components include, among other things, an endpoint service (ES) **864**, and a dynamic DNS resolver **862**.

[0158] The ES **864** can register (e.g., via a control plane) the service **862** as an endpoint service. This registered endpoint service can have a corresponding endpoint **832** in the SDN **810**, which can be a private endpoint service. The ES **864** can include one or more IP addresses as part of the endpoint service. The ES **864** can also include a service DNS

name as part of the endpoint service to simplify customer DNS configuration. The ES **864** can configure a reverse connection enabled (RCE) for the endpoint **832** in support of a service-to-customer connection. In particular, the RCE configuration enables the DNS resolver **862** to connect (e.g., via a private access data plane) with the endpoint **832** such that DNS resolution requests and connection requests can be sent to the endpoint **832** for resolutions and connections to the endpoint **854**.

[0159] In an example, the service private network **820** also includes an ES **822** that enables configuring access of the service **862** to the endpoint **854**. For example, a private access gateway (PGW) **824** can be set up. The PGW **824** can represent a gateway resource that the service **862** creates in the service private network **820** to handle private endpoint traffic. In certain example, only one private access gateway is used for any number of endpoint services registered in a single private network. The ES **822** can configure an RCE for the endpoint **854** in support of the service-to-customer connection. In particular, the RCE configuration enables a DNS proxy within a customer subnet **840** of the service private network **820** to connect (e.g., via a private access data plane) with the customer private network **850** (e.g., with the endpoint **854** and/or a DNS resolver thereof) such that DNS resolution requests and connection requests can be sent to the customer private network **850** for resolutions and connections to the endpoint **854**.

[0160] Each one of the endpoints **832** and **854** can be a private endpoint (one within the service private network **820** and one within the customer private network **850**). Generally, a private endpoint (PE) can be a resource representing a service within the corresponding private network. Each private endpoint can manifest as a VNIC with one or more private IP addresses in the relevant subnet. Customer instances can use the VNIC's private IP address or a corresponding DNS name to access the endpoint service. Multiple private endpoints (in one or multiple customer private network) can represent the same service. The RCE feature provides the data plane to initiate traffic flows. As part of RCE, DNS resolution can be enabled for IP addresses that exist within the customer private network **850** (and/or, similarly, or an on-premises network). When the DNS proxy feature for RCE is enabled, RCE also presents an RCE PE-VNIC in the service private network **820**.

[0161] Based on the architecture of FIG. 8, the customer can choose IP addresses within the customer subnets **840** and **852**. Additionally, or alternatively, the IP addresses are chosen automatically (e.g., by a control plane) from the customer subnets **840** and **852**. In both cases, an IP address (e.g., an overlay IP address, such as "10.10.2.1" from subnet "10.10.2.0/24") can be selected from the customer subnet **840** for the DNS proxy **846**. A first reserved IP address (e.g., "classE-IP1") can be used (e.g., registered by the ES **864**) to represent the DNS proxy **842** in the other network **860** (e.g., for the service **862**). For example, the dynamic DNS resolver stores information **866** about the DNS proxy **842**. This information **866** can enable DNS resolution requests to be sent to the DNS proxy **866** via the endpoint **832**. In one illustration, the information **866** includes an association between the first reserved IP address and the DNS proxy **842** (e.g., the association can indicate that "classE-IP1" represents the DNS proxy **842** having the IP address "10.10.2.1").

[0162] FIG. 9 depicts a sequence diagram for a DNS resolution, according to some embodiments. Steps of the

sequence diagram involve a service **910**, a service endpoint **920**, a DNS proxy **930**, and a customer DNS resolver **940**. The service **910**, the service endpoint **920**, and the DNS proxy **930** are examples of the service **862**, the endpoint **832**, and the DNS proxy **842**, respectively, of FIG. 8. The customer DNS resolver **940** is implemented in a customer private network in support of a customer endpoint of the customer private network, such as the endpoint **854** of FIG. 4. In the interest of clarity of explanation, the steps of the sequence diagram are discussed by continuing the example illustration of FIG. 8.

[0163] The service **910** (e.g., its ES **864**) can send the information **866** (e.g., including the association between a first reserved IP address and the DNS proxy **930**) as configuration information to the service endpoint **930**. This configuration information can be static (e.g., not dynamically updated) and can be stored by the service endpoint **920**. Using the association included in the information **866**, the service **910** (e.g., its dynamic DNS resolver) can send a DNS resolution request via RCE to the service endpoint **920**. The DNS resolution request can include an FQDN of the customer endpoint of the customer private network and can be destined to the first reserved IP address (e.g., "classE-IP1"). Here, the service endpoint **920** can use the configuration information (in particular, that the first reserved IP address corresponds to the DNS proxy **930**) to determine that this request is destined to the DNS proxy **930**. The service endpoint **920** then passes the DNS resolution request to the DNS proxy **930** (e.g., by replacing its destination address with the actual overlay IP address of the DNS proxy **930** such as "10.10.2.1").

[0164] In turn, the DNS proxy **930** sends the DNS resolution request to the DNS resolver **930** via RCE. In response, the DNS proxy receives a DNS resolution (e.g., that the IP address of the customer endpoint is "1.1.1.1"). As explained above, this IP address belongs to a customer subnet of the customer private network. Although not illustrated in FIG. 8 or 9, multiple customer private networks may exist, each including a customer subnet. The customer subnets of the different customers can have overlapping IP ranges.

[0165] The DNS proxy **930** can then mask the IP address of the customer endpoint by using a second reserved IP address (e.g., "classE-IP2") to represent this IP address. The DNS proxy **930** can store information **846** about the customer endpoint (this information **846** is shown in FIG. 8). The information **846** can enable traffic to be sent to the customer endpoint. For instance, this information includes an association between the second reserved IP address and the IP address of the customer endpoint (e.g., the association can indicate that "classE-IP2" represents the IP address "1.1.1.1"). The DNS proxy **930** sends, to the service endpoint **930**, the second reserved IP address to the customer endpoint as part of the DNS resolution. The service endpoint **930** then sends this DNS resolution to the service **910** (e.g., to its dynamic DNS resolver).

[0166] The service **910** (e.g., its dynamic DNS resolver) can determine the second reserved IP address from the DNS resolution and associate it with a third reserved IP address (e.g., "classE-IP3"). This third reserved IP address can be configured for the service **930**. The association can indicate that "classE-IP3 represents classE-IP2 to reach the customer endpoint having the FQDN." The service **910** (e.g., its dynamic DNS resolver) can store information **868** about the customer endpoint (this information **868** is shown in FIG. 8).

The information **868** can enable traffic to be sent to the customer endpoint. For instance, this information **868** includes the association between the second and third reserved IP addresses.

[0167] FIG. 10 depicts a sequence diagram for connecting to an endpoint based on a DNS resolution, according to some embodiments. Steps of the sequence diagram involve a service **1010**, a service endpoint **1020**, a DNS proxy **1030**, and a customer endpoint **940**. The service **1010**, the service endpoint **1020**, the DNS proxy **1030**, and the customer endpoint **1040** are examples of the service **862**, the endpoint **832**, the DNS proxy **842**, and the endpoint **854** respectively, of FIG. 8. In the interest of clarity of explanation, the steps of the sequence diagram are discussed by continuing the example illustration of FIG. 8.

[0168] To establish a connection with and/or exchange traffic with the customer endpoint **1040**, the service **1010** can use the information **868** about the customer endpoint **1040**. In particular, a connection request (and, similarly, data traffic) can be destined to the third reserved IP address (e.g., “classE-IP3”).

[0169] This connection request (and, similarly, data traffic) is received by the service endpoint **1020**. Here, the service endpoint **1020** can store similar information as the information **868** (e.g., including the association between the second and third reserved IP addresses). Given this information, the service endpoint **1020** sends the connection request (and, similarly, data traffic) to the DNS proxy **1030**.

[0170] Given its stored information **846** about the customer endpoint **1040**, the DNS proxy **1030** determines the actual overlay IP address of the customer endpoint **1040** and these the connection request (and, similarly, data traffic) to the endpoint **1040**. On the reverse path, data traffic from the endpoint **854** to the service **862** flows through the DNS proxy **854** and the endpoint **832**, where the opposite re-addressing of the data traffic is performed. Here, the addressing (e.g., change from “classE-IP3” to “classE-IP2” and from “classE-IP2” to “1.1.1.1”) and reverse addressing (e.g., change from “1.1.1.1” to “classE-IP2” and from “classE-IP2” to “classE-IP3”) can be implemented as network address translation (NAT) operations and de-NAT operations, respectively.

[0171] FIG. 11 depicts a flow diagram **1100** implemented on an SDN for DNS resolution, according to some embodiments. As described below, operations of the flow can be performed by a service endpoint of the SDN, where this service endpoint is configured for a non-SDN service (e.g., the endpoint **832** configured for the service **860** of FIG. 8). The non-SDN service belongs to a substrate network, although other types of networks are possible. Some or all of the instructions for performing the operations can be implemented as computer-readable instructions on one or more computer-readable storage media of a system that provides the SDN. While the operations are illustrated in a particular order, it should be understood that no particular order is necessary and that one or more operations may be omitted, skipped, performed in parallel, and/or reordered.

[0172] In an example, the flow diagram **1100** includes operation **1102**, where the service endpoint associates a first reserved IP address with an IP address of a DNS proxy. The DNS proxy is hosted by the SDN for a first private network of a customer. The SDN includes at least a portion of the first

private network (e.g., this first private network can extend to include an on-premises network that is remote from the SDN).

[0173] In an example, the flow diagram **1100** includes operation **1104**, where the service endpoint receives, based on the first reserved IP address and from the substrate network, a DNS resolution request for a DNS associated with the first private network. The DNS resolution request is associated with the non-SDN service (e.g., can be sent by a DNS resolver thereof). The DNS request can include an FQDN corresponding to a customer endpoint of the first private network.

[0174] In an example, the flow diagram **1100** includes operation **1106**, where the service endpoint sends, based on the IP address of the DNS proxy, the DNS resolution request to the DNS proxy. For example, based on association between the first reserved IP address and the DNS proxy’s IP address, the service endpoint determines the DNS proxy’s IP address and sends the DNS resolution request thereto.

[0175] In an example, the flow diagram **1100** includes operation **1108**, where the service endpoint receives, from the DNS proxy, a DNS resolution indicating a second reserved IP address as being associated with the DNS. The second reserved IP address can belong to the DNS proxy. In particular, the DNS proxy can call a DNS resolver of the first private network of the customer to resolve the DNS resolution request (e.g., to look up the FQDN and determine the actual IP address of the customer endpoint). The DNS proxy can then mask the actual IP address by using instead the second reserved IP address and can store an association between these two IP addresses. The DNS resolution returned to the service endpoint can include the second reserved IP address instead of the actual IP address of the customer endpoint.

[0176] In an example, the flow diagram **1100** includes operation **1110**, where the service endpoint sends, to the substrate network, the DNS resolution. For instance, the DNS resolution is returned to the DNS resolver of the non-SDN service.

[0177] In an example, the flow diagram **1100** includes operation **1112**, where the service endpoint associates the second reserved IP address with a third reserved IP address. The third reserved IP address can belong to the substrate network (e.g., reserved for the non-SDN service). The association can be received as part of information from the non-SDN service.

[0178] In an example, the flow diagram **1100** includes operation **1114**, where the service endpoint receives, from the substrate network based on the third reserved IP address, a request for connecting to the customer endpoint of the first private network. This request can be received from the non-SDN service.

[0179] In an example, the flow diagram **1100** includes operation **1116**, where the service endpoint sends, based on the second reserved IP address, the request to the DNS proxy. Based on the association between the third and second IP addresses, the service endpoint can determine the second reserved IP address and accordingly forward the connection request to the DNS proxy having the second reserved IP address.

[0180] Although FIG. 11 does not illustrate the completion of the connection setup, the DNS proxy can send the connection request to the customer endpoint. A response back can be sent from the customer endpoint to the non-SDN

service via DNS proxy and the service endpoint to thereby establish the connection. Data traffic can be exchanged over the connection, whereby the service endpoint and the DNS proxy can be part of the network path between the non-SDN service and the customer endpoint.

[0181] FIG. 12 depicts a flow diagram 1200 implemented by a non-SDN service for DNS resolution, according to some embodiments. As described below, operations of the flow can be performed by the non-SDN service of a network other than an SDN that includes a service endpoint configured for the non-SDN service (e.g., performed by the service 860 of FIG. 8). The other network is described in FIG. 12 as a substrate network, although other types of networks are possible. Some or all of the instructions for performing the operations can be implemented as computer-readable instructions on one or more computer-readable storage media of a system that belongs or forms at least a portion of the substrate network. While the operations are illustrated in a particular order, it should be understood that no particular order is necessary and that one or more operations may be omitted, skipped, performed in parallel, and/or reordered.

[0182] In an example, the flow diagram 1200 includes operation 1202, where the non-SDN service sends, to a service endpoint hosted by the SDN for the non-SDN service, first data indicating an association between a first reserved IP address and an IP address of a DNS proxy. The DNS proxy is hosted by the SDN for a first private network of a customer. The SDN including at least a portion of the first private network.

[0183] In an example, the flow diagram 1200 includes operation 1204, where the non-SDN service sends, to the service endpoint based on the first reserved IP address, a DNS resolution request for a DNS associated with the first private network. The DNS resolution request is associated with the non-SDN service and can include an FQDN of a customer endpoint of the first private network.

[0184] In an example, the flow diagram 1200 includes operation 1206, where the non-SDN service receives, from the service endpoint, a DNS resolution including a second reserved IP address. The second reserved IP address belongs to the DNS proxy.

[0185] In an example, the flow diagram 1200 includes operation 1208, where the non-SDN service associates a third reserved IP address with the second reserved IP address. The third reserved IP address belongs to the substrate network (e.g., by being reserved for the non-SDN service).

[0186] In an example, the flow diagram 1200 includes operation 1210, where the non-SDN service sends, to the service endpoint based on the third reserved IP address, a request for connecting to the customer endpoint of the first private network.

[0187] Although FIG. 12 does not illustrate the completion of the connection setup, the service endpoint can send the connection to the DNS proxy that, in turn, can send it to the customer endpoint. A response back can be sent from the customer endpoint to the non-SDN service via DNS proxy and the service endpoint to thereby establish the connection. Data traffic can be exchanged over the connection, whereby the service endpoint and the DNS proxy can be part of the network path between the non-SDN service and the customer endpoint.

Examples of Cloud Infrastructure

[0188] As noted above, infrastructure as a service (IaaS) is one particular type of cloud computing. IaaS can be configured to provide virtualized set of computing resources over a public network (e.g., the Internet). In an IaaS model, a cloud computing provider can host the infrastructure components (e.g., servers, storage devices, network nodes (e.g., hardware), deployment software, platform virtualization (e.g., a hypervisor layer), or the like). In some cases, an IaaS provider may also supply a variety of services to accompany those infrastructure components (e.g., billing, monitoring, logging, security, load balancing and clustering, etc.). Thus, as these services may be policy-driven, IaaS users may be able to implement policies to drive load balancing to maintain application availability and performance.

[0189] In some instances, IaaS customers may access resources and services through a wide area network (WAN), such as the Internet, and can use the cloud provider's services to install the remaining elements of an application stack. For example, the user can log in to the IaaS platform to create virtual machines (VMs), install operating systems (OSs) on each VM, deploy middleware such as databases, create storage buckets for workloads and backups, and even install enterprise software into that VM. Customers can then use the provider's services to perform various functions, including balancing network traffic, troubleshooting application issues, monitoring performance, managing disaster recovery, etc.

[0190] In most cases, a cloud computing model will require the participation of a cloud provider. The cloud provider may, but need not be, a third-party service that specializes in providing (e.g., offering, renting, selling) IaaS. An entity might also opt to deploy a private cloud, becoming its own provider of infrastructure services.

[0191] In some examples, IaaS deployment is the process of putting a new application, or a new version of an application, onto a prepared application server or the like. It may also include the process of preparing the server (e.g., installing libraries, daemons, etc.). This is often managed by the cloud provider, below the hypervisor layer (e.g., the servers, storage, network hardware, and virtualization). Thus, the customer may be responsible for handling (OS), middleware, and/or application deployment (e.g., on self-service virtual machines (e.g., that can be spun up on demand) or the like).

[0192] In some examples, IaaS provisioning may refer to acquiring computers or virtual hosts for use, and even installing needed libraries or services on them. In most cases, deployment does not include provisioning, and the provisioning may need to be performed first.

[0193] In some cases, there are two different problems for IaaS provisioning. First, there is the initial challenge of provisioning the initial set of infrastructure before anything is running. Second, there is the challenge of evolving the existing infrastructure (e.g., adding new services, changing services, removing services, etc.) once everything has been provisioned. In some cases, these two challenges may be addressed by enabling the configuration of the infrastructure to be defined declaratively. In other words, the infrastructure (e.g., what components are needed and how they interact) can be defined by one or more configuration files. Thus, the overall topology of the infrastructure (e.g., what resources depend on which, and how they each work together) can be

described declaratively. In some instances, once the topology is defined, a workflow can be generated that creates and/or manages the different components described in the configuration files.

[0194] In some examples, an infrastructure may have many interconnected elements. For example, there may be one or more virtual private clouds (VPCs) (e.g., a potentially on-demand pool of configurable and/or shared set of computing resources), also known as a core network. In some examples, there may also be one or more security group rules provisioned to define how the security of the network will be set up and one or more virtual machines (VMs). Other infrastructure elements may also be provisioned, such as a load balancer, a database, or the like. As more and more infrastructure elements are desired and/or added, the infrastructure may incrementally evolve.

[0195] In some instances, continuous deployment techniques may be employed to enable deployment of infrastructure code across various virtual computing environments. Additionally, the described techniques can enable infrastructure management within these environments. In some examples, service teams can write code that is desired to be deployed to one or more, but often many, different production environments (e.g., across various different geographic locations, sometimes spanning the entire world). However, in some examples, the infrastructure on which the code will be deployed must first be set up. In some instances, the provisioning can be done manually, a provisioning tool may be utilized to provision the resources, and/or deployment tools may be utilized to deploy the code once the infrastructure is provisioned.

[0196] FIG. 13 is a block diagram 1300 illustrating an example pattern of an IaaS architecture, according to at least one embodiment. Service operators 1302 can be communicatively coupled to a secure host tenancy 1304 that can include a virtual cloud network (VCN) 1306 and a secure host subnet 1308. In some examples, the service operators 1302 may be using one or more client computing devices, which may be portable handheld devices (e.g., an iPhone®, cellular telephone, an iPad®, computing tablet, a personal digital assistant (PDA)) or wearable devices (e.g., a Google Glass® head mounted display), running software such as Microsoft Windows Mobile®, and/or a variety of mobile operating systems such as iOS, Windows Phone, Android, BlackBerry 13, Palm OS, and the like, and being Internet, e-mail, short message service (SMS), BlackBerry®, or other communication protocol enabled. Alternatively, the client computing devices can be general purpose personal computers including, by way of example, personal computers and/or laptop computers running various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems. The client computing devices can be workstation computers running any of a variety of commercially available UNIX® or UNIX-like operating systems, including without limitation the variety of GNU/Linux operating systems, such as for example, Google Chrome OS. Alternatively, or in addition, client computing devices may be any other electronic device, such as a thin-client computer, an Internet-enabled gaming system (e.g., a Microsoft Xbox gaming console with or without a Kinect® gesture input device), and/or a personal messaging device, capable of communicating over a network that can access the VCN 1306 and/or the Internet.

[0197] The VCN 1306 can include a local peering gateway (LPG) 1310 that can be communicatively coupled to a secure shell (SSH) VCN 1312 via an LPG 1310 contained in the SSH VCN 1312. The SSH VCN 1312 can include an SSH subnet 1314, and the SSH VCN 1312 can be communicatively coupled to a control plane VCN 1316 via the LPG 1310 contained in the control plane VCN 1316. Also, the SSH VCN 1312 can be communicatively coupled to a data plane VCN 1318 via an LPG 1310. The control plane VCN 1316 and the data plane VCN 1318 can be contained in a service tenancy 1319 that can be owned and/or operated by the IaaS provider.

[0198] The control plane VCN 1316 can include a control plane demilitarized zone (DMZ) tier 1320 that acts as a perimeter network (e.g., portions of a corporate network between the corporate intranet and external networks). The DMZ-based servers may have restricted responsibilities and help keep security breaches contained. Additionally, the DMZ tier 1320 can include one or more load balancer (LB) subnet(s) 1322, a control plane app tier 1324 that can include app subnet(s) 1326, a control plane data tier 1328 that can include database (DB) subnet(s) 1330 (e.g., frontend DB subnet(s) and/or backend DB subnet(s)). The LB subnet(s) 1322 contained in the control plane DMZ tier 1320 can be communicatively coupled to the app subnet(s) 1326 contained in the control plane app tier 1324 and an Internet gateway 1334 that can be contained in the control plane VCN 1316, and the app subnet(s) 1326 can be communicatively coupled to the DB subnet(s) 1330 contained in the control plane data tier 1328 and a service gateway 1336 and a network address translation (NAT) gateway 1338. The control plane VCN 1316 can include the service gateway 1336 and the NAT gateway 1338.

[0199] The control plane VCN 1316 can include a data plane mirror app tier 1340 that can include app subnet(s) 1326. The app subnet(s) 1326 contained in the data plane mirror app tier 1340 can include a virtual network interface controller (VNIC) 1342 that can execute a compute instance 1344. The compute instance 1344 can communicatively couple the app subnet(s) 1326 of the data plane mirror app tier 1340 to app subnet(s) 1326 that can be contained in a data plane app tier 1346.

[0200] The data plane VCN 1318 can include the data plane app tier 1346, a data plane DMZ tier 1348, and a data plane data tier 1350. The data plane DMZ tier 1348 can include LB subnet(s) 1322 that can be communicatively coupled to the app subnet(s) 1326 of the data plane app tier 1346 and the Internet gateway 1334 of the data plane VCN 1318. The app subnet(s) 1326 can be communicatively coupled to the service gateway 1336 of the data plane VCN 1318 and the NAT gateway 1338 of the data plane VCN 1318. The data plane data tier 1350 can also include the DB subnet(s) 1330 that can be communicatively coupled to the app subnet(s) 1326 of the data plane app tier 1346.

[0201] The Internet gateway 1334 of the control plane VCN 1316 and of the data plane VCN 1318 can be communicatively coupled to a metadata management service 1352 that can be communicatively coupled to public Internet 1354. Public Internet 1354 can be communicatively coupled to the NAT gateway 1338 of the control plane VCN 1316 and of the data plane VCN 1318. The service gateway 1336 of the control plane VCN 1316 and of the data plane VCN 1318 can be communicatively couple to cloud services 1356.

[0202] In some examples, the service gateway 1336 of the control plane VCN 1316 or of the data plane VCN 1318 can make application programming interface (API) calls to cloud services 1356 without going through public Internet 1354. The API calls to cloud services 1356 from the service gateway 1336 can be one-way: the service gateway 1336 can make API calls to cloud services 1356, and cloud services 1356 can send requested data to the service gateway 1336. But, cloud services 1356 may not initiate API calls to the service gateway 1336.

[0203] In some examples, the secure host tenancy 1304 can be directly connected to the service tenancy 1319, which may be otherwise isolated. The secure host subnet 1308 can communicate with the SSH subnet 1314 through an LPG 1310 that may enable two-way communication over an otherwise isolated system. Connecting the secure host subnet 1308 to the SSH subnet 1314 may give the secure host subnet 1308 access to other entities within the service tenancy 1319.

[0204] The control plane VCN 1316 may allow users of the service tenancy 1319 to set up or otherwise provision desired resources. Desired resources provisioned in the control plane VCN 1316 may be deployed or otherwise used in the data plane VCN 1318. In some examples, the control plane VCN 1316 can be isolated from the data plane VCN 1318, and the data plane mirror app tier 1340 of the control plane VCN 1316 can communicate with the data plane app tier 1346 of the data plane VCN 1318 via VNICs 1342 that can be contained in the data plane mirror app tier 1340 and the data plane app tier 1346.

[0205] In some examples, users of the system, or customers, can make requests, for example create, read, update, or delete (CRUD) operations, through public Internet 1354 that can communicate the requests to the metadata management service 1352. The metadata management service 1352 can communicate the request to the control plane VCN 1316 through the Internet gateway 1334. The request can be received by the LB subnet(s) 1322 contained in the control plane DMZ tier 1320. The LB subnet(s) 1322 may determine that the request is valid, and in response to this determination, the LB subnet(s) 1322 can transmit the request to app subnet(s) 1326 contained in the control plane app tier 1324. If the request is validated and requires a call to public Internet 1354, the call to public Internet 1354 may be transmitted to the NAT gateway 1338 that can make the call to public Internet 1354. Memory that may be desired to be stored by the request can be stored in the DB subnet(s) 1330.

[0206] In some examples, the data plane mirror app tier 1340 can facilitate direct communication between the control plane VCN 1316 and the data plane VCN 1318. For example, changes, updates, or other suitable modifications to configuration may be desired to be applied to the resources contained in the data plane VCN 1318. Via a VNIC 1342, the control plane VCN 1316 can directly communicate with, and can thereby execute the changes, updates, or other suitable modifications to configuration to, resources contained in the data plane VCN 1318.

[0207] In some embodiments, the control plane VCN 1316 and the data plane VCN 1318 can be contained in the service tenancy 1319. In this case, the user, or the customer, of the system may not own or operate either the control plane VCN 1316 or the data plane VCN 1318. Instead, the IaaS provider may own or operate the control plane VCN 1316 and the data plane VCN 1318, both of which may be contained in the

service tenancy 1319. This embodiment can enable isolation of networks that may prevent users or customers from interacting with other users', or other customers', resources. Also, this embodiment may allow users or customers of the system to store databases privately without needing to rely on public Internet 1354, which may not have a desired level of security, for storage.

[0208] In other embodiments, the LB subnet(s) 1322 contained in the control plane VCN 1316 can be configured to receive a signal from the service gateway 1336. In this embodiment, the control plane VCN 1316 and the data plane VCN 1318 may be configured to be called by a customer of the IaaS provider without calling public Internet 1354. Customers of the IaaS provider may desire this embodiment since database(s) that the customers use may be controlled by the IaaS provider and may be stored on the service tenancy 1319, which may be isolated from public Internet 1354.

[0209] FIG. 14 is a block diagram 1400 illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators 1402 (e.g., service operators 1302 of FIG. 13) can be communicatively coupled to a secure host tenancy 1404 (e.g., the secure host tenancy 1304 of FIG. 13) that can include a virtual cloud network (VCN) 1406 (e.g., the VCN 1306 of FIG. 13) and a secure host subnet 1408 (e.g., the secure host subnet 1308 of FIG. 13). The VCN 1406 can include a local peering gateway (LPG) 1410 (e.g., the LPG 1310 of FIG. 13) that can be communicatively coupled to a secure shell (SSH) VCN 1412 (e.g., the SSH VCN 1312 of FIG. 13) via an LPG 1310 contained in the SSH VCN 1412. The SSH VCN 1412 can include an SSH subnet 1414 (e.g., the SSH subnet 1314 of FIG. 13), and the SSH VCN 1412 can be communicatively coupled to a control plane VCN 1416 (e.g., the control plane VCN 1316 of FIG. 13) via an LPG 1410 contained in the control plane VCN 1416. The control plane VCN 1416 can be contained in a service tenancy 1419 (e.g., the service tenancy 1319 of FIG. 13), and the data plane VCN 1418 (e.g., the data plane VCN 1318 of FIG. 13) can be contained in a customer tenancy 1421 that may be owned or operated by users, or customers, of the system.

[0210] The control plane VCN 1416 can include a control plane DMZ tier 1420 (e.g. the control plane DMZ tier 1320 of FIG. 13) that can include LB subnet(s) 1422 (e.g. LB subnet(s) 1322 of FIG. 13), a control plane app tier 1424 (e.g. the control plane app tier 1324 of FIG. 13) that can include app subnet(s) 1426 (e.g. app subnet(s) 1326 of FIG. 13), a control plane data tier 1428 (e.g. the control plane data tier 1328 of FIG. 13) that can include database (DB) subnet(s) 1430 (e.g. similar to DB subnet(s) 1330 of FIG. 13). The LB subnet(s) 1422 contained in the control plane DMZ tier 1420 can be communicatively coupled to the app subnet(s) 1426 contained in the control plane app tier 1424 and an Internet gateway 1434 (e.g. the Internet gateway 1334 of FIG. 13) that can be contained in the control plane VCN 1416, and the app subnet(s) 1426 can be communicatively coupled to the DB subnet(s) 1430 contained in the control plane data tier 1428 and a service gateway 1436 (e.g. the service gateway of FIG. 13) and a network address translation (NAT) gateway 1438 (e.g. the NAT gateway 1338 of FIG. 13). The control plane VCN 1416 can include the service gateway 1436 and the NAT gateway 1438.

[0211] The control plane VCN 1416 can include a data plane mirror app tier 1440 (e.g., the data plane mirror app

tier 1340 of FIG. 13) that can include app subnet(s) 1426. The app subnet(s) 1426 contained in the data plane mirror app tier 1440 can include a virtual network interface controller (VNIC) 1442 (e.g., the VNIC of 1342) that can execute a compute instance 1444 (e.g., similar to the compute instance 1344 of FIG. 13). The compute instance 1444 can facilitate communication between the app subnet(s) 1426 of the data plane mirror app tier 1440 and the app subnet(s) 1426 that can be contained in a data plane app tier 1446 (e.g., the data plane app tier 1346 of FIG. 13) via the VNIC 1442 contained in the data plane mirror app tier 1440 and the VNIC 1442 contained in the data plane app tier 1446.

[0212] The Internet gateway 1434 contained in the control plane VCN 1416 can be communicatively coupled to a metadata management service 1452 (e.g., the metadata management service 1352 of FIG. 13) that can be communicatively coupled to public Internet 1454 (e.g., public Internet 1354 of FIG. 13). Public Internet 1454 can be communicatively coupled to the NAT gateway 1438 contained in the control plane VCN 1416. The service gateway 1436 contained in the control plane VCN 1416 can be communicatively coupled to cloud services 1456 (e.g., cloud services 1356 of FIG. 13).

[0213] In some examples, the data plane VCN 1418 can be contained in the customer tenancy 1421. In this case, the IaaS provider may provide the control plane VCN 1416 for each customer, and the IaaS provider may, for each customer, set up a unique compute instance 1444 that is contained in the service tenancy 1419. Each compute instance 1444 may allow communication between the control plane VCN 1416, contained in the service tenancy 1419, and the data plane VCN 1418 that is contained in the customer tenancy 1421. The compute instance 1444 may allow resources that are provisioned in the control plane VCN 1416 that is contained in the service tenancy 1419, to be deployed or otherwise used in the data plane VCN 1418 that is contained in the customer tenancy 1421.

[0214] In other examples, the customer of the IaaS provider may have databases that live in the customer tenancy 1421. In this example, the control plane VCN 1416 can include the data plane mirror app tier 1440 that can include app subnet(s) 1426. The data plane mirror app tier 1440 can reside in the data plane VCN 1418, but the data plane mirror app tier 1440 may not live in the data plane VCN 1418. That is, the data plane mirror app tier 1440 may have access to the customer tenancy 1421, but the data plane mirror app tier 1440 may not exist in the data plane VCN 1418 or be owned or operated by the customer of the IaaS provider. The data plane mirror app tier 1440 may be configured to make calls to the data plane VCN 1418 but may not be configured to make calls to any entity contained in the control plane VCN 1416. The customer may desire to deploy or otherwise use resources in the data plane VCN 1418 that are provisioned in the control plane VCN 1416, and the data plane mirror app tier 1440 can facilitate the desired deployment, or other usage of resources, of the customer.

[0215] In some embodiments, the customer of the IaaS provider can apply filters to the data plane VCN 1418. In this embodiment, the customer can determine what the data plane VCN 1418 can access, and the customer may restrict access to public Internet 1454 from the data plane VCN 1418. The IaaS provider may not be able to apply filters or otherwise control access of the data plane VCN 1418 to any outside networks or databases. Applying filters and controls

by the customer onto the data plane VCN 1418, contained in the customer tenancy 1421, can help isolate the data plane VCN 1418 from other customers and from public Internet 1454.

[0216] In some embodiments, cloud services 1456 can be called by the service gateway 1436 to access services that may not exist on public Internet 1454, on the control plane VCN 1416, or on the data plane VCN 1418. The connection between cloud services 1456 and the control plane VCN 1416 or the data plane VCN 1418 may not be live or continuous. Cloud services 1456 may exist on a different network owned or operated by the IaaS provider. Cloud services 1456 may be configured to receive calls from the service gateway 1436 and may be configured to not receive calls from public Internet 1454. Some cloud services 1456 may be isolated from other cloud services 1456, and the control plane VCN 1416 may be isolated from cloud services 1456 that may not be in the same region as the control plane VCN 1416. For example, the control plane VCN 1416 may be located in "Region 1," and cloud service "Deployment 17," may be located in Region 1 and in "Region 2." If a call to Deployment 17 is made by the service gateway 1436 contained in the control plane VCN 1416 located in Region 1, the call may be transmitted to Deployment 17 in Region 1. In this example, the control plane VCN 1416, or Deployment 17 in Region 1, may not be communicatively coupled to, or otherwise in communication with, Deployment 17 in Region 2.

[0217] FIG. 15 is a block diagram 1500 illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators 1502 (e.g., service operators 1302 of FIG. 13) can be communicatively coupled to a secure host tenancy 1504 (e.g., the secure host tenancy 1304 of FIG. 13) that can include a virtual cloud network (VCN) 1506 (e.g., the VCN 1306 of FIG. 13) and a secure host subnet 1508 (e.g., the secure host subnet 1308 of FIG. 13). The VCN 1506 can include an LPG 1510 (e.g., the LPG 1310 of FIG. 13) that can be communicatively coupled to an SSH VCN 1512 (e.g., the SSH VCN 1312 of FIG. 13) via an LPG 1510 contained in the SSH VCN 1512. The SSH VCN 1512 can include an SSH subnet 1514 (e.g., the SSH subnet 1314 of FIG. 13), and the SSH VCN 1312 can be communicatively coupled to a control plane VCN 1516 (e.g., the control plane VCN 1316 of FIG. 13) via an LPG 1510 contained in the control plane VCN 1516 and to a data plane VCN 1518 (e.g., the data plane 1318 of FIG. 13) via an LPG 1510 contained in the data plane VCN 1518. The control plane VCN 1516 and the data plane VCN 1518 can be contained in a service tenancy 1519 (e.g., the service tenancy 1319 of FIG. 13).

[0218] The control plane VCN 1316 can include a control plane DMZ tier 1320 (e.g., the control plane DMZ tier 1320 of FIG. 13) that can include load balancer (LB) subnet(s) 1322 (e.g., LB subnet(s) 1322 of FIG. 13), a control plane app tier 1524 (e.g., the control plane app tier 1324 of FIG. 13) that can include app subnet(s) 1526 (e.g., similar to app subnet(s) 1326 of FIG. 13), a control plane data tier 1528 (e.g., the control plane data tier 1328 of FIG. 13) that can include DB subnet(s) 1530. The LB subnet(s) 1522 contained in the control plane DMZ tier 1520 can be communicatively coupled to the app subnet(s) 1526 contained in the control plane app tier 1524 and to an Internet gateway 1334 (e.g., the Internet gateway 1334 of FIG. 13) that can be contained in the control plane VCN 1516, and the app

subnet(s) **1526** can be communicatively coupled to the DB subnet(s) **1330** contained in the control plane data tier **1328** and to a service gateway **1336** (e.g. the service gateway of FIG. 13) and a network address translation (NAT) gateway **1338** (e.g. the NAT gateway **1338** of FIG. 13). The control plane VCN **1516** can include the service gateway **1536** and the NAT gateway **1538**.

[0219] The data plane VCN **1518** can include a data plane app tier **1546** (e.g., the data plane app tier **1346** of FIG. 13), a data plane DMZ tier **1548** (e.g., the data plane DMZ tier **1348** of FIG. 13), and a data plane data tier **1550** (e.g., the data plane data tier **1350** of FIG. 13). The data plane DMZ tier **1548** can include LB subnet(s) **1522** that can be communicatively coupled to trusted app subnet(s) **1560** and untrusted app subnet(s) **1562** of the data plane app tier **1546** and the Internet gateway **1534** contained in the data plane VCN **1518**. The trusted app subnet(s) **1560** can be communicatively coupled to the service gateway **1536** contained in the data plane VCN **1518**, the NAT gateway **1538** contained in the data plane VCN **1518**, and DB subnet(s) **1530** contained in the data plane data tier **1550**. The untrusted app subnet(s) **1562** can be communicatively coupled to the service gateway **1536** contained in the data plane VCN **1518** and DB subnet(s) **1530** contained in the data plane data tier **1550**. The data plane data tier **1550** can include DB subnet(s) **1530** that can be communicatively coupled to the service gateway **1536** contained in the data plane VCN **1518**.

[0220] The untrusted app subnet(s) **1562** can include one or more primary VNICS **1564(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **1566(1)-(N)**. Each tenant VM **1566(1)-(N)** can be communicatively coupled to a respective app subnet **1567(1)-(N)** that can be contained in respective container egress VCNs **1568(1)-(N)** that can be contained in respective customer tenancies **1570(1)-(N)**. Respective secondary VNICS **1572(1)-(N)** can facilitate communication between the untrusted app subnet(s) **1562** contained in the data plane VCN **1518** and the app subnet contained in the container egress VCNs **1568(1)-(N)**. Each container egress VCNs **1568(1)-(N)** can include a NAT gateway **1538** that can be communicatively coupled to public Internet **1554** (e.g., public Internet **1354** of FIG. 13).

[0221] The Internet gateway **1534** contained in the control plane VCN **1516** and contained in the data plane VCN **1518** can be communicatively coupled to a metadata management service **1552** (e.g., the metadata management system **1352** of FIG. 13) that can be communicatively coupled to public Internet **1554**. Public Internet **1554** can be communicatively coupled to the NAT gateway **1538** contained in the control plane VCN **1516** and contained in the data plane VCN **1518**. The service gateway **1536** contained in the control plane VCN **1516** and contained in the data plane VCN **1518** can be communicatively couple to cloud services **1556**.

[0222] In some embodiments, the data plane VCN **1518** can be integrated with customer tenancies **1570**. This integration can be useful or desirable for customers of the IaaS provider in some cases such as a case that may desire support when executing code. The customer may provide code to run that may be destructive, may communicate with other customer resources, or may otherwise cause undesirable effects. In response to this, the IaaS provider may determine whether to run code given to the IaaS provider by the customer.

[0223] In some examples, the customer of the IaaS provider may grant temporary network access to the IaaS

provider and request a function to be attached to the data plane tier app **1546**. Code to run the function may be executed in the VMs **1566(1)-(N)**, and the code may not be configured to run anywhere else on the data plane VCN **1518**. Each VM **1566(1)-(N)** may be connected to one customer tenancy **1570**. Respective containers **1571(1)-(N)** contained in the VMs **1566(1)-(N)** may be configured to run the code. In this case, there can be a dual isolation (e.g., the containers **1571(1)-(N)** running code, where the containers **1571(1)-(N)** may be contained in at least the VM **1566(1)-(N)** that are contained in the untrusted app subnet(s) **1562**), which may help prevent incorrect or otherwise undesirable code from damaging the network of the IaaS provider or from damaging a network of a different customer. The containers **1571(1)-(N)** may be communicatively coupled to the customer tenancy **1570** and may be configured to transmit or receive data from the customer tenancy **1570**. The containers **1571(1)-(N)** may not be configured to transmit or receive data from any other entity in the data plane VCN **1518**. Upon completion of running the code, the IaaS provider may kill or otherwise dispose of the containers **1571(1)-(N)**.

[0224] In some embodiments, the trusted app subnet(s) **1560** may run code that may be owned or operated by the IaaS provider. In this embodiment, the trusted app subnet(s) **1560** may be communicatively coupled to the DB subnet(s) **1530** and be configured to execute CRUD operations in the DB subnet(s) **1530**. The untrusted app subnet(s) **1562** may be communicatively coupled to the DB subnet(s) **1530**, but in this embodiment, the untrusted app subnet(s) may be configured to execute read operations in the DB subnet(s) **1530**. The containers **1571(1)-(N)** that can be contained in the VM **1566(1)-(N)** of each customer and that may run code from the customer may not be communicatively coupled with the DB subnet(s) **1530**.

[0225] In other embodiments, the control plane VCN **1516** and the data plane VCN **1518** may not be directly communicatively coupled. In this embodiment, there may be no direct communication between the control plane VCN **1516** and the data plane VCN **1518**. However, communication can occur indirectly through at least one method. An LPG **1510** may be established by the IaaS provider that can facilitate communication between the control plane VCN **1516** and the data plane VCN **1518**. In another example, the control plane VCN **1516** or the data plane VCN **1518** can make a call to cloud services **1556** via the service gateway **1536**. For example, a call to cloud services **1556** from the control plane VCN **1516** can include a request for a service that can communicate with the data plane VCN **1518**.

[0226] FIG. 16 is a block diagram **1600** illustrating another example pattern of an IaaS architecture, according to at least one embodiment. Service operators **1602** (e.g., service operators **1302** of FIG. 13) can be communicatively coupled to a secure host tenancy **1604** (e.g., the secure host tenancy **1304** of FIG. 13) that can include a virtual cloud network (VCN) **1606** (e.g., the VCN **1306** of FIG. 13) and a secure host subnet **1608** (e.g., the secure host subnet **1308** of FIG. 13). The VCN **1606** can include an LPG **1610** (e.g., the LPG **1310** of FIG. 13) that can be communicatively coupled to an SSH VCN **1612** (e.g., the SSH VCN **1312** of FIG. 13) via an LPG **1610** contained in the SSH VCN **1612**. The SSH VCN **1612** can include an SSH subnet **1614** (e.g., the SSH subnet **1314** of FIG. 13), and the SSH VCN **1612** can be communicatively coupled to a control plane VCN

1616 (e.g., the control plane VCN **1316** of FIG. **13**) via an LPG **1610** contained in the control plane VCN **1616** and to a data plane VCN **1618** (e.g., the data plane **1318** of FIG. **13**) via an LPG **1610** contained in the data plane VCN **1618**. The control plane VCN **1616** and the data plane VCN **1618** can be contained in a service tenancy **1619** (e.g., the service tenancy **1319** of FIG. **13**).

[0227] The control plane VCN **1616** can include a control plane DMZ tier **1620** (e.g., the control plane DMZ tier **1320** of FIG. **13**) that can include LB subnet(s) **1622** (e.g., LB subnet(s) **1322** of FIG. **13**), a control plane app tier **1624** (e.g., the control plane app tier **1324** of FIG. **13**) that can include app subnet(s) **1626** (e.g., app subnet(s) **1326** of FIG. **13**), a control plane data tier **1628** (e.g., the control plane data tier **1328** of FIG. **13**) that can include DB subnet(s) **1630** (e.g., DB subnet(s) **1530** of FIG. **15**). The LB subnet(s) **1622** contained in the control plane DMZ tier **1620** can be communicatively coupled to the app subnet(s) **1626** contained in the control plane app tier **1624** and to an Internet gateway **1634** (e.g., the Internet gateway **1334** of FIG. **13**) that can be contained in the control plane VCN **1616**, and the app subnet(s) **1626** can be communicatively coupled to the DB subnet(s) **1630** contained in the control plane data tier **1628** and to a service gateway **1636** (e.g., the service gateway of FIG. **13**) and a network address translation (NAT) gateway **1638** (e.g., the NAT gateway **1338** of FIG. **13**). The control plane VCN **1616** can include the service gateway **1636** and the NAT gateway **1638**.

[0228] The data plane VCN **1618** can include a data plane app tier **1646** (e.g., the data plane app tier **1346** of FIG. **13**), a data plane DMZ tier **1648** (e.g., the data plane DMZ tier **1348** of FIG. **13**), and a data plane data tier **1650** (e.g., the data plane data tier **1350** of FIG. **13**). The data plane DMZ tier **1648** can include LB subnet(s) **1622** that can be communicatively coupled to trusted app subnet(s) **1660** (e.g., trusted app subnet(s) **1560** of FIG. **15**) and untrusted app subnet(s) **1662** (e.g., untrusted app subnet(s) **1562** of FIG. **15**) of the data plane app tier **1646** and the Internet gateway **1634** contained in the data plane VCN **1618**. The trusted app subnet(s) **1660** can be communicatively coupled to the service gateway **1636** contained in the data plane VCN **1618**, the NAT gateway **1638** contained in the data plane VCN **1618**, and DB subnet(s) **1630** contained in the data plane data tier **1650**. The untrusted app subnet(s) **1662** can be communicatively coupled to the service gateway **1636** contained in the data plane VCN **1618** and DB subnet(s) **1630** contained in the data plane data tier **1650**. The data plane data tier **1650** can include DB subnet(s) **1630** that can be communicatively coupled to the service gateway **1636** contained in the data plane VCN **1618**.

[0229] The untrusted app subnet(s) **1662** can include primary VNICS **1664(1)-(N)** that can be communicatively coupled to tenant virtual machines (VMs) **1666(1)-(N)** residing within the untrusted app subnet(s) **1662**. Each tenant VM **1666(1)-(N)** can run code in a respective container **1667(1)-(N)** and be communicatively coupled to an app subnet **1626** that can be contained in a data plane app tier **1646** that can be contained in a container egress VCN **1668**. Respective secondary VNICS **1672(1)-(N)** can facilitate communication between the untrusted app subnet(s) **1662** contained in the data plane VCN **1618** and the app subnet contained in the container egress VCN **1668**. The container egress VCN can

include a NAT gateway **1638** that can be communicatively coupled to public Internet **1654** (e.g., public Internet **1354** of FIG. **13**).

[0230] The Internet gateway **1634** contained in the control plane VCN **1616** and contained in the data plane VCN **1618** can be communicatively coupled to a metadata management service **1652** (e.g., the metadata management system **1352** of FIG. **13**) that can be communicatively coupled to public Internet **1654**. Public Internet **1654** can be communicatively coupled to the NAT gateway **1638** contained in the control plane VCN **1616** and contained in the data plane VCN **1618**. The service gateway **1636** contained in the control plane VCN **1616** and contained in the data plane VCN **1618** can be communicatively couple to cloud services **1656**.

[0231] In some examples, the pattern illustrated by the architecture of block diagram **1600** of FIG. **16** may be considered an exception to the pattern illustrated by the architecture of block diagram **1500** of FIG. **15** and may be desirable for a customer of the IaaS provider if the IaaS provider cannot directly communicate with the customer (e.g., a disconnected region). The respective containers **1667(1)-(N)** that are contained in the VMs **1666(1)-(N)** for each customer can be accessed in real-time by the customer. The containers **1667(1)-(N)** may be configured to make calls to respective secondary VNICS **1672(1)-(N)** contained in app subnet(s) **1626** of the data plane app tier **1646** that can be contained in the container egress VCN **1668**. The secondary VNICS **1672(1)-(N)** can transmit the calls to the NAT gateway **1638** that may transmit the calls to public Internet **1654**. In this example, the containers **1667(1)-(N)** that can be accessed in real-time by the customer can be isolated from the control plane VCN **1616** and can be isolated from other entities contained in the data plane VCN **1618**. The containers **1667(1)-(N)** may also be isolated from resources from other customers.

[0232] In other examples, the customer can use the containers **1667(1)-(N)** to call cloud services **1656**. In this example, the customer may run code in the containers **1667(1)-(N)** that requests a service from cloud services **1656**. The containers **1667(1)-(N)** can transmit this request to the secondary VNICS **1672(1)-(N)** that can transmit the request to the NAT gateway that can transmit the request to public Internet **1654**. Public Internet **1654** can transmit the request to LB subnet(s) **1622** contained in the control plane VCN **1616** via the Internet gateway **1634**. In response to determining the request is valid, the LB subnet(s) can transmit the request to app subnet(s) **1626** that can transmit the request to cloud services **1656** via the service gateway **1636**. It should be appreciated that IaaS architectures **1300**, **1400**, **1500**, **1600** depicted in the figures may have other components than those depicted. Further, the embodiments shown in the figures are only some examples of a cloud infrastructure system that may incorporate an embodiment of the disclosure. In some other embodiments, the IaaS systems may have more or fewer components than shown in the figures, may combine two or more components, or may have a different configuration or arrangement of components.

[0233] In certain embodiments, the IaaS systems described herein may include a suite of applications, middleware, and database service offerings that are delivered to a customer in a self-service, subscription-based, elastically scalable, reliable, highly available, and secure manner. An

example of such an IaaS system is the Oracle Cloud Infrastructure (OCI) provided by the present assignee.

[0234] FIG. 17 illustrates an example computer system 1700, in which various embodiments of the present disclosure may be implemented. The system 1700 may be used to implement any of the computer systems described above. As shown in the figure, computer system 1700 includes a processing unit 1704 that communicates with a number of peripheral subsystems via a bus subsystem 1702. These peripheral subsystems may include a processing acceleration unit 1706, and I/O subsystem 1708, a storage subsystem 1718 and a communications subsystem 1724. Storage subsystem 1718 includes tangible computer-readable storage media 1722 and a system memory 1710.

[0235] Bus subsystem 1702 provides a mechanism for letting the various components and subsystems of computer system 1700 communicate with each other as intended. Although bus subsystem 1702 is shown schematically as a single bus, alternative embodiments of the bus subsystem may utilize multiple buses. Bus subsystem 1702 may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. For example, such architectures may include an Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnect (PCI) bus, which can be implemented as a Mezzanine bus manufactured to the IEEE P1386.1 standard.

[0236] Processing unit 1704, which can be implemented as one or more integrated circuits (e.g., a conventional microprocessor or microcontroller), controls the operation of computer system 1700. One or more processors may be included in processing unit 1704. These processors may include single core or multicore processors. In certain embodiments, processing unit 1704 may be implemented as one or more independent processing units 1732 and/or 1734 with single or multicore processors included in each processing unit. In other embodiments, processing unit 1704 may also be implemented as a quad-core processing unit formed by integrating two dual-core processors into a single chip.

[0237] In various embodiments, processing unit 1704 can execute a variety of programs in response to program code and can maintain multiple concurrently executing programs or processes. At any given time, some, or all of the program code to be executed can be resident in processor(s) 1704 and/or in storage subsystem 1718. Through suitable programming, processor(s) 1704 can provide various functionalities described above. Computer system 1700 may additionally include a processing acceleration unit 1706, which can include a digital signal processor (DSP), a special-purpose processor, and/or the like.

[0238] I/O subsystem 1708 may include user interface input devices and user interface output devices. User interface input devices may include a keyboard, pointing devices such as a mouse or trackball, a touchpad or touch screen incorporated into a display, a scroll wheel, a click wheel, a dial, a button, a switch, a keypad, audio input devices with voice command recognition systems, microphones, and other types of input devices. User interface input devices may include, for example, motion sensing and/or gesture recognition devices such as the Microsoft Kinect® motion sensor that enables users to control and interact with an input

device, such as the Microsoft Xbox® 360 game controller, through a natural user interface using gestures and spoken commands. User interface input devices may also include eye gesture recognition devices such as the Google Glass® blink detector that detects eye activity (e.g., ‘blinking’ while taking pictures and/or making a menu selection) from users and transforms the eye gestures as input into an input device (e.g., Google Glass®). Additionally, user interface input devices may include voice recognition sensing devices that enable users to interact with voice recognition systems (e.g., Siri® navigator), through voice commands.

[0239] User interface input devices may also include, without limitation, three dimensional (3D) mice, joysticks or pointing sticks, gamepads and graphic tablets, and audio/visual devices such as speakers, digital cameras, digital camcorders, portable media players, webcams, image scanners, fingerprint scanners, barcode reader 3D scanners, 3D printers, laser rangefinders, and eye gaze tracking devices. Additionally, user interface input devices may include, for example, medical imaging input devices such as computed tomography, magnetic resonance imaging, position emission tomography, medical ultrasonography devices. User interface input devices may also include, for example, audio input devices such as MIDI keyboards, digital musical instruments and the like.

[0240] User interface output devices may include a display subsystem, indicator lights, or non-visual displays such as audio output devices, etc. The display subsystem may be a cathode ray tube (CRT), a flat-panel device, such as that using a liquid crystal display (LCD) or plasma display, a projection device, a touch screen, and the like. In general, use of the term “output device” is intended to include all possible types of devices and mechanisms for outputting information from computer system 1700 to a user or other computer. For example, user interface output devices may include, without limitation, a variety of display devices that visually convey text, graphics, and audio/video information such as monitors, printers, speakers, headphones, automotive navigation systems, plotters, voice output devices, and modems.

[0241] Computer system 1700 may comprise a storage subsystem 1718 that comprises software elements, shown as being currently located within a system memory 1710. System memory 1710 may store program instructions that are loadable and executable on processing unit 1704, as well as data generated during the execution of these programs.

[0242] Depending on the configuration and type of computer system 1700, system memory 1710 may be volatile (such as random-access memory (RAM)) and/or non-volatile (such as read-only memory (ROM), flash memory, etc.) The RAM typically contains data and/or program modules that are immediately accessible to and/or presently being operated and executed by processing unit 1704. In some implementations, system memory 1710 may include multiple different types of memory, such as static random-access memory (SRAM) or dynamic random-access memory (DRAM). In some implementations, a basic input/output system (BIOS), containing the basic routines that help to transfer information between elements within computer system 1700, such as during start-up, may typically be stored in the ROM. By way of example, and not limitation, system memory 1710 also illustrates application programs 1712, which may include client applications, Web browsers, mid-tier applications, relational database management systems

(RDBMS), etc., program data **1714**, and an operating system **1716**. By way of example, operating system **1716** may include various versions of Microsoft Windows®, Apple Macintosh®, and/or Linux operating systems, a variety of commercially available UNIX® or UNIX-like operating systems (including without limitation the variety of GNU/Linux operating systems, the Google Chrome® OS, and the like) and/or mobile operating systems such as iOS, Windows® Phone, Android® OS, BlackBerry® 17 OS, and Palm® OS operating systems.

[0243] Storage subsystem **1718** may also provide a tangible computer-readable storage medium for storing the basic programming and data constructs that provide the functionality of some embodiments. Software (programs, code modules, instructions) that when executed by a processor provide the functionality described above may be stored in storage subsystem **1718**. These software modules or instructions may be executed by processing unit **1704**. Storage subsystem **1718** may also provide a repository for storing data used in accordance with the present disclosure.

[0244] Storage subsystem **1700** may also include a computer-readable storage media reader **1720** that can further be connected to computer-readable storage media **1722**. Together and optionally, in combination with system memory **1710**, computer-readable storage media **1722** may comprehensively represent remote, local, fixed, and/or removable storage devices plus storage media for temporarily and/or more permanently containing, storing, transmitting, and retrieving computer-readable information.

[0245] Computer-readable storage media **1722** containing code, or portions of code, can also include any appropriate media known or used in the art, including storage media and communication media, such as but not limited to, volatile and non-volatile, removable and non-removable media implemented in any method or technology for storage and/or transmission of information. This can include tangible computer-readable storage media such as RAM, ROM, electronically erasable programmable ROM (EEPROM), flash memory or other memory technology, CD-ROM, digital versatile disk (DVD), or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or other tangible computer readable media. This can also include nontangible computer-readable media, such as data signals, data transmissions, or any other medium which can be used to transmit the desired information, and which can be accessed by computing system **1700**.

[0246] By way of example, computer-readable storage media **1722** may include a hard disk drive that reads from or writes to non-removable, nonvolatile magnetic media, a magnetic disk drive that reads from or writes to a removable, nonvolatile magnetic disk, and an optical disk drive that reads from or writes to a removable, nonvolatile optical disk such as a CD ROM, DVD, and Blu-Ray® disk, or other optical media. Computer-readable storage media **1722** may include, but is not limited to, Zip® drives, flash memory cards, universal serial bus (USB) flash drives, secure digital (SD) cards, DVD disks, digital video tape, and the like. Computer-readable storage media **1722** may also include, solid-state drives (SSD) based on non-volatile memory such as flash-memory based SSDs, enterprise flash drives, solid state ROM, and the like, SSDs based on volatile memory such as solid-state RAM, dynamic RAM, static RAM, DRAM-based SSDs, magneto resistive RAM (MRAM)

SSDs, and hybrid SSDs that use a combination of DRAM and flash memory based SSDs. The disk drives and their associated computer-readable media may provide non-volatile storage of computer-readable instructions, data structures, program modules, and other data for computer system **1700**.

[0247] Communications subsystem **1724** provides an interface to other computer systems and networks. Communications subsystem **1724** serves as an interface for receiving data from and transmitting data to other systems from computer system **1700**. For example, communications subsystem **1724** may enable computer system **1700** to connect to one or more devices via the Internet. In some embodiments communications subsystem **1724** can include radio frequency (RF) transceiver components for accessing wireless voice and/or data networks (e.g., using cellular telephone technology, advanced data network technology, such as 3G, 4G or EDGE (enhanced data rates for global evolution), Wi-Fi (IEEE 1302.11 family standards, or other mobile communication technologies, or any combination thereof), global positioning system (GPS) receiver components, and/or other components. In some embodiments communications subsystem **1724** can provide wired network connectivity (e.g., Ethernet) in addition to or instead of a wireless interface.

[0248] In some embodiments, communications subsystem **1724** may also receive input communication in the form of structured and/or unstructured data feeds **1726**, event streams **1728**, event updates **1730**, and the like on behalf of one or more users who may use computer system **1700**.

[0249] By way of example, communications subsystem **1724** may be configured to receive data feeds **1726** in real-time from users of social networks and/or other communication services such as Twitter® feeds, Facebook® updates, web feeds such as Rich Site Summary (RSS) feeds, and/or real-time updates from one or more third party information sources.

[0250] Additionally, communications subsystem **1724** may also be configured to receive data in the form of continuous data streams, which may include event streams **1728** of real-time events and/or event updates **1730** that may be continuous or unbounded in nature with no explicit end. Examples of applications that generate continuous data may include, for example, sensor data applications, financial tickers, network performance measuring tools (e.g., network monitoring and traffic management applications), click-stream analysis tools, automobile traffic monitoring, and the like.

[0251] Communications subsystem **1724** may also be configured to output the structured and/or unstructured data feeds **1726**, event streams **1728**, event updates **1730**, and the like to one or more databases that may be in communication with one or more streaming data source computers coupled to computer system **1700**.

[0252] Computer system **1700** can be one of various types, including a handheld portable device (e.g., an iPhone® cellular phone, an iPad® computing tablet, a PDA), a wearable device (e.g., a Google Glass® head mounted display), a PC, a workstation, a mainframe, a kiosk, a server rack, or any other data processing system.

[0253] Due to the ever-changing nature of computers and networks, the description of computer system **1700** depicted in the figure is intended only as a specific example. Many other configurations having more or fewer components than

the system depicted in the figure are possible. For example, customized hardware might also be used and/or particular elements might be implemented in hardware, firmware, software (including applets), or a combination. Further, connection to other computing devices, such as network input/output devices, may be employed. Based on the disclosure and teachings provided herein, a person of ordinary skill in the art will appreciate other ways and/or methods to implement the various embodiments.

[0254] Although specific embodiments of the disclosure have been described, various modifications, alterations, alternative constructions, and equivalents are also encompassed within the scope of the disclosure. Embodiments of the present disclosure are not restricted to operation within certain specific data processing environments but are free to operate within a plurality of data processing environments. Additionally, although embodiments of the present disclosure have been described using a particular series of transactions and steps, it should be apparent to those skilled in the art that the scope of the present disclosure is not limited to the described series of transactions and steps. Various features and aspects of the above-described embodiments may be used individually or jointly.

[0255] Further, while embodiments of the present disclosure have been described using a particular combination of hardware and software, it should be recognized that other combinations of hardware and software are also within the scope of the present disclosure. Embodiments of the present disclosure may be implemented only in hardware, or only in software, or using combinations thereof. The various processes described herein can be implemented on the same processor or different processors in any combination. Accordingly, where components or modules are described as being configured to perform certain operations, such configuration can be accomplished, e.g., by designing electronic circuits to perform the operation, by programming programmable electronic circuits (such as microprocessors) to perform the operation, or any combination thereof. Processes can communicate using a variety of techniques including but not limited to conventional techniques for inter process communication, and different pairs of processes may use different techniques, or the same pair of processes may use different techniques at different times.

[0256] The specification and drawings are, accordingly, to be regarded in an illustrative rather than a restrictive sense. It will, however, be evident that additions, subtractions, deletions, and other modifications and changes may be made thereunto without departing from the broader spirit and scope as set forth in the claims. Thus, although specific disclosure embodiments have been described, these are not intended to be limiting. Various modifications and equivalents are within the scope of the following claims.

[0257] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosed embodiments (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to,”) unless otherwise noted. The term “connected” is to be construed as partly or wholly contained within, attached to, or joined together, even if there is something intervening. Recitation of ranges of values herein are merely intended to

serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate embodiments of the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0258] Disjunctive language such as the phrase “at least one of X, Y, or Z,” unless specifically stated otherwise, is intended to be understood within the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0259] Preferred embodiments of this disclosure are described herein, including the best mode known to the inventors for carrying out the disclosure. Variations of those preferred embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate and the inventors intend for the disclosure to be practiced otherwise than as specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

[0260] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein.

[0261] In the foregoing specification, aspects of the disclosure are described with reference to specific embodiments thereof, but those skilled in the art will recognize that the disclosure is not limited thereto. Various features and aspects of the above-described disclosure may be used individually or jointly. Further, embodiments can be utilized in any number of environments and applications beyond those described herein without departing from the broader spirit and scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive.

1. A system comprising:

- a substrate network hosting a service;
- one or more processors; and
- one or more memories storing instructions that, upon execution by the one or more processors, configure the system to further comprise a software defined network (SDN), wherein the SDN includes:
 - a first private network configured for a customer; and
 - a second private network configured for the service and including an endpoint and a domain name system (DNS) proxy,

wherein:

the DNS proxy is configured for the customer and is associated with the first private network, and the endpoint is associated with the service and is configured to receive a DNS resolution request associated with the service and sent from the substrate network, send the DNS resolution request to the DNS proxy, receive a DNS resolution from the DNS proxy, and send the DNS resolution to the substrate network.

2. The system of claim 1, wherein the second private network includes a first subnet and a second subnet, wherein the first subnet includes the endpoint, and wherein the second subnet includes the DNS proxy.

3. The system of claim 1, wherein the endpoint is a first endpoint, and wherein the DNS resolution request indicates a DNS corresponding to a second endpoint of the customer, wherein the first private network or an on-premises network of the customer includes the second endpoint, and wherein the endpoint is accessible to the service via the first endpoint and the DNS proxy.

4. The system of claim 1, wherein the DNS proxy is a first DNS proxy, and wherein the SDN further includes a third private network configured for a different customer, wherein the second private network further includes a second DNS proxy configured for the third private network, and wherein the endpoint supports DNS resolutions associated with third private network for the service via the second DNS proxy.

5. The system of claim 4, wherein the first private network and the third private network have overlapping IP addresses, and wherein the service of the substrate network is a multi-tenant service.

6. The system of claim 1, wherein the endpoint is configured to store a first association between a first reserved IP address associated with the substrate network and an IP address associated with the DNS proxy, wherein the DNS resolution request is received based on the first reserved IP address and is sent based on the IP address.

7. The system of claim 6, wherein the DNS resolution indicates a second reserved IP address associated with the DNS proxy, wherein the service is configured to store a second association between the second reserved IP address and a third reserved IP address associated with the substrate network.

8. The system of claim 7, wherein the endpoint is further configured to receive, from the substrate network based on the third reserved IP address, a request for connecting to a different endpoint of the customer, the different endpoint corresponding to the DNS, and send, based on the second reserved IP address, the request to the DNS proxy.

9. A method implemented by an endpoint hosted by a software defined network (SDN) for a service of a substrate network, the method comprising:

associating a first reserved internet protocol (IP) address with an IP address of a domain name system (DNS) proxy, the DNS proxy hosted by the SDN for a first private network of a customer, the SDN including at least a portion of the first private network;

receiving, based on the first reserved IP address and from the substrate network, a DNS resolution request for a DNS associated with the first private network, the DNS resolution request associated with the service;

sending, based on the IP address of the DNS proxy, the DNS resolution request to the DNS proxy;

receiving, from the DNS proxy, a DNS resolution indicating a second reserved IP address as being associated with the DNS, the second reserved IP address belonging to the DNS proxy; and

sending, to the substrate network, the DNS resolution.

10. The method of claim 9, further comprising:

associating the second reserved IP address with a third reserved IP address, the third reserved IP address belonging to the substrate network;

receiving, from the substrate network based on the third reserved IP address, a request for connecting to a different endpoint of the first private network, the different endpoint corresponding to the DNS; and

sending, based on the second reserved IP address, the request to the DNS proxy.

11. The method of claim 9, wherein the SDN includes a second private network and a plurality of customer private networks, wherein each one of the plurality of customer private networks is configured for a different customer, wherein the second private network includes the endpoint and a plurality of DNS proxies, wherein each one of the plurality of DNS proxies corresponds to a different one of the plurality of customer private networks.

12. The method of claim 11, wherein the endpoint is configured to communicatively couple the service to the plurality of DNS proxies.

13. The method of claim 11, wherein the second private network includes a first subnet and a second subnet, wherein the first subnet includes the endpoint, and wherein the second subnet includes the plurality of DNS proxies.

14. The method of claim 11, wherein at least two of the plurality of customer private networks have overlapping IP addresses.

15. (canceled)

16. (canceled)

17. (canceled)

18. (canceled)

19. (canceled)

20. (canceled)

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